6
Converging Values for Avogadro's Number

Perrin's Comparisons

Those familiar with the philosophical literature on Perrin might be wondering why we have postponed to last his appeals to agreement among several different theory-mediated ways of measuring Avogadro's number, both in his 1909 monograph and his 1911 Solvay Conference paper. Perrin became a subject in this literature only after Wesley Salmon, in his 1984 *Scientific Explanation and the Causal Structure of the World*, argued that Perrin's appeal, at the end of a later edition of his 1913 book, to different measurements of N_0 represents an instance of common-origin reasoning.[1] That appeal consisted of a list of 16 different theory-mediated determinations of Avogadro's number, all but one of them within the range from 60×10^{22} to 69×10^{22}, followed by two sentences representing the primary conclusion of the whole book:

> Our wonder is aroused at the very remarkable agreement found between values derived from the consideration of such widely different phenomena. Seeing that not only is the same magnitude obtained by each method when the conditions under which it is applied are varied as much as possible, but that the numbers thus established also agree among themselves, without discrepancy, for all the methods employed, gives to the real existence of the molecule a probability bordering on certainty.[2]

But if this is Perrin's principal argument, why haven't we treated it as such, focusing our attention on it?

There are, moreover, good reasons independently of Perrin's appeal to devote philosophical effort to the nature of the evidence arising from agreeing theory-mediated measurements of constants. Over the course of the twentieth century, this form of evidence has become increasingly important within microphysics. Consider, for example, the following conclusion drawn in the 2006 CODATA Report:

> The good agreement of the highly accurate values of [the fine-structure constant] α inferred from $h/m(^{133}\mathrm{Cs})$ and $h/m(^{87}\mathrm{Rb})$, which are only weakly dependent on QED theory, with the values of α inferred from [the electron

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> magnetic moment anomaly] a_e, muonium transition frequencies, and H and D transition frequencies provide support for the QED theory of a_e as well as the bound-state QED theory of muonium and H and D. In particular, the weighted mean of the two values of α inferred from $h/m(^{133}\text{Cs})$ and $h/m(^{87}\text{Rb})$, $\alpha^{-1} = 137.03599934(69)[5.0 \times 10^{-9}]$, and the weighted mean of the two values of α inferred from the two experimental values of a_e, $\alpha^{-1} = 137.035999680(94)[6.9 \times 10^{-10}]$, differ by only $0.5u_{\text{diff}}$, with $u_{\text{diff}} = 5.1 \times 10^{-9}$. This is a truly impressive confirmation of QED theory.[3]

If we believe that agreeing values from diverse theory-mediated measurements provide such important evidence, then why have we postponed discussing the evidence from Perrin's appeal to such agreeing measurements to the very end? We shall answer that question in the next two sections before turning to Perrin's actual comparisons. This will prepare us for the several different perspectives from which his comparisons need to be considered.

6.1. Some Historical Background

One reason for delaying until now Perrin's focus on multiple values for N_0 comes from the long history of appeals to agreeing values in diverse theory-mediated measurements and the often very different kinds of evidential reasoning drawn from them. Complementary theory-mediated determinations of the values of key parameters go back at least as far as Ptolemy's *Almagest*, initiating a practice in mathematical astronomy that has nowhere been employed to greater effect than in Kepler's *Astronomia Nova*. In that tradition, however, complementary determinations served only to safeguard against being misled by values that depended on potentially erroneous theoretical assumptions. In other words, they served only as cross-checks on one another, not as evidence for their theoretical presuppositions.

The most spectacular instance of complementary theory-mediated measures in the seventeenth century was Huygens's two four-significant-figure determinations of the acceleration of surface gravity, one employing pendulums and the other conical pendulums.[4] Huygens too seems to have viewed these two different approaches as merely cross-checking one another in a historical context in which the value for the acceleration of gravity he obtained in the two ways was notably greater than any direct determination of it that had been made before him.[5] A case can nevertheless be made that Newton saw the measurements as providing the strongest evidence then available for his first two laws of motion insofar as Huygens's conical pendulum measure presupposed a special case of those laws, while his pendulum measure presupposed only Galilean kinematic

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principles.[6] If this is correct, then Huygens's two measures may have been the prototype for taking agreeing theory-mediated measurements to be providing evidence for the theoretical claims presupposed in them.

Regardless, in the aftermath of Newton, agreeing theory-mediated measurements did come to be viewed as providing evidence beyond mere cross-checks on one another. A celebrated, though now comparatively less known, example became the principal test of Newton's universal—i.e., particle-to-particle—gravity during the second half of the eighteenth and first half of the nineteenth centuries. The oblateness of the Earth could be measured directly. It could also be inferred from the variation of surface gravity with latitude by means of Clairaut's equation, which presupposed universal gravity and hydrostatic equilibrium of the Earth. Decades of effort went into resolving the failure of the values of oblateness from these two to agree with one another, culminating finally in the realization that the Earth is not in hydrostatic equilibrium.[7] Unsuccessful though it was, this was clearly an example of the demand for agreeing measurements serving as a test of a theory by virtue of that theory being presupposed in one method of measurement, but not the other.

Maxwell in 1864 provided the first truly spectacular instance of evidence for a fundamental—indeed, "ontological"—theoretical conclusion coming from agreement between comparatively direct measurements of the speed of light and the inferred speed of electromagnetic propagation implied by his equations. Looking to show that light consists of electromagnetic waves propagating through "an elastic medium," Maxwell derived an equation in his famous monograph on the electromagnetic field for the velocity of propagation of those transverse waves:

$$v = \sqrt{\frac{k}{4\pi\mu}}$$

where k is "the ratio of the electromotive force to the electric displacement" and μ is "the ratio of the magnetic induction in a given medium to that in air under an equal magnetizing force." Using experimental results from Weber and Kohlrausch from the 1850s, Maxwell concluded that the velocity of propagation in air is 310740 km/sec. He compared this with speeds of light in air measured using interferometers by Fizeau, 314,859 km/sec, and Foucault, 298,000 km/sec, and the speed in the space surrounding the Earth inferred from the aberration of light, 308,000 km/sec. From this he concluded:

> Hence the velocity of light deduced from experiment agrees sufficiently well with the value of v deduced from the only set of experiments we as yet possess. . . . The agreement of the results seems to show that light

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> and magnetism are affections of the same substance, and that light is an electromagnetic disturbance propagated through the field according to electromagnetic laws.[8]

A good deal of further experimental effort was needed before this proposal became established, and by the time of Perrin's Brownian motion experiments, Maxwell's "elastic medium" had begun falling by the wayside. That made his comparison of the speeds no less a watershed, however, for it did open the way to the science of optics becoming subsumed under electromagnetic theory. Indeed, a preferred value for the speed of light subsequently came to be obtained from Maxwell's equation, not from any direct measurement.

The other famous example, just prior to Perrin, of a fundamental theoretical—again "ontological"—conclusion coming from agreeing measurements involved the determination by J. J. Thomson and then others of the mass-to-charge ratio *m/e* of what was coming after 1900 to be called the electron. Initially, Thomson had developed a second, complementary way of measuring *m/e* for cathode rays during the summer of 1897, employing crossed electric and magnetic fields, as a cross-check on his first way, in the process confirming, contrary to Hertz's earlier claim, that the rays are displaced by electrostatic forces. There was no overlap at all between the sets of values obtained for *m/e* that Thomson reported from the two ways, but strikingly all the values were three orders of magnitude smaller than the value of *m/e* for hydrogen inferred from electrolysis; and the values showed no systematic variation with changes in the gas and the material of the electrodes in the cathode ray tubes. From this, Thomson concluded that cathode rays all consist of the same negatively charged discrete elements with an *m/e* of a distinctive order of magnitude.[9]

Thomson's far more monumental conclusion from his *m/e* measurements was announced in the December 1899 issue of *Philosophical Magazine*. Using a crossed-field method related to the earlier one, he confirmed that *m/e* for both the photoelectric and the thermionic discharges is of the same distinctive order of magnitude as for cathode rays. Same-distinctive-order-of-magnitude-same-particle reasoning led to the conclusion, subsequently supported by measurements of *m/e* for other electrical phenomena by others, of the ubiquity of this carrier of negative electricity. On that basis, Thomson made the following watershed proposal:

> It thus possesses the characteristics of being a fundamental conception of electricity; and it seems desirable to adopt some view of electrical action which brings this conception into prominence. These considerations have led me to take as a working hypothesis the following method of regarding the electrification of a gas, or indeed matter in any state.[10]

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What follows his proposal, over the next three pages, is the twentieth-century conception of electrical action. At the time Thomson proposed it, he was still relying on merely order of magnitude agreement of values for *m/e* in different phenomena. By 1906, however, the value was being measured to three significant figures.

Although not likely known to Perrin, one other example of agreeing measurements from the end of the nineteenth century should be noted. Starting in the 1840s, a good deal of effort went into refining Cavendish's famous 1798 torsion-bar experiment measuring the mean density of the Earth. Even in the 1890s, many of the results from such experiments were still being stated in terms of the mean density of the Earth, but by then it had become at least as common to quote them instead in terms of the fundamental constant of gravity *G*.[11] *G*, however, is just the constant of proportionality in Newton's law of gravity. The agreeing measurements for *G* were thus serving to confirm that law—that is, they were evidence that the three variable quantities entering into it and no others affect the magnitude of gravitational forces. This corresponded to the agreeing measurements of the gas constant *R* earlier in the century, again a constant of proportionality, but for the ideal gas law. As we noted in Chapter 2, the most impressive aspect of van't Hoff's results for dilute solutions was the emergence of this same value *R* for his constant of proportionality. The agreeing measurements for his constant of proportionality thus did more than just confirm van't Hoff's law; as Perrin himself emphasized, and Poincaré as well, they linked dilute solutions to gases.

The main point of this review of historically salient appeals to agreeing measurements was to lay out one reason why we have adopted a cautious attitude toward what to make of Perrin's appeal to alternative determinations of Avogadro's number: *historically, different kinds of evidential reasoning were being drawn from agreeing measurements*. Therefore we should not be too quick to assign one form of evidential reasoning to all the places where Perrin compares his values from Brownian motion with values derived from other phenomena. We had best look at the comparisons one by one, asking in each case what evidential conclusions each can support, before turning to his appeal to the entire collection of them.

As a matter of fact, Perrin's aim in the first of his articles on Brownian motion that compared his value for Avogadro's number with values obtained in other ways was not one of establishing the reality of molecules. The first paragraph of that short October 1908 article is worth quoting in full:[12]

> The number of molecules N that the total gram-molecule contains, the charge *e* of the electron and the quotient α of the mean energy of a molecule by its absolute temperature T are the *universal constants* which are all known as soon

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> as any one of them is known. In effect, in the electrolysis of a monovalent salt, 96550 coulombs are transported by N atoms, which gives
>
> $$Ne = 3 \times 10^9 \times 96550 = 29 \times 10^{13} \text{ (electrostatic units, C.G.S.)}$$
>
> and, on the other hand, according to the kinetic theory, the product 3RT measures the translational kinetic energy (*force vive de translation*) that any N molecules in a gram-molecule possess, so that
>
> $$2N\alpha = 3R = 3 \times 83.2 \times 10^6.$$

Perrin went on then, in fine print, to give a range for Avogadro's number obtained from packing molecules into a volume and a sphere of equivalent dielectric capacity, 40×10^{22} to 100×10^{22}; a value from van der Waals's reasoning, 60×10^{22}, with an error as high as 40 parts in 100; a range from the measurements from J. J. Thomson's laboratory, 43×10^{22} to 96×10^{22}; the Planck value, 61×10^{22} (obtained from the 2.02×10^{-16} value of α measured for blackbody radiation by Kurlbaum), which Perrin immediately offsets with a value Lorentz "found from a different theory of radiation," 77×10^{22}; and, in a footnote attached to the paragraph, a range from the "beautiful" Rutherford-Geiger article a month earlier, 62×10^{22} to 72×10^{22}. Returning to normal print, Perrin states that he has a method that has the advantages of being "*direct and susceptible to unlimited precision*," namely from his vertical-gradient results, and gives as values 71×10^{22} for Avogadro's number, 4.1×10^{-10} for e, and 1.7×10^{-16} for α. Perrin's point in making the comparisons, accordingly, was not to call attention to the extent of the agreement between the various values, but to propose that his value should be regarded as more reliable than the others.

(Perrin's presentation of the values of others in this October 1908 article initiated an unfortunate practice that he continued at least through the eleventh edition of *Les atomes* in 1921, namely presenting them in a way that undercut any legitimate claims they had to being as well-founded as his own. The egregious example in 1908 was his presentation of the Rutherford-Geiger findings. They had announced a robust measure of $2e$, 9.3×10^{-10}, from collecting α-particles, arguing that it was to be preferred to other directly obtained values for e in spite of its departure from them. From this value they obtained values of 4.65×10^{-10} for "the charge carried by a hydrogen atom," 62×10^{22} for Avogadro's number, 1.61×10^{-24} grams for the mass of "the hydrogen atom," and 2.72×10^{19} for Loschmidt's number.[13] A footnote to this list reads, "It is of interest to note that Planck deduced a value of 4.69×10^{-10} E.S. unit from a general optical theory of the nature of temperature-radiation." A few pages earlier they had reviewed two cross-checks to their value for e, the second of which (by Boltwood) had given

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a value of 4.1×10^{-10} based on a half-life of radium of 2,000 years, versus their own value of 1,760 years. Immediately before the preceding list of values, they expressly rejected this value for *e* as too small owing to its assumed value for the half-life, and they never derived values for any further atomic parameters from it.[14] The second value Perrin attributed to them, 72×10^{22}, was calculated by him on the basis of this 2,000-year half-life, with nothing in their article in any way authorizing it.

Though not so egregious, Perrin's listing of the value implied by Kurlbaum's measurement of the constant in Planck's blackbody formula, side-by-side with a value Lorentz had "found from a different theory of radiation,"[15] undercut the well-founded claim to precision for the measurement of the constants in the blackbody formula attainable from measuring them at several different temperatures. And Perrin never notes the close agreement between the Planck and Rutherford values. To be blunt about it, one needs always to check the original source of any value for Avogadro's number besides his own that Perrin quotes. We shall return to this point later, for it has had some unfortunate historical consequences, and not just in the recent philosophical literature.)

6.2. What *Agreeing* Measurements?

We have a second reason for being cautious about Perrin's comparisons, a reason arising from the spread among the numbers being compared and concerns over criteria for *agreement* between measurements. Perrin himself kept revising his values for Avogadro's number. In particular, his announced preferred value from his vertical-gradient experiments was 6.7×10^{23} in May 1908 (subsequently corrected to 5.7×10^{23}), 6×10^{23} in September, 71×10^{22} in October, 70.5×10^{22} in his 1909 monograph, 68.3×10^{22} in the table at the end of his 1911 Solvay Conference paper and his *Les atomes* in 1913. From the granule displacement experiments, Chaudesaigues gave a value of 64×10^{22} in November 1908, while Perrin's preferred value was 71.5×10^{22} in his 1909 monograph, the mean of results from a series of multiple trials. A table in his 1911 paper and 1913 book lists results of the trials, with three entries involving more than 500 displacements recorded, yielding values of 69.5×10^{22}, 72.5×10^{22}, and 68.8×10^{22}; in both 1911 and 1913 he chose the last of these values, for which the number of recorded displacements was greatest, as the value he subsequently compared to others.

We are not questioning Perrin's judgment as an experimentalist in preferring later to earlier values, for the difficult experiments were undergoing continuing development over the years in question. Perrin himself offered a summary of his results from his vertical-gradient measure in 1913 that seems to us a reasonable statement of the overall situation for all his results: "In spite of all these variations

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[in the range of emulsions investigated], the value found for Avogadro's number N remains approximately constant, varying irregularly between 65×10^{22} and 72×10^{22}."[16] This, however, still raises a question about criteria of "agreement" and whether the values can be viewed as in agreement from one comparison to another. Perrin may have said that all the values agree with one another within the respective margins of experimental error; but that does not mean that everyone at the time should have agreed.

Perrin's spread was tight enough not only to deter questions about his values until 1911, but to prompt Ostwald's 1908 remark about Perrin having provided "probably the most accurate value for the molecular constant as yet obtained," 71×10^{22}.[17] By contrast, the spread in values for Avogadro's number obtained in other ways was immense, raising questions about the standard that has to be met in order for different values to agree with one another. Taking the difficulty of many of the experiments and the associated measurement uncertainty into consideration, Wesley Salmon remarked, "Let us say that values lying between 4×10^{23} and 8×10^{23} are acceptable."[18] Salmon was endeavoring to give a philosophical analysis of the evidential force of what he took scientists to be granting were agreeing measurements for N_0. Prior to the issue of the philosophic basis of any such evidence, however, is the issue of whether any, and if so what, claims could legitimately have been drawn at the time from comparisons among the different theory-mediated measurements of N_0. Rather than settling the issue of the proper range for agreement, therefore, Salmon's proposal simply raises the question, on what principled basis should the range for agreement be decided? In the case of J. J. Thomson's *m/e* measurements, taking the range of agreement to be an order of magnitude was justified because all his results were three orders of magnitude less than any prior value for *m/e*. In other words, the range of his measured values was appropriately assessed relative to the smallest prior value. Nothing like that was available, however, in the case of N_0. As Table 2.1 in Chapter 2 showed, the outstanding feature of the prior proposed values for N_0 from kinetic theory was their lack of agreement with one another.

This worry about what properly constituted agreement is our other reason for having postponed until last a discussion of the evidence from the comparisons Perrin made between the values for N_0 derived from Brownian motion and the values derived from other phenomena. It also gives a further reason to consider each such comparison individually, asking what to make of it, rather than turning immediately to the question of what to make of the claimed agreement among so many different ways of measuring it.

We remarked at the beginning of the paragraph before last that Perrin's values went largely unquestioned before 1911. Three individuals ended this in 1911. Millikan published his first oil-drop measurements of electric charge, announcing a value for *e* of 4.891×10^{-10} esu and for N_0 of 59.22×10^{22}.[19] And, working

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under Millikan, Harvey Fletcher published results on the Brownian motion of oil-drops in air, announcing a value for N_0 of 57.5×10^{22}, with a consequent value for e from eN_0 of 5.01×10^{-10} esu, within 2.5 percent of Millikan's value.[20] While Fletcher, like Perrin, obtained his value for N_0 by taking the mean kinetic energy of his drops to match that of molecules, his approach, he emphasized, had a special virtue:

> This shows that *all the distributions spoken of in Part I., and also the average value of the Brownian displacement in one second can be calculated without knowing the density of the drop, the viscosity or pressure of the gas, or the law of motion of a sphere through a viscous medium. The only assumption involved is that the velocity of steady motion is proportional to the moving force,* an assumption that is fully justified both here and elsewhere.[21]

And finally, published in the midst of the 1911 papers by Millikan and Fletcher, Boltwood and Rutherford had announced a value for Avogadro's number of 60.3×10^{22} by collecting a measured number of α-particles sufficient to form a gas and then determining the volume they occupy at standard conditions.[22] Even though none of these authors expressly challenged Perrin's values in 1911, the claims made for precision by Millikan and Fletcher and the comparative directness of the measure by Boltwood and Rutherford did raise questions about whether Perrin's values might be too large by more than 10 percent.

These questions gained added force in 1912 when Svedberg published results from new Brownian motion experiments on diffusion and displacement of gold hydrosol particles, announcing values for N_0 of 58×10^{22} and 62×10^{22}, respectively—both falling below Perrin's range.[23] Then in 1913 Millikan published his legendary oil-drop results, announcing values for e and N_0 of $4.774 \pm .009 \times 10^{-10}$ esu and $6.062 \pm .012 \times 10^{23}$, and adding, "So far as I am aware, there is at present no determination of e or N by any other method which does not involve an uncertainty at least 15 times as great as that represented in the above measurements."[24] The question of what Perrin's comparisons of his values for N_0 with those obtained in other ways yielded in the way of evidence was accordingly undergoing a metamorphosis over the course of the years between his first published result and our self-imposed historical boundary of the emergence of the Bohr model. This question continued to gain force beyond that boundary, for example when Fletcher in 1914 published a new value, $60.3 \pm 1.2 \times 10^{22}$, based on refined measurements for Brownian motion of oil-drops in the air.[25]

By the way, viewed retrospectively from the other side of our boundary, Millikan's error-bands were mistaken, but not his insistence that his value was more reliable than Perrin's. Perrin's range of values extended from 8 to 20 percent higher than our current (2018) value of $60.2214076 \times 10^{22}$. Subsequent textbooks

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have attributed the discrepancy to imprecision in his determinations of the diameters of his granules.[26] This, however, does not make much sense. His values obtained from his vertical-gradient and granule rotation experiments depend on granule diameter cubed, while his values obtained from granule displacement and diffusion depend on it to the first power; yet there is no corresponding variation in Perrin's values. Taken together with the viscous action of the liquid on his granules, Perrin's high values for Avogadro's number seem to us more likely to have resulted from his measured values of their mean kinetic energy not being equal to, but instead consistently a little less than, the mean molecular kinetic energy at the same temperatures. It goes without saying that Perrin's published values for mean granule kinetic energy, and hence for Boltzmann's constant, turned out to be less than subsequent molecular values.

In the light of these considerations and those in section 6.1, we are going to examine Perrin's comparisons with those of others in three steps: first, looking at the evidence each of the individual comparisons he made offered in support of the validities of the respective measures; then reviewing his downplaying of the seeming lack of agreement in those individual comparisons; and finally, in section 6.5, considering his claim that the overall set of values, taken together, provided decisive evidence for the reality of molecules, that is, for the salient presupposition that all the different measures had in common.

As an aid to the reader, Table 6.1 lists the principal published values for Loschmidt's number (the number of molecules in a cubic centimeter of ideal gas at standard conditions) and Avogadro's number (the number of molecules per mole)[27] from between Perrin's first published value in 1908 and the 1914 publication of the fourth revised edition of his 1913 book, along with two benchmark values obtained by Planck in 1900 from blackbody radiation and by J. J. Thomson in 1903 from water-drop measurements that were the forerunners of Millikan's oil-drop values. By way of clarification, to say that values are obtained from k means that they are obtained from a measured value of k in conjunction with the equation kN_0 = the gas constant R; and that they are obtained from e means that they are obtained from a measured value of e in conjunction with the equation eN_0 = Faraday's constant (the quantity of electricity required to liberate one mole of a monovalent element in electrolysis). During the period in question there were no "official" recommended values for either the gas constant or Faraday's constant, and hence the quoted numbers depend on the choices for these constants made by the respective authors. (We too often forget what a landmark effort Raymond Birge's 1929 paper, "Probable Values of the General Physical Constants," truly was.[28])

We shall be discussing most of these individual values in the following sections. Two points about them are appropriate to end this section. On the one hand, the spread of the values from 1908 to 1913 was remarkably small compared

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Table 6.1 Principal Published Values for Loschmidt's and Avogadro's Numbers, 1900–1914

2.76×10^{19}	61.75×10^{22}	Planck	1900	From blackbody radiation $k = 1.347 \times 10^{-16}$
3.9×10^{19}	$[87 \times 10^{22}]$	Wilson, J. J. Thomson	1903	From $e = 3.1 \times 10^{-10}$ (water drop)
$[31.6 \times 10^{18}]$	$[70.8 \times 10^{22}]$	Millikan, Begeman	1908	From $e = 4.06 \times 10^{-10}$ (water drop)
$[3.0 \times 10^{19}]$	6.7×10^{23}	Perrin	1908	From Brownian motion: vertical gradient
2.72×10^{19}	62.0×10^{22}	Rutherford, Geiger	1908	From α-particle $2e = 9.3 \times 10^{-10}$
$[32 \times 10^{18}]$	71×10^{22}	Perrin	1908	From Brownian motion: vertical gradient
$[29 \times 10^{18}]$	64×10^{22}	Chaudesaigues	1908	From Brownian motion: displacements
$[28 \times 10^{18}]$	62×10^{22}	Perrin	1909	From van der Waals, applied to argon
$[31 \times 10^{18}]$	70.5×10^{22}	Perrin	1909	From Brownian motion: vertical gradient
$[31.9 \times 10^{18}]$	71.5×10^{22}	Perrin	1909	From Brownian motion: displacements
$[29 \times 10^{18}]$	65×10^{22}	Perrin	1909	From Brownian motion: rotation
2.70×10^{19}	$[60.4 \times 10^{22}]$	Regener	1909	From α-particle $2e = 9.58 \times 10^{-10}$
$[2.76 \times 10^{19}]$	$[61.9 \times 10^{22}]$	Begeman	1910	From $e = 4.669 \times 10^{-10}$ (water drop.)
2.72×10^{19}	$[61.0 \times 10^{22}]$	Millikan	1910	From $e = 4.65 \times 10^{-10}$ (water drop.)
2.644×10^{19}	59.22×10^{22}	Millikan	1911	From $e = 4.891 \times 10^{-10}$ (oil drop.)
2.69×10^{19}	$[60.3 \times 10^{22}]$	Boltwood, Rutherford	1911	From collecting α-particles to form a gas
$[2.57 \times 10^{19}]$	57.5×10^{22}	Fletcher	1911	From Brownian motion: oil droplets in air
$[2.75 \times 10^{19}]$	$[61.7 \times 10^{22}]$	Warburg	1911	From blackbody radiation
$[30.4 \times 10^{18}]$	68.2×10^{22}	Perrin	1911	From Brownian motion: vertical gradient

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Table 6.1 *Continued*

[30.7×10^{18}]	68.8×10^{22}	Perrin	1911	From Brownian motion: displacements
[30.8×10^{18}]	$69 \times 10^{22} \pm 3\%$	Perrin, Brillouin	1911	From Brownian motion: diffusion
[2.6×10^{19}]	58×10^{22}	Svedberg	1912	From Brownian motion: diffusion
[2.8×10^{19}]	62×10^{22}	Svedberg	1912	From Brownian motion: displacements
2.77×10^{19}	62.0×10^{22}	Planck	1913	From blackbody radiation $k = 1.34 \times 10^{-16}$
2.705×10^{19}	$(60.62 \pm 0.12) \times 10^{22}$	Millikan	1913	From $e = (4.774 \pm 0.009) \times 10^{-10}$ (oil drop)
[2.69×10^{19}]	$(60.3 \pm 1.2) \times 10^{22}$	Fletcher	1914	From Brownian motion: oil droplets in air

As of CODATA 2014, a century later (but prior to the new SI units of 2019 in which values for these became exact), Loschmidt's number = $2.68665664 \times 10^{19}$, Avogadro's = $60.22140857 \times 10^{22}$, and $e = 4.803204673 \times 10^{-10}$ esu. The earliest quasi-official values, listed in Birge (1929), were $e = (4.770 \pm 0.005) \times 10^{-10}$ esu, from which he derived a value of $(60.64 \pm 0.06) \times 10^{22}$ for Avogadro's number (corresponding to a value of 2.705×10^{19} for Loschmidt's number). Numbers enclosed in brackets were not stated in the publications cited, but were instead calculated here from values stated in them. The numbers in the table are not entirely consistent with one another owing to different authors not employing the same values for the gas constant or Faraday's constant.

with that for the values obtained from kinetic theory during the 35 years before Planck's, as listed in Table 2.1. On the other hand, what value was to be preferred among those from 1908 to 1913 was surely not clear. Ostwald in 1908 chose 71×10^{22} from Perrin as "the most probable value."[29] In a letter to Perrin in late 1909, Einstein was already focusing more on the differences than on the agreement:

> A precise determination of the size of the molecule appears to me of the highest importance, moreover, because the radiation formula of Planck can be more sharply proved through this than through measurements of radiations. The Planck theory of radiation leads to a determination of the absolute size of atoms, with *exact* validity claimed. Since the difference
>
> $$61 \times 10^{22} \text{ (Planck)}$$
> $$70.6 \times 10^{22} \text{ (Perrin)}$$
>
> between the Planck value and your value appears critical to me, it must be said at this time that the Planck formula is apparently only theoretically

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> grounded. Why the question of whether Planck's formula is correct appears so important to me, you can see from the papers which I am sending you with this.[30]

Four years after this letter, Bohr, faced with having to select a value for e in order to derive a value for Rydberg's constant for hydrogen from his model of the hydrogen atom, chose 4.7×10^{-10} esu, corresponding to a value of 61.5×10^{22} for Avogadro's number, that is, a value at some remove from Perrin's.[31] The formula for the Rydberg frequency that Bohr employed in this calculation was:[32]

$$\frac{2\pi^2 me^4}{h^3}$$

Insofar as the only means for obtaining a value for m, the mass of the electron, was from the known value of e/m (for which Bohr employed 5.31×10^{17}), in effect the value for e entered into this formula to the fifth power. The value for the Rydberg constant that Bohr published in 1913 was 3.1×10^{15} Hz, within roughly 6 percent of the value measured at the time, 3.29×10^{15}. Had Bohr employed the value for e, 4.23×10^{-10} esu, implied by a Perrin value of 68.5×10^{22} for N_0, his calculated value for the Rydberg constant would have been 1.8×10^{15}, a discrepancy of 55 percent from the measured value. While we have not done the historical research to confirm it, Bohr's calculation appears soon thereafter to have put an end to anyone at the cutting edge of physics after 1913 taking Perrin's value for Avogadro's number to be as well-founded as those no greater than 62×10^{22}.[33]

6.3. Perrin's Comparisons, Individually

The comparison Perrin always made, beginning with his initial May 1908 paper, was with values of Avogadro's number obtained from kinetic theory, typically from the combination of the measured viscosity of gases and van der Waals's equation. The value from kinetic theory Perrin gave varied from 7×10^{23} in that first paper to 6×10^{23} in September 1908 (Einstein's choice in 1905 and 1906) and 60×10^{22} in October, and then (after stating bounds of 45×10^{22} and 200×10^{22} based on different gases) to 62×10^{22} based on argon in his 1909 monograph, the value he continued to cite in 1911 and 1913. As we said in section 5.6 and at the end of section 5.5 in Chapter 5, Perrin needed this comparison in order to link his results for Brownian motion to the molecules of kinetic theory, for this was the only way he had for bridging the gap between the two realms. In terms of the evidence deriving from the comparison, therefore, it provided grounds for equating the mean translational kinetic energies of the granules to the mean translational kinetic energies of molecules in gases. This was the sole empirical evidence that

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Perrin had for *equating* the two! All the other values for Avogadro's number with which he compared his were not connected in any way to kinetic theory. Absent this one, therefore, all he had was a question—how does the granular kinetic energy compare with the kinetic energy attributed to molecules?—to which he had no other recourse for answering except stipulation. This is surely why he came to prefer the value from argon, and continued to do so after 1911, even though it fell closer to the values proposed by Planck, Millikan, and Rutherford and his coauthors than to his. It offered definite support for his equating the energies.

In one respect, nevertheless, we are understating the evidential value of this primary comparison. All of Perrin's Brownian motion values for Avogadro's number were based on a formula at the heart of kinetic theory:

$$\frac{2}{3}NW = RT$$

where W is the mean translational kinetic energy of the *hypothesized* discrete elements, that is, the molecules of Clausius and Maxwell. W, R, and T all represent numbers that are not integers; the only basis for saying that N represents an integer is its being stipulated to do so. Everywhere Perrin turned to this formula to obtain a value for N, the question he was answering was, *given the measured value of the mean translational kinetic energy of the granules, how many granules are required to reach the characteristic energy of a mole*? Since granules are discrete, the answer to this question has to be, strictly speaking, an integer. In other words, Perrin's reasoning indisputably involves a count of discrete items. The same cannot be said of the value for Avogadro's number obtained from the viscosity of gases and van der Waals's equation. Indeed, it cannot be said of all but a few of the other approaches to obtaining values for N_0 we are about to consider. In this respect, then, Perrin's comparison of his values with the kinetic theory value he cited for argon was lending at least a little support for the latter's also numbering discrete items.

The first place Perrin compared his value for N_0 with values not obtained from kinetic theory was in his October 1908 paper, "Grandeur des molécules et charge de l'électron," the paper Ostwald expressly cited. As we said earlier, Perrin there noted both the values J. J. Thomson and his colleagues were obtaining from their efforts to pin down the fundamental charge and the values obtained from radiation via the contrasting theories of Planck (61×10^{22}) and Lorentz (77×10^{22}), contending that his value should be preferred to those.[34] In a footnote he also noted the range of values (between 62×10^{22} and 72×10^{22}) from measurements involving radioactivity reported by Rutherford in his "beautiful memoir" of August 1908.[35] We shall look at these three comparisons in the order Perrin considers them in his 1909 monograph, where radioactivity precedes blackbody radiation.

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6.3.1. Comparison with Values from Ions

The first question Perrin had to answer concerned which values of Avogadro's number put forward by Thomson and others he should compare with his values. In October 1908, Perrin cited four values for e—3×10^{-10} esu from Townsend; 3.4×10^{-10} and 6.8×10^{-10} from Thomson before 1900; and 3.1×10^{-10} from Harold Wilson working under Thomson—giving a range for N_0 from 43×10^{22} to 96×10^{22}. In his 1909 monograph, he cited slightly different values of e, giving a range 45×10^{22} to 85×10^{22}, which he remarked "is the same degree of precision of the determination of van der Waals."[36] Then, after commenting on "the large sources of error" in determining e, he cited the values of 4.05×10^{-10} and 72×10^{22} from new experiments by Millikan and Begeman (1908, employing suspended water droplets) that "appear to have permitted more accuracy."[37] But then Perrin immediately turned to experiments for measuring the charge, carried out independently by Ehrenhaft and de Broglie, that had avoided relying on the condensation of water droplets around ions, yet still had a significant source of uncertainty from the dust-particles they had used varying in size and perhaps shape. Perrin's 1909 conclusion:

> So it is found that the exact value of e cannot differ much from 4.64×10^{-10} (Ehrenhaft) or 4.5×10^{-10} (de Broglie), which makes N 64.10^{22}. In spite of the uncertainty pointed out, I am inclined to consider this method more precise and more easy to perfect than that which depends upon the condensation of drops of water by expansion.[38]

By the time of the first Solvay Conference at the end of October 1911, Millikan and Begeman had not only published refined values for water-drops, but Millikan had published his first oil-drop experiments and his value of 59.22×10^{22}, without error-bands, for Avogadro's number. Perrin's paper "Les preuves de la réalité moléculaire" for the conference went on at some length on the determination of e and the recent work by Millikan and Fletcher. It concluded that "Millikan's experiments demonstrate in a decisive manner the existence of an atom of electricity equal to the charge carried by the hydrogen ion in electrolysis."[39] Perrin repeated this sentence, and much of the discussion preceding it, in his *Les atomes*, where he cited Millikan's 1911 values, 4.89×10^{-10} and 59×10^{22}, saying of the latter that it "is, after all, in remarkable agreement with the value 68×10^{22} given above."[40] In the table at the end of his 1911 Conference paper, the value he gives for Avogadro's number based on "charged spheres (in a gas)" is 64×10^{22},[41] yet in the first five editions of *Les atomes*, this number becomes 68×10^{22}, still further removed from Millikan's.[42] By the 1921 edition of the book, however, the designation has become "charge as microscopic particles," and the value,

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61×10^{22} followed by a question mark.[43] None of these variations in the listed number is explained in the text. Why he never lists Millikan's actual published values in his concluding table we leave to others.

By 1913, Perrin was in the ambivalent position of wanting to stress agreement between the value of Avogadro's number obtained from Brownian motion and the value from *e*, yet at the same time wanting to insist that his value was to be preferred to Millikan's. Perhaps we should, as earlier, just rely on Ostwald's assessment of the comparison in his 1909 book. There, without citing anyone, he referred to a method of calculating the number of molecules from the mean cross-section of gas ions, claiming it agrees with the value proposed earlier (71×10^{22}), with a corresponding value for the fundamental charge in agreement with the unit charge of an electrolytic ion, "about 10^{-19} coulomb" (3.3×10^{-10} esu).[44]

One reason for turning to Ostwald in this regard is his having singled out "the isolation and counting of gas ions, on the one hand, which have crowned with success the long and brilliant researches of J. J. Thomson" in conjunction with Perrin's results on Brownian motion as justifying "the most cautious scientist in now speaking of the experimental proof of the atomic nature of matter."[45] When we turn to the chapter "Conduction in Gases and Radioactivity" that Ostwald added in the new edition of his *Grundriss*,[46] however, we find little or no significance attached to comparing values for Avogadro's number in the two ways. Instead, Ostwald emphasizes three lessons from the work by Thomson and his associates: (1) "gas ions behave from the point of view of the kinetic theory like molecules of ordinary gases, and each ion carries the same charge as a univalent electrolytic ion";[47] (2) the difference between gas ions and the negatively charged particles "a thousand times smaller than the mass of a hydrogen atom" known as electrons, "the individual existence of which has been proved experimentally"—"the last product of analysis in this region";[48] and (3)

> the possibility of individualizing and counting the number of electrical charges and of ions. For, in contrast with most other effects, in which the sum of all the particles present is obtained, and which do not afford any means of *distinguishing between separate particles and a continuous system*, every particle here acts individually and summation occurs later, if at all.[49] [emphasis added]

Putting words in Ostwald's mouth, then, what he found so important in Thomson's research were the atomicity of electric charge, the discreteness of gas ions, and the similarity in diffusion between those ions and the molecules of kinetic theory. These three thus provided "experimental evidence of the discrete or grained nature of matter" quite independently of the results from Perrin and others on Brownian motion. All three Perrin stressed in his 1909 monograph and thereafter, and hence, although he also stressed the "remarkable agreement"

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in the values for N_0 more than Ostwald did, he and Ostwald did not otherwise differ on the significance of the research on the "atom of charge."

That still leaves us with our primary question: taking the agreement between the values for Avogadro's number for granted, what evidence did it, in and of itself, provide? The values from Brownian motion were always answers to the question, *given the mean translational kinetic energy $W_T T$ of the granules, how many of them are needed to match the characteristic energy RT of a mole of gas or dilute solution?* The values from the atom of charge, at least on their face, were answers to the question, *given the fundamental unit of charge e, how many of those units are needed to match the total electricity required in electrolysis to produce a mole of a monovalent gas like hydrogen—that is, to match Faraday's constant?* Obviously, something more was needed for the answers to either of these questions to be about the same thing, namely "molecules."

We have repeatedly noted the two-part additional premise serving that purpose in the case of Brownian motion: *the characteristic energy RT of a gas or dilute solution consists of kinetic energy distributed across discrete elements forming a gas or dilute solution—call them "Maxwellian molecules"; and the mean translational kinetic energy of those elements is the same at any given temperature as that of the granules*. Correspondingly, the two-part additional premise in the case of the atom of charge was: *a monovalent gas like hydrogen consists of discrete, chemically stable elements—call them "chemical molecules," like H_2; and one fundamental unit of charge is needed to produce each such element during electrolysis*. Framed this way, the evidence from agreement between the two measures of Avogadro's number can be thought of as amounting to evidence that "Maxwellian molecules," the molecules of the kinetic theory of heat, and "chemical molecules," the molecules of chemistry, are one and the same.

This way of thinking about the matter, however, fails to capture the historical situation. As Perrin invariably emphasized, Townsend's experiment had provided independent evidence that the *Ne* for a mole of a monovalent gas, like hydrogen, in electrolysis matches the *Ne* of a mole of ions in an ionized gas, like hydrogen subjected to x-rays. This result had been confirmed by others, most notably by Fletcher in 1911, who obtained a value for *Ne* for gaseous ionization of 2.88×10^{14} esu versus 2.896×10^{14} esu for electrolysis.[50] What's more, J. J. Thomson and his associates and Fletcher under different assumptions had provided evidence that ions in a gas are discrete and hence countable—Thomson under the assumption of one gaseous ion per condensed water droplet, and Fletcher, one per oil-drop. (Recall that Ostwald had emphasized that ions could be counted in the quote from him given earlier.) And finally, Townsend's experiment had centered on the diffusion and hence the mobility of gaseous ions in an electric field, so that the ions involved in his determination of *Ne* amounted to the Maxwellian discrete elements of kinetic theory.

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Accordingly, the situation in which the two sets of values for Avogadro's number were being compared was one in which there was independent evidence that the chemically characterized discrete elements in electrolysis and ionization were the same as the dynamically characterized discrete elements of kinetic theory. But then the main thrust of the evidence from agreement between these values of Avogadro's number was not just to add more evidence for that conclusion. Rather, the situation became like that with Huygens's two ways of measuring surface gravity. The agreement was providing mutual support for the premises entering into each of the two kinds of theory-mediated measurement, one derived from the universality of the gas constant and the other from the universality of Faraday's constant. These two kinds of measurement had a key assumption in common: stated most generally (and using Ostwald's words), "the discrete or grained nature of matter." So, Perrin was fully justified in claiming that the agreement was evidence for the "reality of molecules"—*at least as an assumption licensing theory-mediated measurements*—for what the convergence of the values for Avogadro's number between the two approaches was doing was to simultaneously support the validity of the complementary measures.

How much to make of this evidence is another question—surely not as much as with the nine significant-figure agreement we presently have in the case of the fine-structure constant, or even the four significant-figure agreement that Huygens obtained for the strength of surface gravity in the middle of the seventeenth century. The obvious place to question the Brownian motion measurements was the premise that the mean translational kinetic energy of the granules matches that of the molecules in the surrounding liquid. The no less obvious place to question the atom of charge measurements was the premise of one fundamental unit of charge per molecule in electrolysis and per ion in gaseous ionization, with the consequence of one or more units of charge per water (or oil) droplet. The agreement to within 15 or so percent of the values for Avogadro's number obtained over the period from 1908 to 1911 from the two different phenomena reduced the grounds for challenging these premises. And, in the process, it gave grounds for concluding that phenomena of electricity in liquids and gases, on the one hand, and dynamical phenomena in emulsions, on the other, have something fundamental in common. Let this be enough of an answer for the moment. We shall return to the question later, after we have considered the other agreeing measurements that Perrin cited.

A final comment is in order about the agreement between the measurements of N_0 we have been considering so far. Playing complementary theory-mediated measurements off against one another can turn into a continuing process. Once Millikan published values for Avogadro's number with bands of uncertainty undercutting Perrin's values, the situation changed. Two questions then became central: Which value is preferred? And what are the sources of uncertainty? Tight

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bands of uncertainty are typically the basis for answering the first question, as they seemed to have been in many people's eyes in this case. How much effort to devote to pinning down sources of uncertainty or, for that matter, sources of systematic error, varies historically from one case to another. What makes the difference, it would appear, is how dependent ongoing research at the time is on the evidence from the challenged measurements. When Millikan's value for *e* came under dispute in the early 1930s, a great deal of effort was put into pinpointing the source of systematic error in his measurements.[51] Little effort, by comparison, was put into identifying specific sources for the deviation of Perrin's values from Millikan's, either at the time or in the ensuing years. Whether the mean kinetic energy of granules is exactly the same as that of molecules, or only nearly the same, seems at the time not to have mattered all that much to most of those other than Perrin.

6.3.2. Comparison with Values from α-Particles

Turning now to the phenomenon of radioactivity, while research on it had been going on for more than a decade, it was in June 1908, and hence virtually contemporaneous with Perrin's publishing his values from Brownian motion, that Rutherford and Geiger announced their values for Loschmidt's and Avogadro's numbers from α-particle radiation, publishing them two months later. The key step in both this determination and the one by Rutherford and Boltwood three years later was an ingenious experimental design that enabled each α-particle entering a narrow tube to be individually recognized, by an abrupt "throw" of an electrometer needle, and hence each of them to be truly counted one by one, at a rate typically around 4 per minute. From this, the rate of release of α-particles from one gram of radium could be inferred with only minimal assumptions concerning the particles themselves. That rate, given in the first of the two Rutherford-Geiger papers of 1908, was 3.4×10^{10} per second from a gram of radium in equilibrium.

(One of the main questions answered in that paper was whether "the number of scintillations observed on a properly-prepared screen of zinc sulphide is, within experimental error, equal to the number of α-particles falling on it, as counted by the [new] electrical method."[52] The answer was yes, and by virtue of the agreement between the two ways of counting, the count from scintillations provided a cross-check of the count from the new electrical method.)

This new electrical method thus yielded strong evidence that α-particles are discrete (and hence countable). It gave evidence as well that they all carry the same charge. Prior to it, the evidence for discreteness had come from the discrete character of the visible scintillations and, even more so, from extensive prior

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measurements of the charge-to-mass ratio of α-particles in crossed electric and magnetic fields that had yielded a *prima facie* universal value of 5.07×10^3 emu/gm (versus 9.63×10^3 measured for hydrogen); just as with the electron, the well-behaved character of α-particles in crossed fields was providing the principal evidence that they all have the same charge-to-mass ratio.

Given the rate of release of α-particles from radium, the charge per particle could then be determined from the total charge over fixed times accumulating on a collector. The values announced by Rutherford and Geiger in their 1908 paper accompanying the one noted earlier were 9.2×10^{-10} esu from a first experiment and 9.4×10^{-10} from a second, yielding the mean of 9.3×10^{-10} esu, their announced value.[53] This value fell between two and three times most of the values for e that had been announced before, in particular the value announced by Wilson in 1903. That initiated the next key step in their reasoning:

> On the general view that the charge e carried by an hydrogen atom is the fundamental unit of electricity, we conclude that the charge carried by an α-particle is an integral multiple of e and may be either $2e$ or $3e$.
>
> We shall now consider some evidence based on radio-active data, which indicates that the α-particle carries a charge $2e$ and that the ordinarily accepted values of e are somewhat too small.[54]

They turned to two cross-checks (the second of which, based on work by Boltwood, Perrin treated as an independent determination of Avogadro's number, quoting a value from it of 70.6×10^{22} in his 1909 monograph). The upshot was a value for e of 4.65×10^{-10} esu, "a somewhat higher value than those found in the measurements of J. J. Thomson, H. A. Wilson, and Millikan."[55] Rutherford and Geiger then singled out the likelihood of evaporation in the water-drop experiments of each of these three, arguing that their new value for e was more reliable; in a footnote, they cited as well Planck's deduced value of 4.69×10^{-10} esu.

From this value and Faraday's constant they then derived 2.72×10^{19} as their value for Loschmidt's number and 62×10^{22} for Avogadro's. From the combination of their value for e and the prior value for e/m of the α-particle, they obtained 3.84 as an atomic weight for the particle. From this they concluded,

> The atomic weight of the helium atom is 3.96. Taking into account probable experimental errors in the estimates of the value of E/M for the α-particle, we may conclude that *an α-particle is a helium atom*, or, to be more precise, *the α-particle, after it has lost its positive charge, is a helium atom*.[56]

This last remark should remind readers that at the time the question of the structure of an atom—meaning, the question of how it has neutral charge in

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spite of apparently containing negatively charged electrons—was entirely unresolved. Rutherford did not propose his planetary model (on the basis of α- and β-particle scattering results obtained in experiments with Geiger and Marsden) until 1911, and Bohr did not provide an explanation for how such a planetary configuration could survive without releasing continuing radiation and hence energy until 1913.

(We have gone into such detail on the Rutherford-Geiger 1908 value for Avogadro's number because Perrin—in his 1909 monograph, his 1911 Solvay Conference paper, and his *Les atomes*—portrayed it somewhat differently from the way they had. *The Rutherford-Geiger value, on their view, was just a variant on the atom of charge value discussed earlier, but with a new, better founded value for the atom of charge.* Perrin in 1909 chose to present it as giving two complementary methods for obtaining Avogadro's number from phenomena of radioactivity, one of which Rutherford had regarded as a less reliable cross-check of the method by which he and Geiger had reached their values for e and hence N_0.)

This still leaves us with the question of the force of the evidence resulting from the "agreement" between Perrin's Brownian motion values for N_0 and the 1908 value Rutherford and Geiger had obtained from α-particle radiation. Notice first that this evidence hinged on an assumption stated by Rutherford and Geiger that there is such a thing as "the fundamental unit of electricity." For, on its face, the fact that their value for the charge of an α-particle lay between $2e$ and $3e$, given the values for e that had been obtained from ionization, raised the possibility that there is no such thing as a single universal discrete unit of charge. That worry was not addressed until Millikan published his revised water-drop paper in 1910 and his first oil-drop paper in 1911,[57] both confirming the claims Rutherford and Geiger had made in 1908 that the value of e had been underestimated before them and that the positive charge of the α-particle is $2e$. Perrin was thus on sound grounds in saying, both at the Solvay Conference of 1911 and in his 1913 book, "Millikan's experiments demonstrate in a decisive manner the existence of an atom of electricity equal to the charge carried by the hydrogen ion in electrolysis."[58]

Accordingly, the question of the evidence from the "agreement" between Perrin's and the Rutherford-Geiger values for Avogadro's number is best considered as of 1911 rather than at the time of Perrin's initial appeal to it in 1908. By 1911, however, the significance to attach to the Rutherford-Geiger value had changed dramatically because of three further results. First, Erich Regener in Berlin had carried out a version of the Rutherford-Geiger experiments, publishing in 1909 a value of 9.58×10^{-10} esu for the charge of α-particles and hence of 4.79×10^{-10} esu for the elementary charge.[59] Second, Rutherford and Royds in that same year had provided "conclusive proof that the α particle after losing its

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charge is an atom of helium" by verifying that the fingerprint of helium, its spectrum, is observed after a few days of collecting the discharge from radium.[60]

Third, and most striking of all, Rutherford and Boltwood, proceeding from results by Sir James Dewar in 1908 and continuing through 1911 with further results by him and by Rutherford and Boltwood, had obtained increasingly reliable values for the rate of production of helium by radium, as measured by the volume occupied at standard conditions.[61] Coupled with the rate of α-particle production measured by Rutherford and Geiger, this volumetric rate yielded, in Perrin's words, "the number of monatomic molecules N of helium that occupy 22,400 cubic centimeters, and which therefore make up a gramme molecule."[62] The number per cubic centimeter at standard conditions announced by Boltwood and Rutherford in 1911 was 2.69×10^{19}, which translates into a value of 60.3×10^{22}, for Avogadro's number. (Perrin gave instead 64×10^{22} as their value for this at the Solvay Conference in late 1911.)

By 1911, therefore, α-particle radiation had yielded two values for Avogadro's number within 1.3 percent of one another, the first derived from the charge of the particles, and the second from the volume at standard conditions they come to occupy when collected over a period of time. The principal assumption these two had in common was the number of α-particles released per second from one gram of radium at equilibrium (among its isotopes), namely 3.4×10^{10}, as measured by Rutherford and Geiger in 1908, supported by the rate of visible scintillations observed when α-particles strike zinc sulphide, and then measured more exactingly by Boltwood and Rutherford. Both of these methods, and Millikan's oil-drop method as well, involved no assumptions at all from the kinetic theory of heat. The three values—60.3×10^{22}, 62.0×10^{22}, and Millikan's 59.22×10^{22} in 1911 (revised to 60.62×10^{22} in 1913)—all fell well below Perrin's values from Brownian motion, which by 1913 he was listing as 68.3×10^{22} from vertical-gradients, 68.8×10^{22} from granule displacements[63], 65×10^{22} from granule rotations, and 69×10^{22} from diffusion rates. (In the table at the end of his 1911 Conference paper, for which Rutherford was in the audience, and as well at the end of the 1913 and 1914 editions of the book, Perrin gives 62.5×10^{22} for the value derived from the charge of α-particles, said to be a mean of those from Rutherford-Geiger and Regener.)

While the discrete elements that were being counted in each of these three methods were not visible in the way that Perrin's granules were, each of the three was supplemented by strong evidence that discrete elements of one sort or another were in fact being counted. What seems less and less straightforward as we consider all these numbers in detail is what *evidence* could have been gained purely from comparing Perrin's values with these others. The most obvious thought is evidence *against* his assumption that the mean translational kinetic energy of his granules is identical with that of molecules at the same temperature.

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Stopping with that, however, is a mistake. The agreement with Perrin's values was nevertheless close enough to provide evidence that, even if granular kinetic energy is not identical with molecular kinetic energy, it is close enough to support his claim that thoroughly random agitation by discrete elements in the liquid is the source of Brownian motion. That in turn, just as Ostwald concluded, provided the first empirically grounded evidence at least for the kinetic theory of heat, if not for the molecular-kinetic theory of heat. The comparison with these other values, in other words, *did* strengthen the evidence Perrin's Brownian motion experiments provided for that theory. Adding still further to that evidence were Fletcher's 1911 Brownian motion experiments involving diffusion in an electric field and hence kinetic theory, which, joined with the Millikan and Rutherford-Geiger values for e, gave a value for Avogadro's number around 61×10^{22}. The comparison of values could therefore have legitimately been viewed at the time as giving less reason to quibble with Perrin's kinetic energy values and more reason to take his results for what they were, the most compelling empirical evidence that had yet been obtained for the kinetic theory of heat.

Notice here, however, that the evidence in the comparison was not reciprocal. Perrin gained support from Millikan and Rutherford-Geiger, while they gained little or no support from Perrin. But then they scarcely needed it, for their results stood on their own, owing especially to the tight agreement among their values for e and N_0, the mutually corroborating values for the latter from the two different measures based on radioactivity, and the tight uncertainty bands Millikan was achieving. From an evidential point of view, the only limitation of the atom-of-charge and α-particle determinations of N_0 was their lack of bearing on kinetic theory.

One other aspect of those determinations, largely absent from Perrin's, should not go unnoticed. In our remarks about the initial atom-of-charge determinations, we contrasted the molecules of kinetic theory with the molecules of chemical theory. Consider now the *questions* prompted by the different determinations of N_0. The Brownian motion determinations did not so much prompt new questions as they underscored long-standing questions about the sensitivity of collision-cross-section and mean-free-path calculations to assumptions about the shapes and sizes of the molecules of kinetic theory. By contrast, the atom-of-charge and α-particle determinations raised a host of questions about the molecules of chemistry. Six different items were being counted in those determinations: (1) elementary units of charge e, (2) α-particles, (3) atoms of radium transmuted, (4) gaseous ions, (5) "monoatomic molecules" of helium, and (6) molecules of gases, like H_2, released in electrolysis. The questions concerned the relationships among these six. Questions about the structure and unity of molecules, if not about the structure of atoms, had been around throughout most of the nineteenth century. In introducing a variety of elements of specificity that

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had been absent before, the common count of the number of "molecules" per mole from these six that had emerged by 1911 put those questions into an entirely new form.

6.3.3. Comparison with Values from Blackbody Radiation

The one further individual comparison Perrin made of his values for Avogadro's number and an independently determined value was with Planck's from 1900. This determination of N_0 from blackbody radiation raised questions of a very different sort, for it was not altogether clear, especially to Perrin, how and why the value quoted by Planck in 1900, 61.75×10^{22}, had to be a count of the number of molecules in a mole. The number itself came from measurements of blackbody radiation determining the values of the two constants, h and k, in Planck's equation:

$$E_\nu = \frac{8\pi h\nu^3}{c^3}\frac{1}{\exp(h\nu/kT)-1}$$

In deriving this formula Planck had assumed (1) a discrete rather than continuous distribution of frequencies, and (2) that Boltzmann's probabilistic analysis of the second law of thermodynamics applied to individual monochromatic resonators radiating and absorbing electromagnetic waves. The constant k enters as a parameter in the expression for the entropy of such a system. Planck reformulated Boltzmann's definition of entropy to show that, for a gas in thermal equilibrium with radiation, the total entropy could be the sum of the entropy of the parts only if $k = R/N_0$.[64] Planck's value for Avogadro's number then followed from the value of k obtained from the blackbody radiation measurements, 1.346×10^{-6} erg/deg K.[65] That the equation successfully captured the distribution of energy across the spectrum as a function of temperature was beyond dispute. So too was the robustness of the measured value of k, relative at least to the precision of the measurements, for both h and k could be re-evaluated at many different temperatures.

The questions centered on Planck's derivation of his equation and its *prima facie* departure from classical physics. The questions were multiplied when Einstein in 1907 re-derived Planck's equation—under the more radical assumption that all energy takes the quantized form of integral multiples of $h\nu$—in proposing a new approach to the specific heats of solids at low temperatures. The quantization of energy, which was in keeping with the quantization of light proposed in his 1905 paper on the photoelectric emission, entailed that individual atoms cease to vibrate, and hence cease to contribute to the heat capacity,

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at temperatures below the requisite *hv* threshold. By 1912 Peter Debye had extended Einstein's approach to multiple degrees of freedom of atom vibration and had shown that, with this extension, it matches low temperature specific heat data for solids to high accuracy.[66] On the one hand, then, were two major theoretical results agreeing with otherwise recalcitrant experimental findings, both of them tied to the two constants *h* and *k*. On the other hand were their radical departures from and hence *prima facie* incompatibility with classical physics.

Perrin cited Planck's determination of N_0 in his October 1908 paper and his 1909 monograph, but only side-by-side with a classical theory of the infrared spectrum that had been put forward by Lorentz and had yielded, as the value of Avogadro's number, 77×10^{22}, in contrast to Planck's 61.75×10^{22}.[67] Perrin's paper for the 1911 Solvay Conference—befitting the preoccupation of the Conference with the Planck and Einstein theories—summarized Planck's determination of N_0 without mentioning Lorentz's, or for that matter Einstein's theory.[68] *Les atomes* devotes a full chapter to "Light and Quanta," though again only summarizing Planck's result without fully deriving it. Perrin's most telling remarks on the comparison with Planck's value came in a much later edition of the book:

> The agreement between this value and those already found is indeed marvelous. And at the same time we have acquired yet another means for determining accurately the molecular magnitudes.
>
> In spite of the importance of these results, we cannot conceal the fact that Planck's theory presents great difficulties. It will be (and already has been) profoundly modified, and we shall certainly see Planck's postulates replaced by others more comprehensible and in more accurate agreement with experiment (Bohr).[69]

Insofar as Perrin had provided no hint of any discrepancy between Planck's theory and experiment, it is unclear what he had in mind, unless a value for Avogadro's number more in keeping with his own.

What then are we to make of the evidence deriving from the comparison of Perrin's values with Planck's? In his book Perrin first presents Planck's theory of the spectrum, cites its agreement with experiment, and then, in turning to Planck's value, says, "But a still more striking verification lies in the agreement found between the values already obtained from Avogadro's number and the value that can be deduced from Planck's equation."[70] This view of the evidence deriving from the comparison is in full accord with the view Planck expressed in the final paragraph of his 1900 paper, which we repeat here:

> If the theory is at all correct, all these relations should be not approximately, but absolutely, valid. The accuracy of the calculated value is thus essentially

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> the same as that of the relatively worst known, the radiation constant *k*, and is thus much better than all determinations up to now. To test it by more direct methods should be both an important and a necessary task for future research.[71]

The evidence deriving from the comparison in this case therefore went only one way, from other values for N_0 to support for Planck's theory, and not reciprocally. Do notice, however, that by 1911 Planck should surely have been stressing the values for N_0 coming from gaseous ionization and radioactivity, not the values coming from Brownian motion.[72]

* * *

To recapitulate, the four individual comparisons Perrin made at the time with values of Avogadro's number determined from other phenomena each yielded some evidence supporting the validity of at least one of the respective measures—especially if the various values were taken to agree with one another to within the margins of uncertainty of the respective determinations. The key unsupported presupposition in Perrin's determinations of Avogadro's number is that the mean translational kinetic energy of the granules is identical with that of the molecules in the substrate. His comparison of his values with the one he states for argon gave independent support for this presupposition by countering the claim that the granule velocities, based on direct observation, are too small for the kinetic energies to match.

So too did the values of Avogadro's number from determinations of the electric charge, from the Rutherford-Geiger (1908) and Regener (1909) values through those from Millikan and his associates from 1908 to 1913. The key open-to-challenge presupposition in the determinations of Avogadro's number from the electric charge was that the number of basic charges needed to liberate a mole of gas in electrolysis is identical with the number of molecules of the gas. Perrin's Avogadro's values gave independent support for this presupposition; and the values from these two together gave evidence as well that the molecules of kinetic theory are the same as the molecules formed in electrolysis, that is, the molecules of chemistry. The only comparisons in which the evidence went only one way were (1) with Planck's blackbody value of Avogadro's number as a consequence of novel aspects of his derivation of his blackbody equation, and (2) with the Boltwood-Rutherford determination, which, unlike the other determinations of Avogadro's number at the time, seemed to involve no salient open-to-challenge presuppositions calling for independent support other than that α-particles remain discrete when they lose their positive charge.

All of this evidence, however, is predicated on taking the various values to agree with one another to within the margins of uncertainty of the respective

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determinations. One worry, of course, was that those margins of uncertainty were anything but clear at the time, if only because so many distinct, *prima facie* well-founded novel determinations of Avogadro's number had emerged in the brief five-year period we (like Perrin) are considering. A more serious worry was that *all* the values of Avogadro's number with which Perrin compared his own were no greater than 62×10^{22}, the value he had stated for argon—that is, roughly 10 percent below the lowest of the values, 68.2×10^{22}, that he had quoted on the basis of his determinations of the mean translational kinetic energies of the granules. The central question of this chapter concerns what evidence the level of agreement among all the values in question provided for the reality of molecules. The systematic difference between the two sets of values threatened to downgrade the force of this evidence. We would therefore be remiss to move on to the central question of the chapter without at least reviewing how Perrin himself responded to this gap between his values and all these others.

6.4. Perrin on the N_0 Values $\leq 62 \times 10^{22}$

We claim to be adopting the viewpoint of someone writing a review article on Brownian motion and molecular reality at the time, say in the aftermath of the 1911 Solvay Conference. Readers might well ask why we need to be doing this when Jean Perrin wrote not one, but at least three review articles at the time, his 1909 monograph, his *Les atomes* in its five editions during 1913 and 1914, and his 1911 paper "Les preuves de la réalité moléculaire" for the Conference. What need is there for a review article beyond his? Part of our answer is that the 1909 paper and the book are too polemical to be considered review articles; their aim, often expressed in rhetorical questions, was to lay out the burden of argument on anyone still prepared to challenge the reality of molecules. The paper at the Solvay Conference, however, does throughout have the tone of a review article—quite appropriately given its extraordinary immediate audience, to whom it was circulated before its presentation at the Conference itself. Why shouldn't it be *the* review article on the subject at the time?

We grant that Perrin's 1911 paper was a review article, though the topic under review was just what its title indicates, not the state of the molecular hypothesis, but solely the body of evidence in support of it. Perrin does not mention the problem of the specific heats of gases and the questions it was raising about the equipartition of energy, nor does he note the extent to which questions about the structure of atoms and molecules remained intractable. (In the exceptionally brief discussion following his presentation, Lorentz did remark on J. J. Thomson's "spheres of positive electricity" in relation to "the hypothesis of perfect spheres."[73])

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Still, the serious shortcoming of Perrin's 1911 paper as a review article, even on this topic, was its failure even to note the *systematic* difference between his values for Avogadro's number from Brownian motion and all the comparably well-founded values with which he directly compared his in the paper. Instead, he compares them one by one, never raising the somewhat obvious question, once the systematic difference is noted, whether it might be pointing to some systematic error in his results.[74] Nor did anyone in the recorded discussion raise this question, not even Rutherford or Warburg. Our concern in this section—in keeping with our general concern with theory-mediated measurement—is, primarily, with how he managed to avoid this question and the consequences of his doing so, and, secondarily, with why no one raised the question in the discussion.

We saw at the end of the preceding section how Perrin in his book undercut the significance of the value of Avogadro's number coming out of the law of blackbody radiation by questioning Planck's theoretical derivation of the law. As we saw earlier, Perrin had grounds for not endorsing Planck's derivation, if only from the letter he had received from Einstein in 1909 which had invoked the value for N_0 from Brownian motion as a challenge to that derivation. One can argue that a derivation of the blackbody radiation law that was well founded did not emerge until after Compton's evidence for light quanta from the effect named after him and Bose's light-quanta based derivation of Planck's law in 1924 in which the distinction between what came to be known as Bose-Einstein and Maxwell-Boltzmann statistics emerged. That said, however, by 1911 the constants in Planck's formula had become well established, with results over a range of temperatures. At Solvay, Warburg reviewed such results, choosing to express them in terms of their implied value for e, for which he gave a value of 4.69×10^{-10}, corresponding to a value for N_0 of 61.7×10^{22}.[75] Perrin listed Planck's value from 1901 (quoted as 61.6×10^{22}) and then noted Warburg's paper at Solvay with praise; yet when it came to adopting a value from blackbody radiation, he chose 64×10^{22} without explanation, thereby glossing over the large discrepancy with his own values, to which Einstein had called his attention.[76]

The systematic difference with his own values was more difficult to avoid in the case of the ones for N_0 obtained from the values of e. As Table 6.1 shows, by 1911, five careful "direct" determinations of e had been published, based on three different approaches to measuring it: two from α-particles (4.65 and 4.79 times 10^{-10} esu), two from water droplets (4.669 and 4.65 times 10^{-10}), and one earlier that year from oil droplets (4.891×10^{-10}). (Warburg listed all of these except the ones involving water-droplets in his Solvay Conference paper, along with Perrin's value (4.23×10^{-10}); he added as well a blackbody value, 4.32×10^{-10}, implied by a measurement of Kurlbaum's, but ended his paper by challenging its validity.[77] Those at the Conference thus had this list of values in front of them prior to Perrin's presentation.)

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The mean of the values we listed is 4.73×10^{-10}, implying a corresponding mean value for N_0 between 60 and 61.2 times 10^{22}, depending in part on the value adopted for Faraday's constant. The convergence of these values for *e* thus lent as much support to them and their mean as the convergence of Perrin's three determinations from the mean translational kinetic energy of the granules *W* he announced in 1911 had lent to his—that is, 68.2, 68.8, and 69 times 10^{22}, the mean of which is 68.7×10^{22} (implying a corresponding mean value for *e* of 4.21×10^{-10}). Given the respective degrees of convergence of the results for *e* and *W*, this difference in the values for N_0 well in excess of 10 percent between the two approaches implies a systematic error either in the step from *e* or in the step from *W* to N_0, or in both. Faced with the audience he had at the Solvay Conference, along with his insistence on using the results for *e* to support the atomicity of electricity, Perrin could scarcely have avoided this worry in his 1911 paper.

Rather than consider the values of N_0 obtained from *e* in the aggregate, however, he reviews the ones from α-particle radiation entirely separately from the others, and he reviews the latter historically, starting with the efforts by Townsend, J. J. Thomson, and Wilson before 1905 and postponing the new oil-drop experiments to a separate section that begins with a review of the problematic Ehrenhaft results.[78] This has the effect of drawing attention to the uncertainties of the measurements of *e*—for example, those arising from questions about the evaporation of water droplets—instead of the trend after 1908 toward a more definite value coming out of Millikan's laboratory. Only in a short final paragraph at the end of the first of these sections does he note the more recent results for *e*, beginning with those by Przibram using alcohol-droplets ($e = 3.8 \times 10^{-10}$), followed by the water-droplet results, first by Millikan and Begeman in 1908 (citing a value of 4.6×10^{-10} that includes corrections Begeman added in 1910 instead of the originally published value of 4.06×10^{-10}) and then Begeman's new 1910 value (4.7×10^{-10}). Perrin then dismisses all of these values at the end of the paragraph with the remark, "But there is no need to discuss these [numbers] at length, much greater precision becoming possible by the study of individual charged particles."[79]

After then dismissing Ehrenhaft's results that challenged the existence of a fundamental atom of charge,[80] Perrin reviews in some detail Millikan's results in his first oil-drop paper, "The Isolation of an Ion, A Precise Measurement of Its Charge, and the Correction of Stokes's Law." (Insofar as Millikan was an American and his paper had been published in the American journal *The Physical Review*, and then only six months earlier, the detail may have been needed, for several of those at the Conference were surely hearing about it for the first time.) The claim of a precise measurement in the title appears not only in the four significant figures Millikan gives for the value, but in his stated assessment of it:

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> The probable error in the final mean value 4.891×10^{-10}, computed least squares from the numbers in the last column, is four hundredths of one percent. . . . Since, however, the coefficient of viscosity of air is involved in the formula, the accuracy with which *e* is known is limited by that which has been attained in the measurement of this constant. There is no other factor involved in this work which has not been measured with an accuracy at least as great as .2 percent.[81]

The paper is nevertheless less famous for its claiming a precise value for *e* than it is for having shown that the charge on a large number of oil droplets was always an integral multiple of a single charge, independently of the absolute magnitude of that charge.

Perrin's account of the paper begins with this last point, laying out some of the results for individual droplets and citing Fletcher's related paper that had announced a "direct" value for *Ne* for gaseous ions, independently of the values of *N* or *e*, within 0.5 percent of Faraday's constant for electrolysis, the strongest evidence yet that *Ne* is the same for both.[82] Perrin drew an unqualified conclusion, contra Ehrenhaft, from these results: "In short, the experiments of Millikan demonstrate in a decisive manner the existence of an atom of electricity, equal to the charge carried by a hydrogen ion during electrolysis."[83]

Perrin then turns to Millikan's claimed precise value for *e*, opening a new section with the sentence, "But these beautiful experiments do not yet appear to me to give a reliable value for this elementary charge which they have made manifest."[84] The doubt, he goes on to say, stems from Millikan's treatment of the frictional action of the air on oil-droplets, that is, the version of Stokes's law for air that Millikan had employed. He then invokes efforts by Roux to repeat Millikan's experiments in his own lab, but in liquid as well as air, saying that they had yielded a value for Avogadro's number around 67.5×10^{22}, versus the value 59×10^{22}—not the more precise 59.22×10^{22}—that Millikan had announced on the basis of his value for *e*. Perrin then concludes by adopting a value for Avogadro's number from the charge on ions of 64×10^{22}, a mean between Millikan's 59×10^{22} and 69×10^{22} from Brownian motion. Perrin's audience, whether at the Conference itself or readers of the subsequent publication in French and German of its proceedings, were left with no reason to raise the question of whether Millikan's precise value for *e* might be exposing a systematic error in Perrin's Brownian motion values for N_0.[85] Indeed, they would have had to turn to Millikan's paper even to realize the level of precision he was claiming.

(Perrin was justified in citing Fletcher's value of 28.8 for *Ne* while ignoring the values for N_0 (57.5×10^{22}) and *e* (5.01×10^{-10}) given in the same paper, for Fletcher himself cited them only to claim that the theory of Brownian motion in air he had developed in the paper "gives a fairly accurate means of determining molecular magnitudes."[86] The situation changed radically a short time later,

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however, after Millikan had published his new value for *e* in 1913, reflecting a revised account of the friction of air on oil droplets. That revision gave Fletcher all he needed to obtain a "direct" value for N_0 through observing the displacements of individual oil drops, and improvements in the apparatus had made it possible to obtain from 1,000 to 6,000 observations. Fletcher's paper, "A Determination of Avogadro's Constant *N* from Measurements of the Brownian Movements of Small Oil Drops Suspended in Air," gave values of N_0 obtained from 12 individual droplets ranging from a low of 57.9×10^{22} to a high of 64.8×10^{22}, with a corrected weighted mean of $60.3 \times 10^{22} \pm 1.2 \times 10^{22}$.[87] Insofar as Fletcher's value came from Brownian motion and differed systematically from Perrin's, it surely should have provoked questions about the possibility of systematic error in Perrin's determinations of the granular kinetic energy were it not for the fact that Bohr's 1913 paper had focused attention away from N_0 and onto the need for a precise value of *e* in order to draw conclusions from spectra. As we noted earlier, Bohr himself had adopted a value of 4.7×10^{-10} (implying a value below 62×10^{22} for N_0) in his theoretical calculation of Rydberg's constant. A few years later, Sommerfeld adopted a value of 4.77×10^{-10}.[88] We know of no one doing research on atomic structure after Bohr who preferred a value for *e* consistent with Perrin's translational values for N_0, that is, a value no greater than 4.25×10^{-10}. All of this, however, falls outside the historical time-window to which we are confining ourselves here.)

In his subsequent section on "genesis and destruction of atoms"—that is, phenomena of radioactivity—at the Solvay Conference, Perrin quotes the values for *e* announced by Rutherford and Geiger, 4.65×10^{-10}, without any comparison to Millikan's value, and then gives 62×10^{22} as the value it implies for N_0, "with a maximum error of not more than 10 parts in 100." He goes on then to acknowledge Regener's greater value for *e,* 4.79×10^{-10} from α-particles and its corresponding value of 60×10^{22} for N_0, but argues that they are less reliable than the Rutherford-Geiger value. In the table comparing the values for N_0 from the various phenomena, he lists 62.5×10^{22} for the value obtained from the charge of α-particles, immediately preceded by a value of 64×10^{22} from "the charge of spheres in gases."[89] No one looking at this table would see any sign of a systematic difference between his values from Brownian motion and those obtained from measurements of the elementary charge.

Finally, in the case of the 1911 Boltwood-Rutherford determination from the volume of the gas occupied at standard conditions by a known quantity of α-particles, Perrin elects not to quote their published value for Loschmidt's number, 2.69×10^{19}, which corresponds to 60.3×10^{22} for N_0. Instead, he quotes a range of values from which he obtains a range for N_0 of 62.4×10^{22} to 64×10^{22}. He then notes the corresponding experiments carried out by Mme. Curie using

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polonium, the volume obtained from them by Debierne, and the value 65×10^{22} implied for N_0. In the table at the end, he gives 64×10^{22} as the value obtained from this approach. Whether Perrin consciously suppressed the 60.3×10^{22} implied by Boltwood and Rutherford's stated value for Loschmidt's number is an interesting question. Rutherford, who was an attendee at Solvay, is not recorded as saying anything in the discussion of Perrin's paper. Perhaps Rutherford himself felt there was enough uncertainty in their results, side-by-side with the Curie-Debierne results, to withhold any objection to the 64×10^{22} value.[90]

Perrin's account of the Boltwood-Rutherford results was unfortunate historically. Because his 1911 paper and subsequent book have been taken—universally, it would seem—as *the* authoritative review of the value of Avogadro's number at the time, the extraordinary character of their experiment has been largely ignored.[91] Of all the different determinations of N_0 at the time, this one involved the least tenuous assumption linking their observations to the molecular constant, namely merely that the helium gas resulting from the collection of α-particles consisted of the same number of monoatomic molecules as the measured number of α-particles. Boltwood and Rutherford say as much quite forcefully:

> There can be no doubt that the α particle during its flight consists of a helium atom carrying two unit positive charges, and that helium itself is monatomic. The agreement between calculation and theory verifies in a remarkable way the essential correctness of the atomic theory of matter. The number of helium atoms expelled per second by one gram of radium have been directly counted, and the corresponding volume of helium has been experimentally measured. From these two experimental observations, it is possible to deduce with a minimum of assumption the number of atoms of helium in one cubic centimetre of that gas at standard pressure and temperature. That number is 2.69×10^{19}. By Avogadro's hypothesis this gives the number of molecules in a cubic centimetre of any gas at standard conditions.[92]

One should probably not dismiss as a mere accident that their value for Avogadro's number came closer to the post-1940 value than any of the others we have been considering.

Perrin was not alone in contributing to the Boltwood-Rutherford results not receiving the attention they deserve in subsequent histories. Svedberg's *Die existenz der moleküle: experimentelle studien* of 1912 makes no reference to Boltwood and Rutherford at all. Insofar as the Proceedings of the Solvay Conference did not appear in print until 1912 in French and still later in German, Svedberg almost certainly had not read Perrin's Conference paper and its discussion of Boltwood and Rutherford at the time his book went to the printer.

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While Svedberg's main title suggests that his book is a review of the evidence for molecules, akin to Perrin's 1911 paper, it is instead, as its subtitle indicates, a review of experiments from Svedberg's laboratory, with extensive lists of numerical results, on diffusion in emulsions and Brownian motion. It gives values of 58×10^{22} for N_0 and 5.0×10^{-10} for e derived from the diffusion experiments, noting the agreement between them and the values quoted by Millikan and Fletcher versus the difference with Perrin's values (cited as 68.8×10^{22} and 4.17×10^{-10})[93]; and from the Brownian motion experiments, 62×10^{22} and 4.7×10^{-10}, this time noting the agreement with Millikan, Regener, and Rutherford versus Perrin (cited as 69×10^{22}).[94] Perrin does not comment on the latter, but in a footnote in *Les atomes* he cites "the high degree of uncertainty involved in the measurement (and even the definition) of the radii of *invisible* granules" in dismissing the former.[95]

Svedberg thus called attention in 1912 to the difference in values that would seem to prompt questions about the possibility of systematic error in Perrin's results, yet so far as we have been able to determine, no one at the time pursued this question much further. In the 1921 edition of his *Theoretical Chemistry from the Standpoint of Avogadro's Rule and Thermodynamics*, Nernst cites both Perrin's and Svedberg's values of N_0 from diffusion and Brownian motion and then concludes:

> The relatively good agreement of the values of N_0 obtained by these various methods from experiments with suspensions is an important fresh proof of the correctness of the theory of thermal motion; the accuracy of the results obtained by Perrin and Svedberg in the work mentioned above is very remarkable, since it would appear very difficult to obtain precise measurements of N_0 by such methods. It is noticeable that Perrin obtained appreciably higher values throughout than Svedberg.[96]

Four pages later, when it comes to listing his preferred values, Nernst chooses Millikan's values of 4.774×10^{-10} and 60.64×10^{22} for e and N_0.[97] So far as we know, any question of systematic error raised by the difference between Perrin's values and the others simply became moot as Millikan's values came to be preferred.

We draw three lessons in concluding this section. First, Perrin was not the person to be writing a truly careful review article in 1911 or 1913. We should not automatically be taken to be criticizing him in saying this. High-quality measurements were coming out of three laboratories, directed by three extraordinary experimentalists, Perrin, Rutherford, and Millikan. Each of them knew the care taken in obtaining the numbers they were quoting from their own

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laboratory, but each of these three had little way to make such an assessment of the numbers coming out of the other two laboratories. Perrin thus had good reasons for trusting his numbers more than the others, and he should consequently not be viewed solely as protecting his numbers in raising questions about the others. The lesson to be learned is that truly careful review articles are best written by someone able to address all the results being reviewed from a truly neutral perspective.

(Our matter-of-fact statements of the results coming out of these three laboratories do not do justice to the exceptional ingenuity and extended effort that went into the design and development of the experiments that produced them. Millikan's experiments became most famous for this, yet the two-part Rutherford-Geiger paper of 1908 displays a more elaborate design in response to a notably more challenging sequence of questions, and the answers they achieved were no less monumental than Millikan's; and one has only to read any one of Perrin's monographs of 1909 and 1911 or his *Les atomes* to come to appreciate the extraordinary demands of his experiments.)

Second, partly because Perrin's paper and book did become the authoritative review article at the time, but also because so little time had passed between the first well-founded values of Avogadro's number in 1908 and 1911 (or for that matter, 1913), no one was in a position at the time to select a preferred value for it. Planck was referring to the value for *e* in the preface to his 1913 edition of *The Theory of Heat Radiation*, but the phrasing he used there, after citing Rutherford, Regener, Perrin, Millikan, and Svedberg, applies equally well to the value for N_0, "between the values of Perrin and Millikan"[98]—that is, between 59×10^{22} and 69×10^{22}, or in other words, the one significant figure claimed by Poincaré. It was entirely reasonable for anyone outside of these three laboratories, at least as of 1911, to take all the values between these two as agreeing with one another.

Third, one needs to appreciate how radically the capacity to assess theory-mediated measurements of microphysical constants changed between 1911 and 1929, the year of Birge's "Probable Values of the General Physical Constants." The change came not from more precise or better experimentation, but from so many more constants being linked to one another, starting with Bohr's linking of *h, e*, *e/m*, and Rydberg's constant in 1913, but rapidly proliferating thereafter. The only effective way to detect systematic error is to play different measures of the same constant or quantity off against one another. If nothing more, this section has brought out how limited they were in effecting this approach before Bohr. Our monograph should accordingly be viewed as a study in theory-mediated measurement in the years leading into an extraordinary increase in the capacity for drawing evidence out of such measurements.

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6.5. Perrin's Comparisons, Collectively

The four comparisons we have examined individually—with values for N_0 obtained from the viscosity of gases, from gaseous ionization, from radioactivity, and from blackbody radiation—are not the only comparisons Perrin came to cite. Starting with his 1909 monograph and then continuing with his 1911 Solvay Conference paper and his 1913 book, he also cited comparisons with values derived from such phenomena as the blueness of the sky, diffusion of dissolved substances, the mobility of ions in water, and, in 1911 and 1913, density fluctuations in concentrated emulsions and critical opalescence. One reason for our not considering these individually is that the uncertainty bands associated with them were far larger than with the values obtained from ionization, radioactivity, and blackbody radiation. Another reason, besides wishing not to try readers' patience more than we already have, is that Perrin himself focused on individual comparisons only when defending his value for N_0 against others. Perrin's preoccupation was instead with the evidence for the existence of molecules deriving from the *aggregate* general agreement among so many different ways of obtaining values for Avogadro's number.

A lesson we intend to be learned from our discussion of the individual comparisons is that any comparison in the aggregate threatens to gloss over details that, when taken into consideration, may limit, if not undercut, the evidence that can legitimately be drawn from the whole. That granted, the many different, highly diverse determinations of Avogadro's number did agree with one another to a remarkable extent, at least in comparison with the uncertainty attaching to all such determinations before 1900. Poincaré's 1912 summary of the situation that we quoted at the very beginning of this monograph is the statement from which we are now going to proceed:

> The brilliant determinations of the number of atoms computed by Mr. Perrin have completed the triumph of atomism. What makes it all the more convincing are the multiple correspondences between results obtained by entirely different processes. Not too long ago, we would have considered ourselves fortunate if the numbers thus derived had contained the same number of digits. We would not even have required that the first significant figure be the same; this first figure is now determined; and what is remarkable is that the most diverse properties of the atom have been considered.[99]

Our question is, what evidence could appropriately have been drawn from this? Keep in mind, as we proceed, Poincaré's further remark, "each new discovery in physics reveals a new complexity of the atom."[100]

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A place to start is with the elements of logical structure that all of the determinations of Avogadro's number had in common. Each case involved a theory-mediated measurement of some primary quantity from which the value of N_0 was then inferred as a further step. For example, the primary quantity in all of Perrin's Brownian motion measurements was the mean kinetic energy of the granules—translational in three of the measures and rotational in the fourth. The primary quantities in the other determinations included the fundamental unit of charge *e*, the number of α-particles emerging from a gram of radium per second, the volume occupied at standard conditions by a gas formed by a specific number of α-particles, and the two constants in Planck's equation, *h* and *k*, with the latter initially representing a mean energy per resonator. In each case the integrity or, if you prefer, the validity of the theory-mediated measure of the primary quantity could be confirmed by testing what we called its *stability* in Chapter 4—that is, not merely its repeatability in the same conditions, but obtaining the same value for the quantity as the values of the other parameters entering into the measure (or in other words, the conditions) are varied. The stability of any theory-mediated measure provides the first layer, so to speak, of evidence in support of the premises entering into it. When the measure is not all that stable, the premises underlying it are subject to question.

Measures of all the primary quantities we have just listed had been confirmed to be stable by 1912. The same cannot be said for the measures involved in determinations of N_0 based on the viscosity of gases, the blueness of the sky, diffusion of dissolved substances, the mobility of ions in water, density fluctuations in concentrated emulsions, and critical opalescence. Until those measures were shown to be stable, the most they could contribute to the evidence coming out of the many determinations of Avogadro's number had to amount to nothing beyond ancillary support, specifically in the form of a *prima facie* promise for still more cases of stable measures yielding agreeing values for N_0.

Continuing with technical terms introduced in Chapter 4, when two different stable theory-mediated measures of the same quantity yield, to within experimental precision, the same values, we follow Perrin in calling them *convergent*. Thus he developed three distinct convergent measures of the mean translational kinetic energy of granules, one based on vertical-gradients, a second on granule displacements, and a third on diffusion coefficients. What made these distinct was a difference in the premises constitutive to each of them. As we said in the case of Huygens's two measures of surface gravity, convergent measures of the same quantity provide mutual support for the differing premises constitutive to the measures—specifically, stronger evidence for the reliability of their premises for purposes of measurement than provided by stability alone. Among the other primary quantities with stable measures listed earlier, only *e* had convergent measures as of 1912, principally from the oil-drop measure of Millikan and

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the cumulative charge measure of Rutherford and Geiger, independently confirmed by Regener. In principle, however, the Einstein-Debye equation for the variation of the specific heat of solids with temperature could have been turned into a convergent measure for *h* and, of course, many other measures of *h* were soon to emerge.

In conjunction with these considerations, we are going to interpret Poincaré's one-significant-figure claim in the manner of Planck's 1913 comment in regard to *e*, namely as N_0 falling within the bounds defined by the two convergent sets of stable measurements: 59×10^{22}, corresponding to Millikan's 1911 value on the low end; and 69×10^{22}, corresponding to Perrin's translational kinetic energy values from the same year on the high end.

A second feature of the primary quantity measures was that their derivations all included reference to something discrete and hence amenable to being counted. What this was varied from measure to measure, as did the strength of the evidence that it truly is discrete. Granules are visibly discrete, and Rutherford and Geiger were counting anywhere from one to five distinct "jerks" of their electrometer needle in each minute, which markedly strengthened the evidence from scintillations that the radiation coming off of radium consisted of discrete items. Millikan always insisted that he had been the one to finally show that electricity consists of discrete unit charges, and at the very least, as Perrin himself contended, Millikan's oil-drop experiments strengthened the evidence that had come from counting droplets presumed to form around ions and measurements of the common charge-to-mass ratio of various discharges of negative electricity. Planck's discrete resonators and the discrete packets of energy associated with them had the least independent evidence as of 1912, and hence were most controversial, in spite of the great stability of the measures of *h* and *k* that presupposed them.[101] The variability of the evidence for the claim of discreteness is not so important, however, as the indispensability of such a claim to all the measures. For, while the discrete elements entering into each measure differed, they were each crucial to the further claim that the value for N_0 derived from their measured primary quantities amounted to a count of discrete items in its own regard.

A third feature that the logical structure of the determinations of N_0 had in common was a premise linking the primary quantity measured in each to what was said to be Avogadro's number. As we have seen in the case of the Brownian motion determinations, this premise—*that the mean translational kinetic energy of the granules W is the same as the mean translational kinetic energy of the molecules of the liquid surrounding them*—had a very different status from the premises underlying the measures of the primary quantity. Not only was there no independent evidence for it, but nothing akin to the stability and convergence of the measures for the energy of the granules provided any support for it. The stability and convergence of those measures provided support for the

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premises underlying them, but so far as those measures were concerned, the premise relating W to N_0 was something further tacked on and hence something that went beyond them. This is why we have distinguished between the primary quantity being measured in each case and N_0. *The determinations of the latter in every case represented a different kind of theory-mediated measurement because of their relying on a further premise tacked on in order to link the primary quantity measured to what was said to be Avogadro's number.*

A more precise way of making this point is that the premise linking the primary quantity measured to N_0 was in no way *constitutive* of the measure of the primary quantity in the sense of being indispensable to it; this premise was constitutive only to the further inference from the primary quantity to N_0. As a consequence, this premise was open to question regardless of how stable, convergent, or otherwise well-founded the measures of the primary quantities happened to be. The least open to question was the premise linking the number of α-particles that had entered a volume and the number of monatomic helium molecules in the volume. That premise requires little beyond α-particles remaining discrete and otherwise intact when, in Rutherford's words, they lose their positive charge. Still, absent more empirically grounded detail about what it is for an α-particle to lose its charge, a gap did remain between counting α-particles and having a count of molecules of helium. Similarly, it was one thing to infer the number of fundamental unit charges required to match Faraday's constant and another to claim that one unit charge in electrolysis yields one H_2 molecule of hydrogen. Why not instead one atom of hydrogen or one unit weight of hydrogen? Tightening the uncertainty bands on the value of the fundamental unit of charge was not ever in itself going to close this question.

We should take stock, then, before turning to the question of the evidence deriving from the extent of the agreement among these different determinations of Avogadro's number. On the one hand, there were clear grounds for concluding that theory-mediated measurements had yielded reasonably precise values of the following quantities by late 1911:

(1) The mean translational kinetic energy W of granules in Brownian motion in dilute emulsions at a given temperature, and with it, though slightly more provisionally, the mean translational kinetic energy per degree Kelvin, W_T, in Brownian motion.
(2) The value of the fundamental discrete unit of charge e, whether negative or positive, in ions and in such discharges as cathode rays, the photoelectric effect, and both α-particle and β-particle radioactivity.
(3) The volume occupied at standard conditions by the electrically neutral helium gas formed by the number of α-particles released and then collected over any given period of time.

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(4) The mean energy per degree Kelvin k per discrete absorber-emitter of radiation in the thermodynamically equilibrium state of blackbody radiation, and with it the value of the new constant h, subsequently named "Planck's constant."

While the values determined for each of these "primary" quantities were theory-mediated in one way or another, in no case did the theory-mediation include, constitutively, any assumption from molecular-kinetic theory, and hence any assumption requiring the reality of molecules.

Consequently, on the other hand, the following further assumptions were correspondingly respectively needed to obtain values of Avogadro's number N_0 from these quantities:

(1) The mean translational kinetic energy of granules in Brownian motion at a given temperature is equal to the mean translational kinetic energy of the molecules in the substrate in which the motion is occurring.
(2) The number of unit charges required to produce a mole of a monovalent gas in electrolysis is equal to the number of molecules in a mole.
(3) The number of α-particles collected in a given time to form a volume of helium gas is equal to the number of monoatomic molecules of helium forming that gas, which in turn is the same as the number of molecules forming any other gas in the same volume at the same temperature and pressure.
(4) The mean energy per degree Kelvin per discrete absorber-emitter in blackbody radiation is equal to the mean energy per degree Kelvin of the molecules forming an ideal gas.

Each of these further assumptions can legitimately be regarded as the "weak-link" in the respective determinations of Avogadro's number—this, for two reasons. First, there was no evidence tied specifically to any one of the determinations to support the further assumption in question. Second, as a corollary to this, each assumption, taken in isolation, was presupposing, and hence on its face begging the question of, the reality of molecules.

But now the answer to our question about the evidence from the aggregate of the agreeing measurements of N_0 is obvious: the aggregate provided support, indeed virtually the only empirically grounded support, for these further premises linking each primary measured quantity to N_0. All those premises had one element in common, the number of "molecules" in a mole. What was thrown into question whenever the value of N_0 obtained in any one approach having a robust measure of its primary quantity failed to be in agreement with the others

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was not the measurement of this primary quantity, but the premise yielding the number of "molecules" in a mole. We saw such a question arise earlier in the case of the emerging discrepancy between the Perrin values and, for instance, the Millikan and Rutherford-Geiger values. Were Perrin's measured mean translational energies of the granules really equal to the mean translational energies of the molecules of the surrounding liquid?

A good way always to pin down the positive thrust of any evidence is to ask what a failure of the results to meet expectations would have been evidence of. If lack of agreement raised doubts about the linking premise, then, correspondingly, agreement gave support to it—the closer, the stronger. Accordingly, agreement among the values obtained from diverse phenomena for N_0 was the principal evidence at the time supporting the individual inferences from the primary quantities measured in each realm of phenomena to what was claimed to be the value of Avogadro's number.

Another way to see this is to consider what happened between 1908 and 1912. Attempts to derive a value for Avogadro's (or Loschmidt's) number went back at least as far as 1865. As Meyer made clear in the last chapter of his 1899 edition of *The Kinetic Theory of Gases*, none of the approaches to doing so had managed to achieve well-behaved measurements of the primary quantities from which, the hope was, a value for Avogadro's number could be derived. Their weak links lay in their measures of their primary quantities. Between 1908 and 1912, stable measures for at least four distinct primary quantities emerged: (1) the mean kinetic energies of granules in Brownian motion; (2) the fundamental unit of charge; (3) the number of α-particles per unit volume at standard conditions; and (4) the mean energy k per degree Kelvin governing in part how the spectrum varies with temperature. Moreover, at least with the first two of these, convergence among complementary stable measures had been achieved. As a consequence, the weak link in deriving a value for Avogadro's number from each of these measured quantities lay in their respective premises relating them to it. Agreement among the different derived values for Avogadro's number provided evidence substantiating that remaining weak link. Order of magnitude agreement with still other approaches that continued to have their own weak links and hence greater degrees of uncertainty could then serve to reinforce that evidence.

From the point of view of theory-mediated measurement, this was the principal evidence supplied by Perrin's aggregate of comparisons. But that was not the only evidence coming out of them. For example, the agreement among the measures was evidence that they really were measuring a single definite quantity, nominally the number of "molecules" in a mole. That was important because the

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primary count in each of the four cases was not of the number of molecules, but of four different items:

(1) Given their mean kinetic energy *WT*, the number of granules required to match the characteristic energy *RT* of a mole of gas or dilute solution;
(2) Given the magnitude of the fundamental unit of charge *e*, the number of these units required to match the total electricity *F* needed to yield a mole of a monovalent gas in electrolysis or, as had been shown to be equivalent to this, the total electrical charge in a mole of gaseous ions;
(3) Given the rate of production of α-particles from a gram of radium, the number of these positively charged particles required to yield a standard volume, 22,400 cubic centimeters, of helium gas at standard conditions.
(4) Given the mean energy per degree Kelvin *k* per radiator-absorber of electromagnetic radiation, the number of radiator-absorbers required to maintain thermodynamic equilibrium with the energy *RT* of a mole of gas.

Agreement among these four numbers was evidence that they are each counting a different *aspect*, in the classical philosophical sense of the term, of something beyond them that they have in common. Failure of agreement—again the tell-tale indicator—would have been clear evidence that they could not be counts of one and the same thing.

Viewing agreement among these four in this way explains the scare quotes and adverb in our peculiarly guarded phrasing, "nominally the number of 'molecules' in a mole." These four at the time were raising at least as many questions about what molecules are as they were answering—all specific variants of the generic question, *how do these amount to aspects of one and the same thing?* For those at the time, what "molecule" was supposed to refer to was not a matter of dispute, namely stable chemical unities combining fixed proportions by weight of elements—like NO, NO_2, H_2O, and He—that, at least in gases, move as integral wholes, interacting with one another in accord with the laws governing bodies in motion. Some questions raised by this characterization had received nothing but conjectural answers for a long time. For example, what holds the parts of molecules together to form a unity, and how is it related to the release or uptake of energy during chemical processes? What are the sizes and shapes of molecules, and how do they interact, through force-fields surrounding them, as proposed by Maxwell,[102] or through impact? The aspects of molecules in the four ways of counting them shed little or no light on these questions, and added new more perplexing ones. How do molecules absorb and emit electromagnetic waves in a spectrum that varies with temperature? Do they contain separate charges that nullify one another, and if so how? What happens chemically to a molecule that

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includes radium when an α-particle is emitted? As Poincaré said, each new discovery during the period from 1897 to 1912 had revealed a new complexity.

At the same time, possibilities for designing novel experiments to attack these questions had increased because of the many aspects of molecules that had newly been measured and hence could be *taken* as known in further experiments, and also because of ties among diverse phenomena revealed by those measurements. The theoretical connection among the gas constant R, Avogadro's number N_0, and Boltzmann's constant k had been experimentally confirmed as a consequence of each of them being measured independently of the other two. In the process, the three of them had become linked not only to the fundamental unit of charge e, Faraday's constant and the total charge in a mole of ionized gas, but also to the new "quantum" constant h governing the variation with temperature of both the blackbody radiation spectrum and, with Debye's (1912) refinement of Einstein's theory, the measured specific heats of solids. Measurements of the charge-to-mass ratio of the electron, the hydrogen ion, and the α-particle, combined with the value of e, had determined their masses, and the agreement between the masses of the latter two and the masses of the hydrogen and the helium atom inferred from Avogadro's number had established, at least to reasonable approximation, not only the masses of hydrogen and helium molecules, but by virtue of the long-standing system in chemistry of atomic and molecular weights—or, as some had called them, "equivalent" weights[103]—the masses of molecules of substances generally for which chemical formulas were known. Finally, with these masses in hand, the number of molecules in any quantity of liquid of known chemical composition—whether uniform or in given relative concentrations of different components—could be determined.

Granted, the residual uncertainties in the values determined for many of these quantities limited the precision to which they could be considered known to one or two significant figures. Contrast that, however, with the situation in 1899, as summarized in the last chapter of Meyer's *The Kinetic Theory of Gases* and J. J. Thomson's paper proposing his new conception of electricity, where the attempts to assign values to many of these quantities were still falling so far short that it made no sense to estimate residual uncertainties.

Moreover, still further experimental prospects for getting at questions about molecules were continuing to emerge at the time of Poincaré's 1912 remark. Sommerfeld announced that summer that Max von Laue, then at Munich but until 1909 Planck's assistant, had shown that crystals diffract x-rays—a discovery that won him the Nobel Prize a mere two years later. Within months, the Braggs had extended von Laue's discovery to determine the spectra of x-rays and the structure and lattice dimensions of various crystals—which won them the Nobel Prize a year after von Laue. In 1911 Rutherford, in a two-page abstract that went

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largely ignored at the time, had proposed on the basis of α- and β-particle scattering "a type of atom which consists of a central electric charge concentrated at a point and surrounded by a uniform spherical distribution of opposite electricity equal in amount."[104] In the summer of 1913, Bohr published his model of the hydrogen atom, in the process linking *h, e*, and *e/m* for electrons to the Rydberg constant.[105] And later that year, in the same volume of *Philosophical Magazine*, Henry Moseley, also at Manchester alongside Rutherford and Bohr, took advantage of the work by the Braggs to establish a relationship between measured spectra of x-rays emitted by different chemical elements and their atomic number, linking both again to *h, e*, and *e/m* for electrons; this led to atomic number replacing atomic weight as the more fundamental parameter and in the process effectively established Rutherford's model once and for all.[106]

A key point to appreciate about these landmark developments in 1913 was the extent to which they presupposed and built off of the measured values of the various quantities considered in the paragraph preceding the last two. In taking some exception to van Fraassen on the subject of grounding in section 5.3 of Chapter 5, we pointed out how little Perrin's Brownian motion results, taken by themselves, contributed toward empirically grounding molecular-kinetic theory. We now see, however, that the aggregate of the various measurements of Avogadro's number had by 1912 or so done an enormous amount toward grounding the theory. Recall that Ostwald's demand for what van Fraassen calls grounding was for (1) experimentally measured values of sufficient key parameters of the theory to enable true testing of it, and, with that, (2) step-by-step experimentally driven further development of it, to become possible. In the same vein, van Fraassen says, "a theory *needs to be informative enough to make testing possible at all.*"[107] Even though many of the "difficulties" with molecular-kinetic theory that Ostwald had stressed remained unresolved in 1912, not to mention new difficulties that were emerging, and many questions about molecules remained unanswered, his demand had clearly been met. And, of course, van Fraassen's requirement had been met as well, though not by Perrin's results alone, but by the whole group of determinations in aggregate with one another.

6.6. "Les preuves de la réalité moléculaire"?

So far we have singled out three respects in which Perrin's comparisons among the different determinations of Avogadro's number, taken collectively, yielded evidence in their own right: (1) evidence for the "weak-link" premises licensing each inference from its measured primary quantity to a value for N_0; (2) evidence that the diverse determinations, each with its own "weak-link" premise, were nevertheless measuring—that is, yielding counts of—a single quantity;

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and (3) evidence showing that a host of separate quantities—some of them presumed to be constants and others variable—are systematically related to one another not just in theory, but in measured numerical fact, thereby providing empirical grounding for molecular-kinetic theory beyond any it had ever had before. Perrin was accordingly not just engaging in rhetoric when he ended his 1909 monograph, his 1911 Solvay Conference paper, and his *Les atomes* by citing more than a dozen different determinations of Avogadro's number and insisting that the "convergence" among the values provided the most important evidence of all.

That, however, still leaves us with the question of whether Perrin was overreaching in saying at the end of his book that, *by virtue of this convergence among the values*, "the real existence of the molecule is given a probability bordering on certainty."[108] Van Fraassen's dismissal of this proposal occurs in two steps. In the first he argues that the convergence of values is indeed important, but that importance lies in its fulfilling a key part of the requirements for empirically grounding a theory. His statement of those requirements, we remind the reader, is adapted from Weyl:

- *Determinability*: any theoretically significant parameter must be such that there are conditions under which its value can be determined on the basis of measurement.
- *Concordance*, which has two aspects:
  - Theory-Relativity: this determination can, may, and generally must be on the basis of the theoretically posited connections.
  - Uniqueness: the quantities must be "uniquely coordinated," there needs to be concordance in the values thus determined by different means.[109]

On first thought, therefore, van Fraassen's construal of the convergence of values for N_0 appears to be the same as ours, save for our having spelled out in more detail the evidence deriving from meeting the uniqueness part of his concordance requirement. But this thought is wrong. Central to van Fraassen's account of how molecular-kinetic theory became grounded is that the "theoretically posited connections" enabling the measurement of N_0 and related parameters were extensions to—his phrasing is "enrichments" of—that theory. On his account, the historical core of molecular-kinetic theory was not "informative enough to make testing possible at all."[110] "The appropriate, and typical, response in that case is to start *enriching* the theory so that it becomes more informative, informative enough to allow the design of experiments in which this empirical determination of the values [of its parameters] does become possible."[111] But then, once grounding is achieved, it is the *whole* theory that becomes tested, with claims about the "reality" of its posits really amounting to nothing more

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than statements about the empirical adequacy of models lending specificity to the otherwise abstract theory. Therefore, in particular, the convergence of values Perrin cited was not providing evidence, in van Fraassen's view, for the "reality of molecules" if for no other reason than that it was just putting molecular-kinetic theory in a position where it could finally be properly tested.

Needless to say, we agree with van Fraassen that the determinations of N_0 involved theory-mediated measurements, and hence theoretical relations between it and experimentally more accessible quantities. And we agree as well that those theory-mediated measurements had to meet requirements in order not to be question-begging and consequently vacuous. We, however, have gone to quite some pains to show *how little* the theory-mediated measurements of the four primary quantities in the determinations of N_0 we have examined in detail depended on molecular-kinetic theory. Rather than re-arguing the point here, we refer the reader to Chapter 4 where we showed how each of Perrin's theory-mediated measurements of the kinetic energy of granules did not require any claim from molecular-kinetic theory *constitutively*. It is because the measurements of the primary quantities were independent of molecular-kinetic theory that a further premise was needed in each of the four approaches to obtain a value for N_0 from the measured value of the primary quantity.

Departing still farther from van Fraassen, notice how these further premises—the "weak links," as we have called them—offered remarkably little information about "molecules." Because of this, they were "weak" in a second sense—*they presupposed as little about the realm of molecules as they could and still yield a value for* N_0. But that, on our view, was the great virtue of the four determinations of N_0 we have discussed in detail. They left so many questions about the microphysical realm without answers, instead opening the way for subsequent experimental research to answer them, even one by one. Van Fraassen regards the empirical grounding of molecular-kinetic theory achieved during the years we have been considering as making possible experimental testing of that theory as a whole. We regard it as having made possible a systematic experimentally based further development of that theory, which it so badly needed given the unresolved difficulties that had encumbered it for decades and the new ones Poincaré referred to that were then emerging.

The remark about presupposing as little as they could and still yield a value merits further comment insofar as it is central to theory-mediated measurement in general. The key presupposition in any such measurement is the one linking the less accessible target quantity to its more accessible proxy. Having it be as limited in its theoretical commitments as possible while still yielding a stable measured value of the target quantity has the virtue of making the measurement more robust—that is, secure from future theory change—than it would be otherwise. The stability of the measure, in the technical sense we have employed in this

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book, can provide evidence supporting this theoretical link between the quantities, at least for purposes of measurement over the range for which measured values are obtained. This gives stable theory-mediated measures an integrity in their own right independently of the evidence supporting the theory from which they are derived.

The difficulty during the late nineteenth century in obtaining values of Avogadro's number from such proxies as the coefficients of viscosity and diffusion came from having to assume so much about molecules and their interaction, such as their sphericity, or even weaker, their having a definite effective size *s* in terms of which mean free paths could be determined. By contrast, Perrin's determinations of Avogadro's number did not require any assumption about the millions of molecules in the immediate vicinity of each granule and how their interactions were creating local, extreme high-frequency pressure fluctuations that over time were restoring kinetic energy to the granules lost in their viscous interaction with the surrounding fluid. This does not mean that more general theory is irrelevant to theory-mediated measurement. Its role, in addition to the derivation of the link between the two quantities in the first place, is to address questions about why the measure is stable, what its envelope of applicability might be, what sources of potential systematic error it might incorporate, and ultimately what is truly being measured. These questions, one should note, are subject to continuing inquiry as theory develops further, especially as other measures of the same target-quantity emerge, often involving quite different general theory.

Because this difference between us and van Fraassen concerning the enrichment of kinetic theory is at the core of our disagreement with him, we had best spell out a specific instance of it before we turn to his second step in dismissing Perrin's claim about the reality of molecules. We agree with him that two major enrichments to molecular-kinetic theory occurred with the advent of statistical mechanics when Avogadro's hypothesis took the form of a universal constant N_0 tied to the gas constant R and Boltzmann then tied the constant k not only to these other two, but to entropy. So, we are not disagreeing with van Fraassen about either the character or the importance of the theoretically posited connections that occurred with the enrichment of molecular-kinetic theory made with the advent of statistical mechanics. Our point, developed in section 6.3, is that the actual determinations of N_0 and k cited by Perrin managed to bypass presupposing the relation $N_0 = R/k$. They did so through e providing values for N_0 independent of that relation, and Planck's blackbody relation doing so as well for k. As a consequence, each of the three quantities was measured independently of the theoretical relation among them, and consequently the values so obtained then provided evidence supporting that theoretical relationship—indeed, supporting it "locally," so to speak, without having to import more of the complex apparatus

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of statistical mechanics! And in doing so they introduced an empirical constraint on the further development of statistical mechanics at the time.

We will allow van Fraassen to state the second step in his dismissal of Perrin's proposal in his own words:

> It is still possible, of course, to also read these results as providing evidence for the reality of molecules. But it is in retrospect rather a strange reading—however, much encouraged by Perrin's own prose and by the commentaries on his work in the scientific and philosophical community. For Perrin's research was entirely in the framework of the classical kinetic theory in which atoms and molecules were mainly represented as hard but elastic spheres of definite diameter, position, and velocity. Moreover, it begins with the conviction on Perrin's part that there is no need at his late date to give evidence for the general belief in the particulate character of gases and fluids. On the contrary (as Achinstein saw) Perrin begins his theoretical work in a context where the postulate of atomic structure is taken for granted.[112]

Van Fraassen goes on to propose an alternative to the usual reality-of-molecules interpretation of what Perrin accomplished with which we have some sympathy; but let us postpone considering that until Chapter 7.

We have already shown in detail how Perrin's theory-mediated measurements did not presuppose classical kinetic theory, much less a hard elastic sphere model of atoms and molecules. The sole place where that model enters the discussion was when he compared his values for Avogadro's number with the far more widely varying values obtained from the measured viscosities of gases. Save for the more recent value of N_0 derived from argon cited by Perrin, the values from the viscosities of gases he considered corresponded to those in the last chapter of Meyer's 1899 *Kinetic Theory of Gases*. As we have seen, the more telling comparisons Perrin made were with the values obtained by Millikan and Fletcher, by Rutherford and Geiger, by Boltwood and Rutherford, and by Planck. None of those values presupposed the hard elastic sphere model of molecules. Indeed, as we remarked near the end of section 5.4 in Chapter 5, those values were regarded by some at the time as a starting point for reconsidering and perhaps refining that model, at least as it was being deployed in the analysis of viscosity and diffusion. From that point of view, the virtue of the values of N_0 we have been emphasizing was their very independence from classical kinetic theory!

The main point we want to address in the quotation from van Fraassen is his "rather a strange reading" remark. Before turning to it, however, we should again lay out a specific instance in which the values for Avogadro's number, rather than presupposing kinetic theory, were providing a basis for reconsidering it. Of the

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different measures we have been emphasizing, the one that would seem most to have presupposed statistical thermodynamics—more specifically, Boltzmann's probabilistic theory of entropy (which of course had nothing to do with hard elastic spheres)—was Planck's equation, whether obtained via his derivation of it or Einstein's. During the decade after its publication, Planck's *equation* had been repeatedly confirmed experimentally. E. Warburg and H. Rubens each gave reports on its verification at the 1911 Solvay Conference, immediately preceding Planck's own presentation. More than just the equation had been confirmed experimentally, however. That the coefficient of temperature k in the equation stands for Boltzmann's constant had also been confirmed through establishing that the value for it measured from blackbody radiation spectra agrees with values obtained for the mean energy per degree Kelvin from other phenomena. The most direct comparison yielding this confirmation was with the mean energy per degree Kelvin of Perrin's granules. Less direct was a comparison of the value for Avogadro's number inferred from R divided by the measured value of the coefficient k—the very point Planck was making when he derived a value for N_0 in his original paper.

Indeed, as we noted earlier, Warburg ended his report on the verification of Planck's equation at the Solvay Conference by comparing values for e derived from blackbody radiation with values derived from the other measurements we have been emphasizing: in esu, 4.69×10^{-10} (along with his challenged 4.32×10^{-10}) from blackbody radiation; 4.65×10^{-10} and 4.79×10^{-10} from α-particle radiation; 4.89×10^{-10} from Millikan; and 4.23×10^{-10} from Perrin's results for Brownian motion.[113] The agreement among these numbers not only confirmed that the coefficient k in Planck's equation represents Boltzmann's constant; *in doing so it also rendered that assertion independent of any theoretical derivation of this equation*! To put the point more generally, the experimental verification of Planck's equation rendered even its validity independent of statistical thermodynamics.

The importance of this should not be underestimated. One of the principal issues at the Solvay Conference was the proper theoretical derivation of Planck's equation. Einstein's derivation had quantized energy universally, Planck's had not, and some of the attendees, as we saw earlier in the quote from Perrin's book about his attitude toward these derivations, would have preferred a derivation requiring no quantization at all.[114] (In a letter following the Conference, Einstein remarked, "I was able to convince Planck to a large extent about my views, now that he has resisted them for years."[115]) The experimental verification of Planck's equation and the signification of the constant k in it had freed the equation from all derivations of it, in the process turning it into a constraint on further theorizing. To put the point slightly differently, the equation and the signification of k in it had become an established fact; the question then concerned what it was

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telling us about the physical world. Notice, however, that what had done this was not the measurement of k in blackbody radiation by itself, but the comparison with other measurements, whether of Avogadro's number or constants related to it like the fundamental unit of charge.

Notice as well the seeming element of circular reasoning between this argument for k being Boltzmann's constant independently of the derivation of Planck's equation and our earlier argument that the relationship among R, N_0, and k was being established empirically independently of its derivation within kinetic theory. There we relied on k being Boltzmann's constant to use the value obtained for it from blackbody radiation as grounds for saying that, numerically, $k = R/N_0$. But here we have appealed to the relation among these three along with the not exceptionally good match between the measured number and the mean energy per granule in Brownian motion as grounds for saying that, numerically, the value of k obtained from blackbody radiation matches, to within the precision of the experiments, the value determined independently of blackbody radiation for Boltzmann's constant.

This was not a careless mistake on our part. We intentionally chose the two examples in order for the seeming circularity between them to illustrate how the converging values for N_0 and the quantities related to it from diverse theory-mediated measures were able to contribute to ongoing research at the time. Metaphorically speaking, the values from the different measures could be *played off against one another*. Thus, when the issue under consideration concerned the derivation of Planck's equation, the other determinations of k allowed the equation to stand independently of any derivation of it, leaving a free range for considering alternative derivations. If, instead, the issue was something like the equipartition of energy in classical kinetic theory, the determinations of R, k, and N_0 independently of one another allowed the equation $k = R/N_0$ to stand independently of its derivation within kinetic theory. The overall collection of converging determinations did not make them all independent of any theoretical derivations of the equations constitutive of their theory-mediated measurement. Rather, it lessened the dependency of all of those constitutive equations on the theoretical derivations of them, thereby allowing them to be played off against one another when considering changes to the theories by means of which they were originally derived.

This was one way of playing off the converging theory-mediated measurements of the quantities in question against one another. A second, illustrated earlier in section 6.2, was to use differences among the values obtained from the quantities to expose sources of systematic error in the measurements. Thus, for example, we saw in that section a clustering of values for N_0 between 59×10^{22} and 62×10^{22} that had been obtained by Rutherford and Geiger, Boltwood and Rutherford, Millikan, and Planck by 1911—all markedly less than the values

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Perrin had obtained, yet greater than Fletcher's 57.5×10^{22}—from Brownian motion. That raised questions not only about which values should be preferred, but also about what Perrin's consistently large values might be indicating about his measurements. What saying that converging measurements can be played off against one another in more than one way really means is that they can provide diverse forms of evidence in ongoing research, depending on the nature of the questions on which that research is focusing.

Returning finally to van Fraassen's "strange reading" remark about Perrin's results being evidence for the reality of molecules, we agree that it would be strange—not just in retrospect, but even at the time, long before Heisenberg and Schrödinger—to take any of the various determinations of Avogadro's number as evidence for the reality of molecules "as hard but elastic spheres of definite diameter, position, and velocity." We take ourselves to have shown that, regardless of the extent to which such a picture of molecules remained in the minds of physicists at the time,[116] that picture in no way entered constitutively into the determinations of N_o or the quantities related to it that we (and Warburg) have emphasized. Indeed, judging from Jeans's 1916 edition of his *The Dynamical Theory of Gases*, the determinations of N_0 led to clear experimental evidence that molecules are *not* "hard but elastic spheres of definite diameter."

In dismissing van Fraassen's argument against taking those determinations to have been evidence for the reality of molecules, however, we have not explained how and why it was appropriate at the time to take them to be evidence for it. We turn now to that task, focusing on Perrin's paper at the Solvay Conference, "Les preuves de la réalité moléculaire."

Something that does appear strange in retrospect is why Perrin was invited to give *this* paper, more than twice as long, at 98 pages, as any other paper at the Conference save for Sommerfeld's 60 pages, and occupying 22 percent of the Proceedings volume, yet prompting the least discussion of any. The letter of invitation to attend the Conference, part of which we quoted in Chapter 1, made clear that the reality of atoms and molecules was not what was going to be at issue:

> It appears that we find ourselves at present in the midst of a new evolution of the principles on which the classical molecular theory and the kinetic theory of matter have been based.
>
> On the one hand, this theory leads to a logical development of a radiation formula whose validity is contradicted by the results of experiments; on the other, there follows from the same theory certain results on the specific heat (constancy of the specific heat of a gas with the variation of temperature, the validity of Dulong and Petit's law up to the lowest temperature), which are equally refuted by many measurements.

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> As Planck and Einstein in particular have shown, these contradictions disappear if one places certain limits (doctrine of energy quanta) on the motion of electrons and atoms in the case of their oscillations around a position of rest. But this interpretation, in turn, is so far-removed from the equations of motion of material points employed until now, that its acceptance would necessarily and incontestably lead to a vast reformation of our current fundamental theories.
>
> . . . A great step in the direction of the orderly continuous development of atomistics would have already been taken if one could clearly establish as to which of our molecular and kinetic interpretations are in accord with observations and which others should undergo a complete transformation.
>
> . . . The subjects of the conference treatises will be the following:
>
> 1 —The derivation of the Rayleigh formula for radiation.
> 2 —Comparison of the kinetic theory of ideal gases with the results of experiments.
> 3 —Application of the kinetic theory to emulsions.
> 4 —The kinetic theory of specific heats due to Clausius, Maxwell, and Boltzmann.
> 5 —The formula for radiation and the theory of quanta of energy ("Quantenhypothese").
> 6 —Specific heat and the theory of quanta.
> 7 —Application of the theory of quanta to a series of problems in physics.
> 8 —Application of the theory of quanta to a series of problems in physical chemistry and chemistry.
>
> For each of these questions we are asking a particularly competent member to write a preliminary paper. These papers, written in French, in German, or in English, will be printed and distributed to the different members, if possible by the end of September; later they will be joined in one volume together with accounts of the discussions prompted by them.[117]

While Perrin was on the list of invitees from the outset, the one topic not in the original list proposed by Nernst, but added by Solvay or one of his advisors, was the "application of the kinetic theory to emulsions."

In keeping with the invitation, Perrin's subtitle, in parentheses, was "(*Études spéciale des émulsions*)"; and slightly more than half his paper concerns Brownian motion results that he and his colleagues had produced, including Léon Brillouin's results on granule diffusion not yet available at the time of Perrin's 1909 monograph. The one other slight departure from that monograph involved the range of values determined for Avogadro's number: 68.3×10^{22} (versus 70.5×10^{22}) from vertical-rarefaction and 68.8×10^{22} (versus 71.5×10^{22}) from granule

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displacement, along with Brillouin's 69×10^{22} from granule diffusion. The only question during the brief discussion of Perrin's paper concerning his results raised the possibility of his numbers being affected by radiational pressure acting on the granules, a possibility that he immediately dismissed. (Virtually all the rest of the discussion concerned the proper magnitude of the value of *e*, with Einstein calling attention to recent results that had shown Ehrenhaft's much smaller measured values of *e* to be a consequence of experimental error, and Madame Curie citing the agreeing values published by Rutherford and Regener.)

Insofar as Perrin's 1909 monograph had been translated into German and English in 1910 and was surely known to the attendees, one can ask what this part of his paper contributed to the Conference. So far as we can see, the answer seems to have been (1) the strong evidence for the thoroughgoing irregularity of Brownian motion and hence too of the pressure fluctuations driving it, and (2) the consequent evidence for the granular character of matter, along the lines we made explicit in Section 5.6 of Chapter 5. This was not a small contribution insofar as one of the topics at issue at the Conference was Boltzmann's probabilistic interpretation of entropy and the second law of thermodynamics. Thus we find Einstein opening the discussion of his own paper, at the end of the conference, insisting that "another principle whose validity we must retain unconditionally is Boltzmann's definition of entropy through probability."[118] The sole experimental evidence that Einstein cited in arguing for this proposal was the Gaussian distribution that Perrin had obtained in his experiments on granule displacements![119]

The remaining 35 pages of Perrin's paper present, in some analytical detail, values for Avogadro's number obtained from other phenomena. These included the recent oil-drop results by Millikan (who was not present), up-to-date results from α-particle radiation by Rutherford (who was present) and his colleagues, and results from blackbody radiation. As in Perrin's 1909 monograph and subsequent *Les atomes*, this discussion ends with proposed values for N_0, 67×10^{22}, and *e*, 4.3×10^{-10} esu, and a section entitled "*Réalité des molecules*" that includes a table[120] listing the converging values for N_0 followed by a pronouncement:

> We are struck with admiration before the miracle of such precise agreement from the basis of such different phenomena. That we not only find the same magnitude obtained by each method while varying experimental conditions as much as possible, but also that the numbers so defined without ambiguity by so many methods coincide: this gives molecular reality as much certainty as the principles of Thermodynamics can have.[121]

Notice that the final phrase in this conclusion, expressed to other leading figures in the field, differs a little from the one in the corresponding passage in his *Les*

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atomes that we quoted at the opening of the chapter—"gives to the real existence of the molecule a probability bordering on certainty."

In keeping with the plural "proofs" in the title of Perrin's Solvay Conference paper, the others present doubtlessly took each of the several agreeing determinations of Avogadro's number to have provided a proof of some sort or other. The passage we just quoted, however, suggests that his definitive proof of the reality of molecules, overriding any shortcomings in any of the prior ones, was the convergence of the diverse determinations. This raises two questions. How might the others at the Conference have construed this convergence as providing the strongest evidence? And, given that most of them, if not all,[122] had already accepted the existence of molecules, what might they have thought Perrin's proofs were contributing to the Conference?

Perrin's presentation of each of the determinations was careful and thorough, and several of those contributing to them—Einstein, Langevin, Rutherford, and Planck—were present. So, the conferees should have had little trouble seeing that the measurements of what we have called the primary quantities in the determinations we have emphasized were at most only weakly dependent on molecular-kinetic theory, if not entirely independent of it. That would have highlighted the premises linking those primary quantities to the claimed count of molecules in a mole: (1) that the kinetic energy per degree Kelvin of granules in Brownian motion matches that of molecules; (2) that the ions in a gas are just atoms, or molecules, with one or more unit charges; (3) that one fundamental unit of charge yields one molecule of a monovalent substance in electrolysis; (4) that α-particles retain their discreteness, and hence become molecules, when they lose their positive charge to form helium gas; and (5) that the coefficient of temperature in Planck's equation is identical with the Boltzmann constant of statistical mechanics, giving the mean energy per molecule per degree Kelvin. But then they would have viewed the convergence of the corresponding values for N_0, just as we said in section 6.5, as responding to any remaining uncertainty in any of these five considered separately from the others. And with that the evidence for the reality of molecules would have been seen as derivative from the strong evidence for the discreteness of the unit charge, the discreteness of α-particles, the discreteness of $h\nu$ quanta of energy in radiation, and the discreteness required of localized elements in the fluid in order for their thoroughly random, mutually uncorrelated motions to sustain Brownian motion.

In section 6.5 we put scare quotes around the word "molecules" in essentially the same construal of the evidence for their existence. We did this because, construed in this way, it offered so little evidence pertaining to features of molecules that had been postulated within molecular-kinetic theory at one time or another in conjunction with proposed explanations of phenomena. As we said there, the evidence left so many questions unanswered precisely because

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the determinations of N_0 singled out were so minimally tied to molecular-kinetic theory.

Perrin included three determinations in his Conference paper beyond the ones we have been emphasizing that depended more strongly on that theory, namely those derived from viscosity, critical opalescence, and the blueness of the sky, with both of the latter tied to fluctuations in fluids, that is, to "regimes of permanently varying inequalities in the properties of the microscopic portions of matter in equilibrium."[123] From viscosity Perrin listed two values in his table, ca. 45×10^{22} in parentheses based on the volume of monoatomic mercury and 62×10^{22} based on van der Waals's equation and argon. No sooner did he give those values in the body of his paper, however, than he indicated that errors in the values of mean free paths, owing to the spherical-molecule assumption in the formula tying them to viscosity, rendered these values for N_0 uncertain.[124]

The values from both critical opalescence and the blueness of the sky (as a consequence of the diffraction of light) were both based on an equation derived by Keesom and later independently by Einstein for the intensity of light diffused by one cubic centimeter of a fluid.[125] The value Perrin listed in his table for the former was 75×10^{22}, while for the latter, 60×10^{22} with a question mark attached. In the derivation of the first of these two, Perrin had spoken of "approximation within 15 percent."[126] So, these two values, like the two obtained from viscosity, were likely viewed by others at the Conference as being rather uncertain compared to those from the determinations we have been emphasizing. They offered a promise of tying the latter to determinations more dependent on molecular-kinetic theory and hence giving more information about molecules themselves, but further research was going to be required for that promise to be realized.[127]

That the clearest evidence for the existence of "molecules" left so many questions about them unanswered was likely regarded at the Conference as not all that surprising and in many respects a virtue—for consider the topics at issue. The equipartition of energy principle at the heart of the classical kinetic theory of specific heats was under challenge from Einstein's quantum theory for solids, in which the vibration of individual atoms disappears at low temperatures from the absence of sufficient energy to exceed an $h\nu$ quantum threshold for those statistically at low energies, as well as from Nernst's proposal that rotational degrees of freedom of non-spherical diatomic molecules in gases drop out at low temperatures for the same reason.[128] Needless to say, the challenge here was not merely to the equipartition principle, but to the classical mechanics of motion as well, at least so far as atoms and molecules were concerned.

The equipartition principle deployed in the classical statistical thermodynamics of radiation was also under challenge by the failure of the Rayleigh-Jeans blackbody equation and the success of Planck's equation. Jeans's effort at the Conference to save the principle by deriving Planck's equation instead

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from hypothetical proposals concerning the process of energy exchange between matter and radiation also put at issue how neutrally charged atoms and molecules nevertheless release (and absorb) electromagnetic radiation, presumably through some sort of interaction with the ether or motions of charges, in particular electrons, within them. In rejecting such proposals in his paper, "On the Application to Radiation of the Theorem of the Equipartition of Energy," that opened the Conference, Lorentz concluded that a satisfactory formula for radiation and explanation of Planck's constant *h* were going to require "considerations of an entirely different order" from those of classical theory.[129] Reinforcing the need to look for new directions were Kamerlingh Onnes's short paper on the newly discovered phenomenon of superconductivity and papers considering possible further applications of the quantum principle, like Langevin's on the kinetic theory of para- and ferro-magnetism.[130]

In short, what was at issue at the Conference concerned how to move away from classical molecular-kinetic theory and statistical thermodynamics without abandoning them entirely. In such a circumstance it was a virtue, not a shortcoming, of the convergent determinations of Avogadro's number emphasized at the end of Perrin's paper that they were so minimally tied to classical theory; then the values for N_0 and the various other microphysical quantities tied to it could provide an empirical basis for further theorizing without imposing constraints that might have turned into obstacles to be overcome. Theory-mediated though they were, the determinations of these quantities did not presuppose stances on the theoretical points at issue at the Conference and hence were not going to require revision as these issues came to be resolved. The values, at least to the level of precision then achieved, could be regarded as established facts. It was, moreover, the convergence of the values from diverse phenomena that gave them their factual status, for because of what we have called the "weak-link" premises, any one determination by itself would have been regarded by those at the Conference as precarious until it was supported by others.

The mention of constraints on further theorizing points to the answer to our other question about the attitudes toward Perrin's paper at the Conference: What might the others have seen Perrin's "proofs of the reality of molecules" as contributing to it? It also points to why it was appropriate for Perrin's going to such length in his paper, covering not only his determinations of N_0 from Brownian motion, but also those from other phenomena, laying out in considerable detail the considerations in every case leading to the equations mediating the determinations. Not just the values of the quantities, but the equations used in deriving them from measured phenomena, were going to be serving as at least provisional constraints on further theorizing.

We have already cited an example of this with Einstein's invoking Perrin's finding of thoroughgoing randomness in his defense of Boltzmann's probabilistic

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account of entropy at the Conference. The detail to which Perrin went made clear to the other attendees just how robust were the various equations and values going into what we have called the measurement of the primary quantities in each determination of N_0, and equally how limited were the further claims made by the additional "weak-link" premises licensing the values for N_0 obtained from those quantities. In doing so, Perrin's paper, though not the only one at the Conference to focus on experimental results, made clear to all several important, by then more firmly established, empirical results pertaining to microphysics.

One of these results, tied at most loosely to classic molecular-kinetic theory, was that matter consists of chemically stable discrete elements—"molecules," as it were—numbering $(64 \pm 5) \times 10^{22}$ per mole. Moreover, a *mole* is constituted not just by pneumatic conditions of temperature and pressure, but also by chemical composition. Thus the mole density of a gas composed of NO_2 begins to decrease when conditions reach a point where N_2O_4, nitrogen peroxide, begins to form. But then the number of chemically stable discrete elements forming that gas must begin to decrease as well. The step from "molecules" to the molecules of chemistry was thus quite small as a consequence of the converging values for Avogadro's number, even without invoking the monoatomic molecules of chemically inert helium in the Boltwood-Rutherford measure. To put the matter only slightly differently, the converging values of N_0 had licensed the ideal gas law to be written not just as $p = (n/V)RT$, where (n/V) is *mole-density*, but also as $p = (N/V)kT$, where (N/V) is the number of chemically defined molecules per unit volume.

Furthermore, while a great many questions were left open about these molecules, the convergent determinations of N_0 had provided strong evidence for the links between them and such phenomena as the thoroughly random Brownian motion; ionization, electrolysis, and the fundamental unit of charge; α-particle radioactive decay and the transformation of radium into other elements; and blackbody radiation. The general question left for further theorizing was how to preserve those results and links among parameters while moving away from classical molecular-kinetic theory. All this, needless to say, underscores just how far removed the actual historical situation was from van Fraassen's claim that "Perrin's research was entirely in the framework of the classical kinetic theory in which atoms and molecules were mainly represented as hard but elastic spheres of definite diameter, position, and velocity."

Such spherical molecules continued to appear in works presenting *mathematical* statistical mechanics in the years after 1912, but this was because the actual shapes of molecules and the force-fields involved in their interactions remained unknown, and impact of elastic spheres was mathematically tractable; and, thanks in part to the successful determination of N_0, comparison of the dynamical theory of elastic spherical molecules with experimental results increasingly

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underscored how limited this theory was for linking any specific quantitative feature of phenomena like viscosity to specific features of chemically different molecules.[131]

To summarize, those at the Conference would scarcely have found Perrin's "proofs" strange at all. Equally, however, they would not have regarded them as addressing the question, Do molecules *really* exist at all? For no one at the Conference was questioning this. Rather they would have seen the "proofs" as laying out what about the molecular realm could be regarded as having been experimentally established during the preceding decade—or, perhaps better put, sufficiently established to be taken as constraints in ongoing research on the topic of the Conference, "the theory of radiation and the quanta." Such constraints were needed, even somewhat desperately, given how *prima facie* irreconcilable quantization seemed to be with classical physics in 1911. Perrin's paper was not the only one at the Conference to be presenting such constraints, but it did the most toward doing so. Its exceptional length was accordingly not inappropriate.

6.7. On Eliminating All Reference to the Invisible

As a final step in his argument that the empirical content of Perrin's results extended only to the grounding of molecular-kinetic theory, van Fraassen cites Perrin's remark near the end of his 1909 monograph, "we ought always to be in a position to express all the visible realities without making any appeal to elements still invisible."[132] Perrin followed the remark first by noting how "the constant N is simply a completely known numerical factor, figuring in the enunciation" of each equation employed in its determination, and then showing how to drop the term by using any one of these equations to substitute for N_0 in all the others, yielding equations in which "*only evident realities occur*."[133] Perrin made the same proposal at the end of both his 1913 book and his 1911 Solvay Conference paper. What are we to make of it?

As we have been doing, let us focus on this proposal at the Solvay Conference and how its attendees might have taken it. Perrin's specific phrasing there, which immediately followed the remark we quoted earlier about molecular reality having as much certainty as the principles of thermodynamics can have, was:

> Nevertheless, and as greatly necessary as the existence of molecules or atoms may be, we should always be in a position to express visible reality without making appeals to as yet invisible elements. And indeed this is very easy. It suffices to eliminate the constant N among the p equations that helped us determine it, in order to obtain (p–1) relations in which only sensible properties figure and which express profound connections among phenomena at first

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> sight as completely independent as the viscosity of gases, Brownian motion, the blue of the sky, the blackbody spectrum, or radioactivity.[134]

Perrin proceeded to illustrate this by combining Einstein's equation for granule diffusion in dilute emulsions with Planck's equation:

$$\xi(\lambda,T)=\frac{8\pi ch}{\lambda^5}\frac{1}{e^{\frac{ch}{\lambda T}\frac{6\pi a\zeta D}{\theta}}-1}$$

where c is the speed of light and θ is the temperature at which the value for the diffusion coefficient D is measured. Perrin then added, "This ensures that we can control, by means of spectrophotometric measures, points pertaining to emulsions!"[135]

Surely most of those at the Conference would have regarded this equation and even more so the remark following it as parody, ridiculing those like Duhem and Mach who were still continuing to deny the admissibility of molecules and atoms in physical theory. Just consider for a moment what sort of research would have revealed this relation between granule diffusion and blackbody radiation as a robust empirical result independent of any thought of molecules and atoms. Or consider the challenge to theoreticians to offer an explanation of it without appealing to microphysics.

That Perrin himself may well have intended it to be parody, at least at the Conference, is reinforced by his closing paragraph, immediately following the exclamation point quoted in the preceding:

> But we will not have the tactlessness, under the pretext of rigor, to rid our equations of the elementary magnitudes that enabled us to obtain them. This would not amount to taking away the scaffolding rendered useless to the completed building, but to concealing the pillars that make up its structure and its beauty.[136]

Although we have not managed to establish the direct connection, we find it difficult to believe that Perrin's choice of words here was not drawn from Ostwald, as reflected in a passage from his 1907 *Monist* paper that we chose not to quote earlier (that had been published in German earlier that year in the opening number of *Rivista di Scienza*):

> In the scientific language of to-day these two fundamentally different kinds of hypotheses are called by the same name, "hypotheses." I am in favor of leaving the name hypothesis for the unverifiable assumptions, since indeed most hypotheses of the science of to-day are of this character. The other assumptions

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> which, like scaffolding, serve the purpose of the particular investigation on hand, and in the course of the work are replaced once or oftener according to our need for new usable assumptions until the condition sought for is actually found,—these assumptions made for the purpose of positive work, I call prototheses. A protothesis, therefore, is set up at the beginning of an investigation and disappears at its close if the work is successful; while a hypothesis is established when the work can be carried no farther. For this reason it is customary that in the production of scientific work the different prototheses which the investigator has employed are for the most part not mentioned at all, for it has been usual to communicate only those assumptions which the investigation has finally proved to be correct or at least approximate. Silence is kept with regard to the unsuccessful prototheses just as the scaffolding is removed after a building is completed. Only very rarely, as for instance in Kepler's reports of his astronomical researches, do we learn a little about unsuccessful prototheses. On the other hand the hypotheses in the narrower sense occupy a large place in literature. Because inasmuch as they have reference to scientifically inaccessible things, they can be neither proved nor refuted, an endless pro and con is the usual result; further, because among the problems there are also pseudo-problems which do not refer to demonstrable things, these problems are unsolvable and are dragged through science as unanswered questions which can not be disposed of, but not until they are recognized to be merely pseudo-problems.[137]

Ostwald himself, who had not been invited to the Conference, must have taken Perrin's words to be responding to his when he finally read the Conference proceedings, whether in the original French or the official German translation.

Parody though they may well have been, Perrin's closing remarks contained two important further points not stressed during his review of the different determinations of N_0. First, his reference to "elementary magnitudes" reminded his audience that not just Avogadro's number, but other terms as well appearing in the equations, involved "invisible elements." Even if the unit charge was not being identified with the electron, counts of invisible ions associated with water and oil droplets entered into some of the determinations, as did counts of α-particles. Getting rid of the latter, for example, would have required replacement of the rate of α-particle production in the equation by the number of "throws" per minute of an electrometer in an extraordinarily recondite experimental setup. But then too the number of electromagnetic cycles per second of light of any specific color was scarcely visible either. The challenge posed by the invisibility of molecules and the like in 1911 was not one of inaccessibility, for enough experimental access had been gained to answer *some* questions about them. The

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challenge was one of designing further experiments to gain still greater access. We should not be surprised that Laue and the Braggs were awarded Nobel Prizes, each within two years of their providing such further access by means of x-ray diffraction over the next two years.

Second, the emphasis Perrin placed on the wide range of "at first sight completely independent" phenomena yielding agreeing values for N_0 amounted to more than just an argument that the values were well founded. The more ubiquitously N_0 shows up in equations describing different phenomena, the stronger are the grounds that it represents a *fundamental* constant of nature.

This last point, however, can be turned on its head. Ostwald had introduced the word "mole" only a couple of decades earlier to stand for a fundamental unit in chemistry. What made this unit appropriate was the realization that the chemical makeup of a gas can change with temperature and pressure—for example, the formation of nitrogen peroxide in a gas composed initially of NO_2 as the temperature is lowered. To accommodate this, the gas law needed to be reformulated as a proportionality among pressure, absolute temperature, and, rather than density, mole density or its equivalent. The mole nevertheless had a somewhat artificial character as a unit, tied as it was to the system of atomic (or "equivalent") weights, and its somewhat arbitrary choice for the unit weight—not to mention the still unfolding system of chemical formulas. (This artificiality came even more into question two years later with Mosley's evidence that atomic number, not atomic weight, is the fundamental chemical parameter.) The diverse multiple determinations of N_0 served historically to remove worries about the artificiality of the mole. In particular, the two robust determinations from radioactivity, one (involving volume) dependent on R and the other (involving charge) not dependent on R, but on Faraday's constant, gave evidence that N_0 provides a physical characterization of something universal to a mole, namely its containing, at least so far as precision of measurement then allowed, a specific number of chemically stable independent discrete elements—molecules, as it were. The more diverse were the phenomena that yield increasingly convergent values, the greater was the evidence that N_0 represents a fundamental constant that provides such a count, and hence the stronger were the grounds that the mole is a well-constituted physical unit.

Both of these final two points that we have teased out of Perrin's closing remarks reinforce his claim that quantities like N_0 pertaining to invisible elements in the equations he had presented did not comprise mere "scaffolding" to be discarded in the future. They were going to remain as "pillars," regardless of how the reformulation of classical molecular-kinetic theory and statistical thermodynamics might unfold, not to mention molecular theory within chemistry.

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6.8. A Parting Comment, Regarding *réalité moléculaire*

Our conclusion, therefore, is that the converging measures of Avogadro's number, though still at the time less convergent than one might want, provided much stronger evidence as of early 1913 for the "reality of molecules" than Perrin's results for Brownian motion alone did. While the Gaussian distribution of displacements in Brownian motion provided strong evidence for the discontinuous or "granular" character of the liquid substrate, a clear lacuna remained between this conclusion and the claim that the discontinuous character in question is from chemical molecules, or even Maxwell's "molecules." We grant that Perrin's results left molecular-kinetic theory as the only available explanation for both the perpetual and stochastic character of Brownian motion. To infer the "reality of molecules" from this, however, is to engage in inference to the best explanation. Regardless of what one thinks of inference to the best explanation, to invoke molecular-kinetic theory as an explanation of Brownian motion is to presuppose the reality of the molecules being invoked and then take the explanation they provide as added evidence for them—which is more or less what van Fraassen claimed about Perrin's reasoning.

By contrast, neither the "reality of molecules" nor the molecular-kinetic theory of heat has to be presupposed if the evidence provided by the converging measures of Avogadro's number is construed as a form of same-effect-same-cause reasoning—specifically as same-magnitude-same-quantity-being-measured reasoning. Consider the three least controversial measures that were yielding, to within reasonable precision, the same magnitudes: the number of unit charges, that is, electrons, required to yield a mole of hydrogen, or any other monovalent element, in electrolysis; the number of granules in Brownian motion required to match the total energy at any given temperature of a mole of gas (or liquid solution); and the number of α-particles required to form a mole of helium. Insofar as all three yield a numerical value characteristic of a *mole*, the proposed inference plus inductive generalization yields the conclusion of the existence of some distinctive quantity—a certain number of units of something—characteristic of any mole of any substance. Suppose we call this quantity *m-units*, with the three measures taken to be indirect measures of it.

As we noted earlier, the ideal gas law implies that this quantity has a systematic relationship to chemical formulas. Suppose, for example, we take a mole of nitrogen dioxide, NO_2, and lower its temperature to the point where it consists of half (by volume) nitrogen dioxide and half nitrogen peroxide, N_2O_4. Then the pressure-volume relationship of the mixture is going to entail a mole density of two-thirds of what it was when the gas was pure nitrogen-dioxide. But this in turn is going to entail a reduction in the number of *m-units* to two-thirds of what it was originally—half of which is a measure of the number of *m-units* of NO_2,

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and the other half, of the number of *m-units* of N_2O_4. This point generalizes: the *molar density* of any gas, pure or mixed, is going to entail a corresponding value for *m-unit density*.

Nothing in this reasoning so far requires the *m-unit* quantity to be a count of individual, discrete items. Each of the quantities taken to be indirect measures of it, however, are counts of individual discrete items: the fundamental negative charge, which has a definite measured (rest) mass associated with it and hence can no less appropriately be called an electron; granules in Brownian motion; and the α-particle, which again has a definite measured mass associated with it. Extending the sort of same-cause-same-effect reasoning we have been developing, the fact that these three are counts of individual discrete items gives grounds for concluding that the *m-unit* quantity too is a count of individual discrete items; or put more conventionally, it gives grounds for taking Avogadro's number to represent a count of the number of these chemically stable individual, discrete items in a mole, each with its characteristic chemical formula. The value of Avogadro's number then suffices to determine the mass of each such item from its chemical formula and the system of equivalent weights that had been worked out over the course of the nineteenth century. Moreover, we have reached these conclusions without invoking the evidence for the discontinuity or "granular" character of matter provided by Brownian motion, and hence that evidence adds further support for taking Avogadro's number to represent a count of something.

Insofar as no one had any claim to knowing anything more about the hypothetical entities called "molecules" than what we have listed in the preceding, and before 1908 the absence of a credible value for Avogadro's number had left them with no legitimate claim to knowing some of what we have listed, the reasoning we have offered provides grounds for calling the items counted by *m-units* "molecules." This reasoning required no assumptions whatever about the sizes and shapes of such molecules, nor about how they interact with one another. In particular, the molecular-kinetic theory of heat did not enter into the reasoning at all, nor, contrary to van Fraassen, did any assumption to the effect that what we are now calling molecules are "hard but elastic spheres of definite diameter, position, and velocity." Nor did the reasoning in any way conflict with the claim Nernst made so forcefully in the 1893 edition of his text that we quoted more than once in Chapter 2, a claim that he repeated word-for-word in the 1908 edition: "At present scarcely anything definite is known regarding either the nature of the forces which bind the atoms together in a molecule and which hinder them from flying apart in consequence of the heat motion, or regarding their laws of action." In the 1921 edition, Nernst again repeated this sentence, but appending to it a further clause: ". . . in this case also the application of the quantum theory is helping to throw light on the problem."[138]

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Construed as we have been proposing, therefore, the converging measures of Avogadro's number gave clear grounds for *taking* molecules to be real while still leaving open the many questions about their specific features that we argued in section 2.7 of Chapter 2 are needed for molecular theory to explain any specific features of macroscopic phenomena. In the process these questions had gained more legitimacy than they had had before, and the related results on the fundamental unit of charge, ionization, and α- and β-particle radiation had added new constraints on answers to these questions, as well as some promising leads. In fact, the questions had gained more than just added legitimacy. They had gained an added demand to find experimental answers to them, for a failure to obtain such answers would prompt revisiting of the reasoning we have proposed, in particular the steps from the converging measures to their being indirect measures of a further quantity, *m-units*, and from their being counts of their respective individual discrete items to taking *m-units* to be a count of individual discrete items.

In particular, this last step to *m-units* being a count of individual discrete items involves a reach beyond the available evidence. Nothing in the evidence we have examined *requires* a mole to consist of $(64 \pm 5) \times 10^{22}$ *individuals* with sufficient integrity to persist in and of themselves over time. Even granting that atoms exist as individuals and chemical compounds consist of atoms in combination with one another, nothing in the evidence precludes those combinations from existing evanescently, with atoms dissociating from one another and combining with different atoms all the time at an extraordinarily small time-scale. That such a process can be represented as if each combination of atoms persists over an extended time-scale, yet still as nothing but a heuristic conceptualization consistent with all the measured results, has not been ruled out by the evidence. The shortcomings of trying to further represent the combinations as spheres or having spheres of influence underscore that the evidence has not yet managed to reveal sufficient specifics of the putative molecules to assure their existence as individuals.

This, however, is not to say that the step to *m-units* being a count of individual discrete items is unwarranted for purposes of ongoing research. Rather, it is to say that it places a great deal of stress on experimentally pursuing answers to questions about the integrity of molecules as individuals, especially on, in Nernst's words, "the nature of the forces which bind the atoms together in a molecule and which hinder them from flying apart in consequence of the heat motion."

On our construal, accordingly, the converging measures by no means established the reality of molecules once and for all. For, what the term "molecule" was being taken to designate remained, in spite of the added constraints, not only highly underspecified, but perhaps even failing to refer to definite persisting

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individuals at all. One might regard the extent to which the referent of "molecule" remained underspecified as a shortcoming of the evidence deriving from the agreeing measurements considered in this chapter. This, however, would be to miss the point. The evidence is so strong for the values of the molecular magnitudes in question precisely because it presupposes so little about molecules themselves, and hence is less in jeopardy of being overturned by future research than it would have been had it presupposed more. It thus contrasts sharply with the evidence achieved in the nineteenth-century efforts to obtain values of these very magnitudes, depending as it did on some form of assumption of sphericity. In this regard our construal seems closer to Alan Chalmers's response to van Fraassen than to Wesley Salmon's original invocation of common-cause reasoning for the case of the converging values for Avogadro's number; for Salmon's construal seems to require so much more of molecular-kinetic and chemical-molecular theory than either Chalmers's or ours does.[139]

All of this is in keeping with the force of Newton's same-effect-same-cause rule of reasoning, for it too, along with his other rules, licensed conclusions only to be *taken* to be true—his phrase—for purposes of ongoing research until experimental grounds should arise prompting their reconsideration.[140] Newton deployed these rules expressly with the goal of "arguing more securely"[141]—that is, of achieving comparatively secure grounds for some conclusions about gravity while leaving questions about its mechanism open to future research. Our construal of the force of the evidence in this chapter is accordingly not without precedent.[142] Nor, for that matter, is its presumption of persistence. For this presumption is fully akin to that of Clausius, Maxwell, and Boltzmann to the effect that all molecules of, say, oxygen are the same and hence interchangeable—a presumption that the discovery of isotopes showed to be not quite true. In both of these cases, the presumptions amounted to a research strategy, for not making them was tantamount to giving up hope that ongoing research was going to get much of anywhere. Here too Newton has given a warrant for the strategy, this time with his first Rule of Reasoning, which amounts to requiring that no complication be added to theory unless expressly called for by some specific empirical finding.

Notes

1. Salmon (1984), pp. 221ff.
2. Perrin (1990), p. 215f. The English translation, unlike the French original, does not set the final section of the book, called "Conclusions," aside as a separate section, thereby understating its intended force. Also, the 1913 and 1914 editions of *Les atomes*, and the 1916 English translation of the 1914 fourth edition, ended with a comparison of

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13, not 16, values for Avogadro's number, all but two of them between 60×10^{22} and 75×10^{22}. See Perrin (1913), p. 289; (2014), p. 299; (1914), p. 293; and (1916), p. 206.

3. Mohr, Newell, and Taylor (2008), p. 86.
4. Huygens (1986). See also Smith (2007). In citing Huygens's pendulum measure, we should note that this was done first with cycloidal pendulums, but then subsequently with small-arc circular pendulums for which the length-to-period relationship matches that of the corresponding cycloidal pendulum in the asymptotic limit as the arc approaches 0. Correspondingly, his conical pendulum measure initially involved constant height pendulums, but then he added a measure implemented on a conical pendulum in which the bob is constrained to the surface of a paraboloid. For more details, see Yoder (1988).
5. Koyré (1968).
6. See Smith (2007) and (2001a), especially p. 334.
7. See Smith, "The Question of Mass in Newton's Law of Gravity," in preparation; also, Todhunter (1962), especially Volume 2; Sabine (1825); and Heiskanen and Vening Meinesz (1958).
8. Maxwell (1865), p. 499. Ironically, the "official" value of the speed of light for a time became the one obtained from Maxwell's equation—for the obvious reason that the quantities entering into it can be measured to higher precision than can be achieved by any measurement directly of light.
9. Thomson (1897). See also Smith (2001b).
10. Thomson (1899), p. 563. For more details, see Smith (2001b).
11. See Poynting (1894) and Zenneck (2007).
12. Perrin (1908d), p. 594.
13. Rutherford and Geiger (1908b), p. 171.
14. Ibid.
15. See Perrin (1910), p. 89, for a description of how Lorentz obtained his value.
16. Perrin (1990) and (1916), p. 105; in the original French, (1913), p. 150, or (2014), p. 201.
17. Ostwald (1912), p. 486.
18. Salmon (1984), p. 224.
19. Millikan (1911), p. 396.
20. Fletcher (1911), p. 106f.
21. Ibid., p. 95. Italics in the original.
22. Boltwood and Rutherford (1911); this paper had first appeared in German in March 1911. The actual value announced in the paper was 2.69×10^{19} for Loschmidt's number (p. 599); our value for Avogadro's number in the text was obtained from it.
23. Svedberg (1912), pp. 83 and 136.
24. Millikan (1913), p. 141.
25. Fletcher (1914), p. 453.
26. See, for example, Loeb (1934), p. 405; there he also cites Millikan's subsequent proposal that Perrin's reliance on Stokes's law contributed to the discrepancies as well.
27. To remind the reader, the product of Loschmidt's number and the volume occupied by a mole of a perfect gas at standard conditions—22,400 cm^3 to three significant figures, 22,413.96954 as of 2018—gives a value for Avogadro's number.

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28. Birge (1929). Birge's values there (p. 59): for Faraday's constant, 2.8927×10^{14} abs es units; for the gas constant, 8.3136×10^{7} ergs per deg mole.
29. Ostwald, (1912), p. 510.
30. Einstein to Perrin, November 11, 1909, quoted from Nye (1972), p. 135f.
31. Bohr (1913), p. 9. Bohr's calculated value differed from the measured value of Rydberg's constant by 6 percent, largely because of his values for *h* and *e* both being too small and the calculation of Rydberg's constant depending on *e* to the fifth power. See Smith (2001b) for more details.
32. The Rydberg constant *R* is normally now stated as the inverse of a wave length. Bohr's formula thus corresponds to *cR*, that is, the product of *R* and the speed of light. Bohr's derivation assumed an infinite mass for the nucleus of the hydrogen atom. In subsequent years this led to a distinction among three Rydberg constants, R_{∞}, R_H for the hydrogen atom, and R_{He+} for the helium ion. That these three values calculated from the Bohr model are observationally distinct and agree to the extent that they did with the measured values for the latter two became significant evidence for it.
33. The Braggs had correspondingly adopted a value for the mass of a hydrogen atom 11 percent greater than Perrin's value (equivalent to a value of 61.5×10^{22} for Avogadro's number) in their landmark paper on x-ray diffraction crystallography published in the same summer as Bohr's paper; see Bragg and Bragg (1913), p. 437.
34. Perrin (1908d).
35. Rutherford and Geiger (1908a, 1908b). Rutherford, by the way, was awarded the Nobel Prize in Chemistry in 1908; perhaps Perrin cites his work only in a footnote because he had not appreciated the relevance of Rutherford's work to his until then.
36. Perrin (1910), p. 82.
37. Ibid., p. 83. The experiments in question followed the method of Harold Wilson, using water droplets in an electric field; see Millikan and Begemann (1908).
38. Perrin (1910), p. 85.
39. Perrin (1912a), p. 237. As the title of the Proceedings indicates, almost all of the other papers at the Conference concerned radiation, quanta, and the problem of specific heats. The one other paper tied to molecular-kinetic theory was by M. Knudsen, reviewing experimental results pertaining to ideal gases.
40. Perrin (1990), p. 185, (1916), p. 176; in the original French, (1913), p. 250, or (2014), p. 272.
41. Perrin (1912a), p. 249.
42. Perrin (1913), p. 289 or (2014), p. 299; (1914), p. 293; in English (1916), p. 206.
43. Perrin (1990), p. 215.
44. Ostwald (1912), p. 535f, p. 510, and p. 538.
45. Ibid., p. vi.
46. Ibid., pp. 533–548.
47. Ibid., p. 536.
48. Ibid.
49. Ibid., p. 538.
50. Fletcher (1911), p. 107.

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51. See Birge (1942), pp. 116–119, where the source of systematic error in Millikan's value is discussed and reasons are given for abandoning his measure in favor of the value for *e* based on the grating space of crystals obtained from wave-lengths of x-rays determined with ruled gratings and the angles at which these x-rays are reflected by a crystal. This gave him a preferred value of Avogadro's number, $(6.0228 \pm 0.0011) \times 10^{23}$, from which he then derived his preferred value for *e*, $(4.8025 \pm 0.0010) \times 10^{-10}$ esu.
52. Rutherford and Geiger (1908a), p. 161.
53. Rutherford and Geiger (1908b), p. 168.
54. Ibid.
55. Ibid., p. 170.
56. Ibid., p. 172.
57. Millikan (1910) and (1911).
58. Perrin (1912a), p. 237.
59. Regener (1909).
60. Royds and Rutherford (1963), p. 135.
61. Dewar (1908); Boltwood and Rutherford (1909); Dewar (1910); and Boltwood and Rutherford (1911).
62. Perrin (1990), p. 210f, and (1916), p. 201ff; in the original French, (1913), p. 285ff, or (2014), p. 295f.
63. In the 1923 edition of *Atoms* but not in the French editions of (1913) and (1914) and not the English edition of (1916), the value quoted for granule displacements became 64×10^{22}; see section 4.8 and note 63 of Chapter 4 for the explanation.
64. See Kuhn (1978), p. 111.
65. Planck (1967), p. 82; (1972), p. 13.
66. See Rogers (2005) for details. Refining Debye's account still further was a paper by Born and von Kármán (1912).
67. In particular, see Perrin (1910), pp. 88–90, where he shows how little he had grasped Planck's theory with the remark, "Independently of Lorentz, by a more complicated theory, Max Planck had already arrived at the same formula."
68. Perrin (1912a), pp. 246–288.
69. Perrin (1990), p. 156. Perrin's brief discussion of Einstein's specific heat theory following this quotation ends up deferring the matter to Nernst. The citation of Bohr is a reference to the short final section of the "Light and Quanta" chapter, where Perrin notes Bohr's proposal of quantization in his theory of the hydrogen spectrum without mentioning the Bohr-Rutherford model of the atom. No such passage occurs in the 1913 or 1914 editions of *Les atomes* or the 1916 English translation.
70. Perrin (1990), p. 155 and (1916), p. 153; in the original French, (1913), p. 218, or (2014), p. 249.
71. Planck (1967), p. 89f.
72. In the Preface to the second edition of his *The Theory of Heat Radiation*, published in 1913, Planck remarked (1991), p. vii, "Meanwhile the experimental methods, improved in an admirable way by the labors of *E. Rutherford, E. Regener, J. Perrin, R. A. Millikan, The Svedberg* and others, have without exception decided in favor of

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the value [of *e*] deduced from the theory of radiation which lies between the values of Perrin and Millikan." In 1900 he had deduced values for k, N_0, and e in support of his derivation, and he had compared his value for the last of these, 4.69×10^{-10}, with J. J. Thomson's value, 6.5×10^{-10}.

73. Perrin (1912a), p. 251. Three other topics were raised in the discussion: (1) whether Perrin had taken into consideration the "pressure of radiation" (by Langevin); (2) whether Ehrenhaft's results on charged particles in the air presented a challenge to the unit charge (by Einstein); and (3) the relationship between radiation from radium and from polonium and Regener's measurements (by Madame Curie).
74. The one place we have noted (in Chapter 4 and the preceding) in which Perrin retrospectively raised the possibility of systematic error in his measurements, specifically in those of ξ^2/τ, was in a later edition of *Les atomes*, following Constantin's suggestion that including granules very near the walls was introducing such an error (Perrin 1923, p. 124); this induced him to make some new measurements and then to change his value for Avogadro's number from this measure from 68.2×10^{22} to 64×10^{22}, but without any mention of the new discrepancy arising from this change when compared with his values obtained from his vertical-gradient and Brillouin's diffusion measurements. This revision dates from after 1914.
75. Warburg (1912), p. 84.
76. Perrin (1912a), pp. 248 and 249. Perrin's announced value of 64×10^{22} from blackbody radiation at Solvay appears to have come from averaging the two values for *e* that Warburg had listed, 4.69×10^{-10} and 4.32×10^{-10}, the validity of the second of which Warburg had expressly challenged. In his 1913 book Perrin retained this value for blackbody radiation, but then in the 1921 edition, as can be seen in Perrin (1990), p. 155, he switched to 62×10^{22}, that is, a rounded-off version of Planck's 1900 value, which of course corresponded to Warburg's preferred Solvay value for *e*.
77. Warburg (1912), p. 84f.
78. Perrin (1912a), pp. 228–233.
79. Ibid., p. 233.
80. On the basis of Weiss's more recent results showing that Ehrenhaft's dust particles actually carried a charge around 4.5×10^{-10} esu.
81. Millikan (1911), p. 385. Millikan revisited his approach to the frictional action of the air on oil droplets in his 1913 paper, resulting in a revised value for *e* of 4.774×10^{-10}.
82. Fletcher (1911). Fletcher's value was 2.88×10^{14} in electrostatic units, versus a then accepted value of 2.896×10^{14} for Faraday's constant. (Birge's value for Faraday's constant in 1929 was 2.892×10^{14}.)
83. Perrin (1912a), p. 237.
84. Ibid. Even in the eleventh edition of *Les atomes* in 1921, Perrin was still saying, "The precise value of this elementary charge, which we now know to exist, has yet to be determined" (Perrin, 1990, p. 184), offering the same line of reasoning that he had in 1911.
85. In all the editions of *Les atomes*, Perrin's preferred value is 67×10^{22} rather than the 64×10^{22} of the Solvay Conference paper; see Perrin (1913), p. 251, or (2014), p. 273, (1916), p. 177, and (1990), p. 186.

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86. Fletcher (1911), p. 106.
87. Fletcher (1914), p. 453.
88. Sommerfeld (1923), p. 213.
89. Perrin (1912a), p. 249.
90. Madame Curie was, of course, present at the Conference, so perhaps Rutherford's silence was to avoid raising questions even indirectly about any of her work.
91. We know of only two sources that have given it appropriate recognition: Whittaker (1973), p. 8, where unfortunately a page later he attributes a value of 60.6×10^{22} to Perrin's determination from Brownian motion; and Post (1968), in a footnote on p. 226 with no mention of Boltwood or Rutherford by name.
92. Boltwood and Rutherford (1911), p. 599.
93. Svedberg (1912), p. 83.
94. Ibid., p. 136f.
95. Perrin (1990) and (1916), p. 128; in the original French, (1913), p. 182f, or (2014), p. 225f.
96. Nernst (1923), p. 510.
97. Ibid., p. 514.
98. Planck (1991), p. vii.
99. Poincaré (1963), p. 90.
100. Ibid.
101. According to Kuhn (1978, p. 209f), the question of the indispensability of quantizing something in the case of blackbody radiation was receiving increasing attention in the years preceding the Solvay Conference. In particular, "Ehrenfest produced a fuller proof than any previously given that no radiation law which agreed with Planck's in the high and low frequency limits could be derived without recourse to discontinuity" (p. 210). Kuhn specifically cites Ehrenfest (1911).
102. Namely, in his "The Dynamical Evidence of the Molecular Constitution of Bodies," Maxwell (1875).
103. See Rocke (1984).
104. Rutherford (1963); this abstract was followed, later in the year, by a full paper [Rutherford (1911)] in *Philosophical Magazine* that likewise drew limited favorable response until late 1913. So far as we have been able to determine, if Rutherford called attention to this paper while he was attending the Solvay Conference later that year, it was not during the recorded discussion of any of the papers.
105. Bohr (1913). This is the first of the three papers Bohr published in 1913 on the subject.
106. Moseley (1913). This again is the first of two papers Moseley published on the subject, with the second appearing in 1914.
107. van Fraassen (2009), p. 23. Italics in the original.
108. Perrin (1990), p. 215f; (1916), p. 207; in French, (1913), p. 289f, or (2014), p. 299.
109. van Fraassen (2009), p. 11.
110. Ibid., p. 23.
111. Ibid., p. 10.
112. Ibid., p. 23.
113. Warburg (1912), p. 84.

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114. Lord Rayleigh's letter to the Conference and James Jeans's paper, which preceded Warburg's, Rubens's, and Planck's, were looking for a way to obtain Planck's results within the framework of classical physics, without any quantization.
115. Letter from Einstein to Heinrich Zangger, quoted in Mehra (1975), p. xiv.
116. The most ambitious efforts on the transfer properties of basically spherical molecules along Maxwell's lines, but this time in non-uniform gases, were carried out by Chapman and Enskog in the decade following the first Solvay Conference. For the details of the resulting theory, see Chapman and Cowling (1939); included, pp. 380–390, are a historical summary of the efforts and a bibliography of the principal publications.
117. The version of the letter sent to Poincaré can be found on the site of the Solvay Institutes. The translation in our text has been adapted to fit that version from Jagdish Mehra's translation of Nernst's draft proposed to Solvay, as given in Mehra (1975), p. 6.
118. In de Broglie and Langevin (1912), p. 436. Translation from Straumann (2011); this paper offers a thorough technical discussion of the points at issue on radiation and quanta at the Conference, particularly among Lorentz, Planck, and Einstein.
119. de Broglie and Langevin (1912), p. 439. See, as well, Einstein (1993), pp. 426–437.
120. The values Perrin lists in his table (1912a, p. 249), as well as his proposed preferred value of 67×10^{22}, seem to us biased toward supporting the values he had obtained from Brownian motion. One example of this is his citing in the text Millikan's recently published oil-drop value, 59×10^{22}, in the body of the paper, but then deciding a page later that the preferred value from measurements of e is 63×10^{22}, yet finally in the table giving for this value 64×10^{22}. Another is his choice of 64×10^{22} for the value obtained from the volume of helium obtained from α-particles versus the recently published Boltwood-Rutherford value of 60.25×10^{22}; Rutherford did not comment on this during the recorded discussion. A still more striking example is Perrin's value of 64×10^{22} from blackbody radiation, while earlier in the Conference, Warburg's review of the experimental evidence for Planck's equation had given a value for e that entailed a value of 61×10^{22} for N_0. Of course, the bias we are attributing to Perrin here may instead be our own, knowing as we do, and he did not, that his values from Brownian motion were consistently on the high side versus our current value.
121. Perrin (1912a), p. 249. Translation by Adwait Parker.
122. The exception we are allowing for is Poincaré, who attended the Conference, participated to some extent in the discussions of the individual papers, and even more so in the concluding discussion.
123. Perrin (1912a), p. 216.
124. Ibid., p. 161f.
125. Ibid., p. 222.
126. Ibid., p. 223.
127. By the time of the eleventh edition of *Les atomes*, improved measurements for the diffusion of light had yielded a value of 69×10^{22} for Avogadro's number from the blueness of the sky. See Perrin (1990), p. 139.
128. Nernst's (1912) paper at the Conference focused on departures of Einstein's equation for specific heats of solids at low temperatures from measured results and a

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phenomenological equation proposed by Lindemann to eliminate the discrepancies. In the long discussion following his paper, Nernst (1912, p. 300f) proposed the dropping out of rotational degrees of freedom of molecules at low temperatures in response to a question from Poincaré, remarking on uncertainties in the dimensions and consequently the rotational energies of molecules. As we have noted earlier, the shortcomings in Einstein's formula disappeared in 1912 with Debye's refinement. Nernst's proposal on rotation did not come to full fruition until Dennison (1927).

129. Lorentz (1912), p. 34.
130. As often occurred at the Conference, Langevin's paper raised worries about how to achieve agreement with experiment when he acknowledged in the discussion period that he had adopted Sommerfeld's action principle for the motion of the electron rather than the *prima facie* more appropriate choice of Planck's because it yielded results in disagreement with experiment [de Broglie and Langevin, 1912, p. 405]. Sommerfeld's paper, which preceded Langevin's and had provoked a good deal of discussion, concerned the application of the theory of the quantum of action to non-periodic molecular phenomena, while Langevin's theory concerned a periodic phenomenon.
131. See Jeans (1916), Chapters 10 through 14. A comparison of these chapters with the parallel chapters of Meyer's 1899 edition of *The Kinetic Theory of Gases* reveals a great deal about both what had changed with the advent of a precise value for N_0, and what had not changed.
132. Perrin (1910), p. 91.
133. Ibid., p. 92.
134. Perrin (1912a), p. 249. Translation by Adwait Parker.
135. Ibid., p. 250.
136. Ibid. Perrin used the same word, "scaffolding"—"*échafandages*"—in a shorter version of the paper with the same title at the 1912 annual meeting of the Société française de Physique (1912b, p. 52). His 1909 monograph has no paragraph comparable to the one quoted, but in its place the single sentence, "The discovery of such relationships marks the point where the underlying reality of molecules becomes a part of our consciousness" (1910, p. 92). In his 1913 book he switches to a different metaphor, "In doing so we should not be removing the support from a thriving plant that no longer needed it; we should be cutting the roots that nourish it and make it grow" [(1913), p. 290f, or (2014), p. 300; (1916), p. 207; (1990), p. 216].
137. W. Ostwald (1907), p. 499f. Two pages earlier Ostwald had acknowledged his adopting of Mach's term, pseudo-problem (*Scheinprobleme*).
138. Nernst (1895), p. 237; (1911), p. 280; (1923), p. 327. The paragraph preceding this sentence that defended the further development of molecular theory in spite of the hypothetical status of molecules was dropped in the 1909 and 1921 editions.
139. Chalmers (2011); and Salmon (1984), pp. 211–227.
140. See Smith (2016).
141. Cohen and Koyré (1972), p. 298 (p. 188 in the original Latin, p. 589 in the Cohen and Whitman translation). The Latin phrase is "*tutius disputare licebit*"—"it will be possible to argue more securely."
142. See Harper (2012) and Smith (2014).

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7
Our Initial Issues, Revisited

We began this monograph by posing two issues: (1) *What did the new standing of atomic-molecular theory at the end of the first decade of the twentieth century amount to*? (2) *What did the research by Perrin and others establish about Brownian motion* independently *of the molecular hypothesis and kinetic theory?* In posing these two, we chose not to challenge the "lore" that Perrin's Brownian motion results made a crucial difference to the change in standing of the theory. Our long discussion of the various determinations of Avogadro's number between 1908 and 1913 in Chapter 6 has raised a further question that we should answer as well: (3) *In what respects, if any, did Perrin contribute crucially to the change in the standing of atomic-molecular theory that occurred in those years*?

The second and third questions have been receiving partial answers along the way in Chapters 4 through 6. So, all we need do with them here is to pull together the points from those chapters that together comprise direct answers to the questions as stated—which we shall do in sections 7.1 and 7.2. The first question, by contrast, we have been sidestepping throughout Chapters 4 through 6, restricting our attention to the evidence bearing on chemical and kinetic molecular theory during the years in question. This evidence has turned out to be more complicated than it is usually portrayed. In particular, the question of a preferred value for Avogadro's number changed dramatically between Ostwald's remark in 1909, "The most probable value is $.71 \times 10^{24}$ (Perrin 1908)," and Millikan's statement in 1913 that the "estimated limits of uncertainty" for his newly refined value of 6.062×10^{23} were $\pm\ 0.012 \times 10^{23}$. These complications have led us to answer our question about the new standing in three steps. Section 7.3 will answer it from the perspective of Ostwald in 1909; section 7.4 will answer it, as best we can, from the perspective of someone writing a review article on the question in the aftermath of the publication of the Solvay Conference proceedings in 1912; and section 7.5 will then step back to comment on the different respects in which a hypothesis can legitimately be said to have become accepted within a scientific community.

Our answer to the question about the new standing thus forms the heart of the chapter. This change in standing, however, was inextricably linked to the emergence of a new way of using theory-mediated measurement as a source of evidence pertaining to the microphysical realm. This new approach to evidence had begun with J. J. Thomson's theory-mediated measurements of the charge-to-mass

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ratio of the constituents of cathode rays in 1897 and Planck's appeal to complementary measurements of microphysical constants as evidence for his theory of blackbody radiation in 1900. Over the course of the twentieth century it became increasingly the most decisive form of evidence in microphysics, culminating in the formation of the Committee on Data for Science and Technology—that is, CODATA—and their Task Group on Fundamental Constants now publishing new preferred values for over two hundred constants every four years.

Perrin, and the 1905–1913 period on which we have been focusing, formed a key step in the transition between Thomson and Planck, on the one hand, and the full flowering of this new way of using theory-mediated measurement as a primary source of evidence, on the other. Just what that step was and how it fit into a twentieth-century revolution in theory-mediated measurement is the subject of section 7.6. This section thus completes the "study of theory-mediated measurement" promised in our subtitle by answering the question: (4) *How did the developments in theory-mediated measurement of Avogadro's number and constants linked to it during the period 1905–1913 contribute to the increasing reliance on such measurements for evidence that ensued thereafter*?

7.1. Conclusions Established about Brownian Motion

Throughout the previous three chapters we have focused almost exclusively on the results on Brownian motion coming out of Perrin's laboratory. In doing so, we have followed the lead of Perrin himself, who consistently dismissed the results on Brownian motion obtained by others, especially those by Svedberg.[1] Svedberg had been the first to publish experimental confirmation, faulty though it was, of Einstein's formula, and as we shall see in section 7.3, this confirmation seems to have been what had convinced Ostwald, before he had read Perrin's 1908 paper, that Brownian motion had established the kinetic theory of heat. Both at the Solvay Conference and in *Les atomes*, moreover, Perrin cited the value 2.88×10^{14} for *Ne* that Fletcher had published in 1911 from experiments on Brownian motion in air, noting its close agreement with the established value of Faraday's constant from electrolysis, 2.896×10^{14} esu, but without noting in either place that Fletcher's result implied a 57.5×10^{22} value for Avogadro's number, far below his own values; nor did he note any other respect in which Fletcher's announced results for Brownian motion might have a bearing on his own. (Fletcher's refined results, including the value 60.3×10^{22}, did not appear until 1914, outside the time frame we are considering.)

Perhaps largely because of what Perrin had said about research by others on Brownian motion, others at the time appear to have taken the results from his laboratory to be the definitive ones. To give just one example, Jeans's 1916 edition

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of *The Dynamical Theory of Gases* lists all four of Perrin's Solvay Conference values for Avogadro's number and no values published by any others on Brownian motion from their research.[2] We have thus not been idiosyncratic in taking the results by others at the time on Brownian motion as mostly just complementary to those of Perrin and his younger colleagues, Chaudesaigues and Léon Brillouin. If indeed we put aside issues about the preferred values of fundamental molecular constants, and focus instead on just what was established during the period about Brownian motion itself, the results of others were in fact almost entirely complementary. Arguably, then, we are not being unfair to them in confining our answer to the question of what was established about Brownian motion during the period to the four points established in Perrin's laboratory.

The basic result was the universality of the mean translational kinetic energy of (spherical) granules at any given temperature in dilute emulsions—that is, universality with respect to the size and mass of the granules, the density and viscosity of the liquid, and the range of temperatures examined. Even though the range of temperatures was limited, the most appropriate way to state this finding, on our view, is that Perrin and his colleagues established that the mean translational kinetic energy of granules varies as $W_T T$, where W_T, they showed, is a constant. They showed this, of course, by determining its value, roughly 1.8×10^{-16} ergs per degree K, by means of three different theory-mediated measures. Moreover, this value did not differ ~~that~~ all that much from the corresponding values for $3k/2$ that had been estimated for molecules, such as Planck's 2.02×10^{-16} from blackbody radiation.

As we emphasized in Chapter 4, Perrin does not himself ever quote a value for the mean translational kinetic energy of granules per degree Kelvin, but only what he takes to be the corresponding value (1.77×10^{-16} in 1909, 1.8×10^{-16} in 1911) for molecules. The 10 percent discrepancy between his value for W_T for granules and the values that others were obtaining for the corresponding value for molecules may be raising a question in some readers' minds about whether Perrin really did establish a value for W_T. One possibility is that he somehow managed to introduce a roughly 10 percent systematic error in all his results for Brownian motion, including those for the mean rotational energy of his granules. The four mediating equations he employed had only one parameter in common, the radius of the granules, and that occurs linearly in the measurements based on displacements and diffusion and cubed in the measurements based on vertical-gradients and rotation. We, therefore, are inclined not to attribute the discrepancy to an across-the-board systematic error in Perrin's four measures of W, for we do not see any plausible source for such an error. We are instead inclined to say that his values for W_T really do represent the mean translational kinetic energy per degree Kelvin of (spherical) granules in dilute emulsions, and that the discrepancy represents a real difference between that mean kinetic energy and

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the corresponding mean kinetic energy per degree Kelvin for molecules. Hence, we feel justified in saying that Perrin established the universality of W_T as a constant over the range of Brownian motions his experiments covered.

Perrin's value for W_T had a notable consequence: velocities of granules in Brownian motion obtained from the quotient of observed displacements and the times in which they occur were totally misleading. True root-mean-square granule velocities are orders of magnitude greater than those values. This in turn entailed that the granules were undergoing thousands, if not millions, of changes in motion over the course of the time in which observers could see only one change in motion. With this result, a great many prior findings on Brownian motion were overturned, once and for all.

Related to establishing the constancy of and a reasonable value for W_T was Perrin's result for the rotational kinetic energy of granules and its confirmation, at least for the case investigated, of the equipartition of energy across all degrees of freedom. While this result was important in Perrin's mind because of the long-standing controversy over equipartition of energy, we put less stress on it because it was confirmed only for one kind of granule; and of course, unlike Perrin, we know that the equipartition principle was about to be replaced by the partition function in statistical thermodynamics.

As important as establishing the constancy of W_T in Brownian motion was, we are inclined to agree with Deborah Mayo that still more important was Perrin's establishing that granule displacements versus time have a Gaussian statistical distribution. This, together with the finding that granules must be undergoing thousands, if not millions, of changes of motion per second, established the thoroughgoing randomness of Brownian motion at a scale far smaller than what can be observed. Needless to say, it was by virtue of this feature that "Brownian motion" came to denote a subject in mathematics, so-called random-walk stochastic processes. Brillouin's further confirmation of Einstein's formula for the diffusion of granules then showed that diffusion itself, at least in the case of granules in dilute emulsions, is a consequence purely of a random-walk process, not requiring any forces to drive it.

Finally, by virtue of the previously established indefinite persistence of the motion over time in combination with the preceding results, Perrin established important conclusions about the cause of Brownian motion, albeit conclusions that do not categorically require molecular theory. From a classic hydrodynamic perspective, what maintains Brownian motion while causing the granules to undergo so many changes of motion per second has to be extremely high-frequency pressure fluctuations in the liquid on a spatial scale far smaller than the smallest granule Perrin tested. These pressure fluctuations must themselves persist in the liquid indefinitely over time; and each pressure fluctuation must involve motion of liquid within it, and hence localized kinetic energy. But then, to maintain

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Brownian motion indefinitely, the mean value of the kinetic energy of the liquid over the many pressure fluctuations must be at least as great as that of the mean translational kinetic energy of the granules, and hence it too must be a mean kinetic energy that varies linearly with the absolute temperature of the liquid. This result, we think, can legitimately be taken as decisive evidence for the kinetic theory of heat, at least so far as the liquid substrate.

Still further, as we argued in section 5.6 of Chapter 5, the thoroughgoing randomness of the granule motion on a time scale far smaller than can be observed requires the localized pressure fluctuations in the liquid to be correspondingly random on a no greater time scale. That, however, implies that the individual pressure fluctuations in the liquid must not be correlated with one another, and hence the highly localized fluctuating motions of the liquid must over time have some independence of one another. That in turn suggests—we choose the vague word scientists often use in this kind of situation—that the elements of liquid moving in each pressure fluctuation have at any one instant an individual integrity of their own, and hence in principle can be counted in the manner of discrete elements. At the very least, this is evidence that the liquid substrate is not a continuum at a scale below that of the granules, and hence (using Ostwald's word) can be taken to be granular.

To go further than this requires considerations beyond the results on Brownian motion that were established independently of molecular-kinetic theory. Perrin invoked the range of values for Avogadro's number that had been derived from the viscosity of gases in order to argue, in effect, that the mean kinetic energy of the granules is identical with that of molecules. With respect to the reasoning purely in terms of Brownian motion itself that we have just given, this amounts to arguing that the mean kinetic energy of the individual fluctuations in the liquid that drive Brownian motion matches $3R/2N_0$, and hence the local liquid elements in random motion are just the molecules posited in kinetic theory. After Rutherford and his colleagues had provided a robust value for N_0 under the assumption that α-particles become helium molecules when they lose their charge, a parallel line of reasoning could have been invoked to reach the same conclusion. As we said, however, such reasoning does involve considerations beyond the conclusions that were established about Brownian motion independently of molecular-kinetic theory.

We have derided the idea, expressed by Perrin and many others at the time and still today, that *Brownian motion is molecular motion writ large*—that is, that the proper way to think of it is as if it consisted of very large molecules in motion. According to Perrin's own estimates, molecules in the surrounding liquids are three orders of magnitude smaller in their principal dimension than the granules he tested. This means that the number of molecules required to fill the volume occupied by a single granule is of an order of magnitude greater than 10 million,

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and the number required to cover its surface is of an order greater than 100,000. Moreover, granules in Brownian motion in a dilute emulsion do not interact with one another, and their motion is resisted by forces on the surface arising from the viscosity of the fluid. Indeed, two of Perrin's three measures of the mean translational kinetic energy of the granules expressly presupposed the viscous action of the fluid on them, and the third presupposed their buoyancy in the fluid. In these respects, Brownian motion is a macrophysical, not a microphysical, phenomenon, even though a microscope is required to observe it.

There is, nevertheless, a version of the claim that we have derided that we regard as both warranted and informative: *Brownian motion is molecular motion made manifest*. We have two extraordinary aspects of Brownian motion in mind in granting this: (1) it persists indefinitely in time, a fact about it known long before Perrin began his research; and (2) it is thoroughly irregular to a degree that cannot begin to be seen even in an ultramicroscope, a fact about it that Perrin's laboratory has legitimate claim to having established experimentally. If "making molecular motion manifest" in these respects is what Nernst had in mind in using the phrase "*ocular* confirmations," we have no objections. Nor do we object in any way to the claim that Perrin's efforts left no candidate for explaining Brownian motion other than perpetual motion of invisible, discrete constituents of the underlying liquid substrata.

Perrin also claimed to have shown that the gas law holds for granules in dilute emulsions in full parallel with van't Hoff's result for the osmotic pressure of solutes in dilute solutions. Here, we think, Perrin was overstating what his experimental results had shown. They had indeed shown that the impact of granules on the wall of the container produces a pressure on it proportional to the product of the number-density of the granules and the absolute temperature of the liquid. Unlike van't Hoff, however, Perrin did not measure the osmotic pressure of the granules and therefore did not show that the constant of proportionality in this relationship matches the universal gas constant R. His sole grounds for concluding that it does was from his assumption that the mean kinetic energy of the granules matches that of molecules. Had he, independently of molecular theory, shown that the constant of proportionality amounts to R, somewhat stronger conclusions than those we granted earlier might have been drawn about the pressure fluctuations driving Brownian motion.[3]

7.2. Perrin's Contribution to the New Standing

"Lore" has it—and on this point van Fraassen seems to agree—that Perrin not only did the most of anyone at the time to change the standing of molecular-kinetic theory, but that he did so by obtaining from Brownian motion the first

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reasonably robust value for Avogadro's number. Most of the philosophic literature, however, including van Fraassen, has offered differing reasons for why the determination of a robust value for that constant had such an abrupt and decisive effect on the standing of the theory. On that question we side with van Fraassen's general answer: the value Perrin obtained for N_0 *grounded* molecular-kinetic theory by yielding values for several of its other parameters. Our primary reason for this conclusion is more historical than philosophical, for it comes partly from what Ostwald himself said before and then after 1908, most notably the remark, "The atomic hypothesis is thus raised to the position of a scientifically well-founded theory."[4] Our differences with van Fraassen, which we do regard as fundamental, concern the extent to which molecular-theory itself entered into the theory-mediated measurements that yielded Perrin's value for N_0, or for that matter the theory-mediated measurements of its value made at the time by others. We shall not belabor those differences further here.

Nevertheless, we are uncomfortable, looking at the history, with the conclusion that Perrin's singular contribution was his being first in obtaining a reasonably robust value for N_0. True, Ostwald specifically cited Perrin's value for the "molecular constant," along with the parallel between his verified formula for the vertical variation of the number density of granules and Maxwell's kinetic theory formula for the vertical variation of the density of different species in the atmosphere.[5] But Ostwald wrote this passage before Perrin had begun publishing his results testing Einstein's theory, and also, it would appear, before he read the Rutherford-Geiger paper in August 1908 giving their determinations of e and N_0. What Ostwald would have said about Perrin had he been writing his book a year later surely would have credited him for more with regard to the implications of Brownian motion and probably less with regard to determining the value of N_0. Moreover, those at the Solvay Conference in late 1911 would surely have concluded from Perrin's paper that his value for N_0, while robust, was but one of several independently determined robust values. And in fact, though somewhat suppressed in Perrin's presentation, the cluster of values from the two complementary α-particle determinations by Rutherford and his colleagues and Regener, from the various blackbody spectrum measurements supporting Planck, and from Millikan's oil-drop experiments agreed significantly more closely with each other than they did with Perrin's.

The shift in the standing of molecular-kinetic theory did not occur in a single instant. Any assessment of what each new result contributed to it therefore depends not only on which scientists' attitude one relies on in assessing it, but also on the date one chooses for considering it. Maybe because of our retrospective bias, we are inclined to single out the two determinations of N_0 made by Rutherford, respectively with Geiger (1908), supported by Regener's independent determination, and with Boltwood (1911), as the most decisive. For the

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"weak link" in them, especially in the latter, seems to us to have involved the least challengeable step from the primary quantity measured to a count of the number of molecules per mole. Granted that its conclusion applied directly only to helium and hence that other determinations of N_0 were needed to support its inductive extension to other molecules. Still, at the very least, the Rutherford determinations and Regener's, followed immediately by the support Millikan's result lent them, had lessened the importance of Perrin's determinations by the time of the Solvay Conference.

If anyone at the Conference challenged Perrin's insistence that his numbers were to be preferred to the others, it would appear to have been outside of the formal sessions, for we have found no recorded discussion of the gap between his numbers and those of the rest in the proceedings in French published in 1912. Arnold Eucken, however, raised the issue in the Appendix, "The Development of Quantum Theory from the Autumn of 1911 to the Summer of 1913," that he added as editor to the proceedings in German, published in 1914. In particular, he gave, in bold, two recent values for Avogadro's number from blackbody radiation, 59.7×10^{22} and 62.0×10^{22}, plus Millikan's 1913 value, 60.62×10^{22}. He went on then to note the earlier Rutherford-Geiger and Regener values, 62×10^{22} and 60×10^{22}, remarking "the gap between numbers found from Perrin remains."[6] Eucken was not an official participant in the Conference, though as a research assistant to Nernst he might have been present. Regardless, whether from Nernst or otherwise, he came to see the gap with Perrin's values as enough of an issue that he could single it out without needing to explain it further.

In point of fact, even though Perrin continued to stress his determinations of the molecular constants in the various editions of his *Les atomes*, others in need of values for any of these constants had already chosen not to use his in the year when its first edition appeared in print. To give two examples from the summer of 1913, Bohr adopted a value for *e* in his legendary paper on the structure of the hydrogen atom of 4.7×10^{-10} instead of Perrin's Solvay Conference value of 4.2×10^{-10};[7] and the Braggs had adopted a value for the mass of a hydrogen atom of 1.64×10^{-24} grams instead of Perrin's 1.47×10^{-24} in their seminal paper on the lattice structure of crystalline NaCl and related crystals.[8] Moreover, as we noted earlier, in the 1916 edition of his *The Dynamical Theory of Gases*, Jeans cites all four of Perrin's Solvay Conference values for Avogadro's number (68.3, 68.8, 65, and 69, all times 10^{22}) and then adopts for his book the value 61.2×10^{22}.[9] So, Perrin's values for the molecular constants had ceased being viewed as a still-standing reliable contribution scarcely before the ink of the first edition of *Les atomes* had dried.

This, however, does not mean that Perrin made no *singular* contribution to the change in the standing of the theory. While Einstein may have preceded him

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in championing the idea of settling molecular-kinetic theory once and for all, Perrin became the spokesperson for doing so after he began getting converging results on Brownian motion. Perrin was the first, in a short note in May 1908, to announce a value for Avogadro's number derived from Brownian motion ("supposing the deduction of the preceding equations rigorous"); he confirmed the value in the case of granules twice as large as in the first series in a September note; he followed this with a refined value and values implied by it for *e* and other constants in October in a note that cited the Rutherford-Geiger results as well as Planck's; and this was then followed in November by Chaudesaigues's note announcing the Gaussian distribution of the displacements.[10] Throughout this period, Perrin was notably more preoccupied than any of the others with using their measured values more to establish the reality of molecules than to establish a value for constants. In this regard, Perrin was unique in the emphasis he put on agreeing values of Avogadro's number and what they collectively imply about the existence of molecules.

Chaudesaigues's note we will consider in a moment. Perrin's October note, though it barely extended to a third page, nevertheless deserves further mention, and not just because it was the only publication of Perrin's that Ostwald cites. It opens (see Figure 7.1) by pointing out that, by virtue of Faraday's constant and the gas constant, a determination of any one of the three molecular magnitudes N_0, α, and e yields values for the other two. It then goes on to argue that his determination from Brownian motion is the "most direct and amenable to unlimited precision," and displays in bold his values for the three.[11] Because he singles out values obtained by other means—in particular, the 1903 Wilson-Thomson and 1908 Rutherford-Geiger values for *e*—he is calling attention to the possibility of playing off values obtained by different methods in pursuit of greater precision. Planck had cited the interconnection of the values for the magnitudes in 1900, but Perrin's 1908 note appears to be the first place where a comparison of values obtained by different approaches occurs side by side with a preoccupation with improving precision.

In addition to this, none of the other determinations of N_0 involved motion in any significant way, and hence none of the others had such direct bearing on kinetic theory as Perrin's. Ostwald credited Svedberg's testing of Einstein's theory with having given "a fairly satisfactory proof of the kinetic theory of heat," which theretofore had "remained an arbitrary though useful hypothesis."[12] Perrin's tests of Einstein's theory were far more definitive than Svedberg's, even without Brillouin's verification of Einstein's formula for diffusion; and with it included, Perrin's and his colleagues' results provided the most direct and decisive evidence for the kinetic theory of heat that had ever been put forward. So, this was part of Perrin's singular contribution. It may also be a proper spelling out of the substantive content lying behind the thought expressed by many at the time that,

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En résumé, les gaz de l'atmosphère non liquéfiables dans l'air liquide sont constitués par du néon et de l'hélium, et notre procédé nous a permis de trouver des lignes nouvelles dans le spectre du néon pur.

PHYSIQUE. — *Grandeur des molécules et charge de l'électron.* Note de M. JEAN PERRIN, transmise par M. J. Violle.

Le nombre N de molécules que contient toute molécule-gramme, la charge e de l'électron et le quotient α de l'énergie moyenne d'une molécule par sa température absolue T sont des *constantes universelles* qui sont connues dès qu'une d'elles est connue. En effet, dans l'électrolyse d'un sel monovalent, 96550 coulombs sont transportés par N atomes, ce qui donne

$$Ne = 3.10^9.96\,550 = 29.10^{13} \text{ (unités électrostatiques C. G. S.)},$$

et, d'autre part, d'après la théorie cinétique, le produit 3RT mesure la force vive de translation que possèdent à un instant quelconque les N molécules d'une molécule-gramme, en sorte qu'on a

$$3N\alpha = 3R = 3.83,2.10^6.$$

La théorie cinétique donne au moins l'ordre de grandeur de N en s'aidant de considérations qu'on peut, je crois, résumer comme il suit : d'une part, dans un liquide, les molécules (qu'on suppose sphériques) ne peuvent être plus serrées que des boulets dans une pile de boulets, et de là résulte que N doit être plus grand que 4.10^{22}; d'autre part, le volume *vrai* des molécules ne peut qu'être supérieur à celui des sphères *parfaitement* conductrices qui, substituées à ces molécules, donneraient la même constante diélectrique au fluide, et de là résulte que N doit être plus petit que 204.10^{22}; enfin, d'après la théorie de Van der Waals, le volume *vrai* de N molécules est le douzième du volume critique de la molécule-gramme, et de là résulte pour N la valeur 60.10^{22}, déjà sans doute assez approchée, sans qu'on puisse dire au juste à combien près (un écart de 40 pour 100 en plus ou en moins n'étonnerait pas).

Les recherches du deuxième groupe utilisent le pouvoir qu'ont les ions d'un gaz de condenser la vapeur d'eau (C.-T.-R. Wilson). On peut alors obtenir e, soit en divisant la charge présente dans le gaz par le nombre de gouttelettes qui la portent (Townsend, J.-J. Thomson), soit en observant l'action d'un champ électrique sur ces gouttelettes chargées (Harold A. Wilson). Townsend a ainsi trouvé, en unités électrostatiques C. G. S., 3.10^{-10} (ions des gaz de l'électrolyse); J.-J. Thomson a trouvé $3,4.10^{-10}$ (ions des deux signes dus au radium) et $6,8.10^{-10}$ (ions négatifs produits par la lumière); Harold A. Wilson a trouvé $3,1.10^{-10}$ (ions négatifs produits par les rayons X). Les valeurs correspondantes de N sont comprises entre 43.10^{22} et 96.10^{22}.

Je discuterai ailleurs ces expériences qui, fixant un ordre de grandeur inconnu, ont

Figure 7.1a. Perrin ~~of~~ fundamental constants, 1908.

in Nernst's words, Brownian motion had provided "*ocular* confirmations of the kinetic theory of the molecular world."[13]

Still, singular that it was, it was only part of Perrin's unique contribution to the change in standing of the theory. No less important at the time, and probably more important in the years following the 1911 Solvay Conference, was

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imposé la notion capitale de *corpuscules* bien plus petits que les atomes (J.-J. Thomson), mais qui, je crois, n'ont pas pu être très précises (¹).

En troisième lieu je dois citer les résultats de Max Planck et H.-A. Lorentz. Le premier, dans sa belle théorie électromagnétique du rayonnement noir, a trouvé pour α, à partir des mesures de Kurlbaum, la valeur $2{,}02.10^{-18}$, qui fait N égal à $\mathbf{61}.10^{22}$; mais, à partir des mêmes mesures, par une théorie différente, Lorentz a trouvé N égal à $\mathbf{77}.10^{22}$.

J'ai suivi, pour déterminer N, une méthode qui me semble *directe et susceptible d'une précision illimitée*. Elle se fonde sur ce que des grains égaux se répartissent dans une émulsion étendue comme feraient des molécules de même masse obéissant aux lois des gaz parfaits (²). Cela entraîne l'équation :

$$2{,}3 \log_{10} \frac{n_0}{n} = \frac{N}{RT} \frac{4}{3} \pi a^3 g (d - \sigma) h$$

où n et n_0 désignent les concentrations des grains en des niveaux distants de h, a le rayon des grains et $(d - \sigma)$ l'excès de leur densité sur celle de l'eau.

Trois séries d'expériences m'ayant prouvé, comme je l'ai dit, que des grains de tailles très différentes conduisaient sensiblement à la même valeur de N, je me suis efforcé d'en faire une particulièrement soignée, dont je donnerai ailleurs le détail; T était égal à $273 + 20$, h à 30^{μ}, $(d - \sigma)$ à $0{,}2067$, a à $0^{\mu}{,}212$; enfin, pour avoir $\frac{n_0}{n}$, j'ai compté un à un 13000 grains en 16000 lectures. En conséquence de ces mesures, je crois pouvoir donner

$$N = \mathbf{71}.10^{22},$$

ce qui entraîne pour l'électron

$$e = \mathbf{4{,}1}.10^{-10}$$

et pour la constante d'énergie moléculaire

$$\alpha = \mathbf{1{,}7}.10^{-16}.$$

(¹) Enfin Rutherford, en un beau Mémoire (août 1908) qui vient seulement aujourd'hui à ma connaissance, obtient pour e, par deux groupes de mesures de radioactivité, deux nombres qui placeraient N entre $\mathbf{62}.10^{22}$ (qu'il regarde comme le plus probable) et $\mathbf{72}.10^{22}$ (ceci en admettant égale à 2000 ans la constante de temps du radium).

(²) *Comptes rendus*, t. CXLVI, 1908, p. 967; t. CXLVII, 1908, p. 530.

Figure 7.1b. Perrin on fundamental constants, 1908, continued.

the evidence he supplied for the thoroughgoing randomness of Brownian motion and, with it, of the fluctuations in the liquid maintaining it as well. Joined with Brillouin's measurements of granule diffusion, this showed that diffusion itself involves nothing more than what we have now come to call a random-walk

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process. On the importance of this point we agree with Mayo. Earlier we noted both Einstein's (1911) and the Ehrenfests' (1912) appeal to Perrin's results when defending Boltzmann's probabilistic interpretation of entropy and with it statistical, in contrast to classical, thermodynamics. The statement by the latter bears repeating:

> A "visible" state S_A at time t_A completely determines the "visible" state for any subsequent time t_B.
>
> This statement is clearly not a direct expression of an experimental fact. Apart from a very small group of aerodynamical processes, we always deal with turbulent motions of the gas, where it is impossible to follow the "visible" state with our measuring instruments. Furthermore, even the best insulation against thermal conductivity and radiation is completely unsatisfactory unless we deal with very short periods of observation.
>
> The above statement is therefore a postulate, which goes considerably beyond the possibility of an experimental check. Our observations cannot tell us anything about what the sequence of state of an actual gas quantum would be over a very long period of time and whether it would satisfy the principle of determinacy.
>
> As soon as we include a microscope among the instruments of observation of the "visible" state, we discover the Brownian motion. For such a more precise "visible" state the principle of determinacy does not seem to hold any more. On the other hand, it has been shown that this phenomenon is surprisingly well suited to a statistical interpretation.[14]

This is undoubtedly the most justified respect in which those at the time could speak of Brownian motion as molecular motion made manifest.

The only other robust determination of N_0 at the time tied to Boltzmann's theory was Planck's, and the way he obtained it was part of what was behind others, including Perrin, questioning his derivation of the empirically well-founded blackbody equation. Consequently, for the very reason that Einstein cited it, Perrin's evidence for the thoroughgoing randomness of Brownian motion, which (as we saw) was independent of his determinations of N_0, made its own unique contribution at the Solvay Conference, where of course the question of how properly to derive Planck's formula was at the forefront.

As we acknowledged in Chapter 5, Perrin's emphasis on the visibility of Brownian motion—in words, diagrams, photographs, and films—had much the rhetorical effect on broader audiences that he intended. Of course, the visibility of the granules was indispensable to his theory-mediated measurements, which indeed were a major contribution. Equally, establishing the Gaussian distribution of their displacements required individual granules to be followed visually

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over time. What tends to get lost in this emphasis on visibility, however, is that the true motion of the granules is not visible at all, and as a consequence its full stochastic character is not within the reach of vision or film, but is instead a matter of inference.

7.3. The New Standing of Molecular-Kinetic Theory: Ostwald's "Conversion"

Part of the "lore" surrounding Perrin and his research on Brownian motion is its having been instrumental in Ostwald's reversing his stance on the reality of molecules in the 1909 edition of his *Grundriss der allgemeinen Chemie*. Even a text in applied mathematics on the subject labeled "Brownian motion" ends its introductory presentation with the quote from Ostwald's Preface, "this evidence now justifies even the most cautious scientist in speaking of the *experimental* proof of the atomic nature of space filling matter. What has up to now been called the atomistic hypothesis is thereby raised to the level of a well-founded theory"[15]—a statement we too have quoted more than once. The emphasis put on Ostwald's conversion is nevertheless misleading in two respects. First, in large part because of his prior view, Ostwald had long since ceased having influence within the physics community on the question of molecules, and hence his shift may have mattered more to himself than to anyone in it.[16] Second, Perrin and his research almost certainly was not a factor in Ostwald's shift in stance toward molecules in the new edition, for this shift had to have occurred before he was aware of Perrin's results. We will briefly explain both of these and then justify why Ostwald's conversion is nonetheless worthy of consideration before turning to the question of the new standing in his eyes.

While Ostwald was not the primary instigator of the "energetics movement," especially in comparison with Georg Helm, he did come to champion it in the first half of the 1890s. As a consequence he became a target for the strong criticisms of this movement, especially by such figures within German physics as Boltzmann and Planck.[17] Ostwald's continuing association with the movement over the next decade, and his continuing opposition to "mechanistic materialism," had led him to be regarded increasingly as a figure on the periphery of the physics community, if not outside it entirely.

Even so, his continuing insistence on the hypothetical status of molecules seems not to have been dismissed to the same extent. To give one significant example, Boltzmann responds to Ostwald's views in a single passage in his *Lectures on Gas Theory*, first rejecting in two paragraphs the claim that atoms lose their integrity and individuality when forming a chemical compound, and then adding a third paragraph:

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> There is no question that for the calculation of natural processes the mere equations, without their foundations, are sufficient; likewise, empirically confirmed equations have a higher degree of certainty than the hypotheses used in deriving them. But on the other hand, it appears to me that the mechanical basis is necessary to illustrate the abstract equations, in the same way that geometric constructions illuminate algebraic relations. Just as the latter are not made superfluous by mere algebra, so I believe that one cannot completely dispense with the intuitive representation of the laws valid for the action of macroscopic masses provided by molecular dynamics, even if he doubts the possibility of knowledge of the latter, or indeed the existence of molecules. A clear understanding is just as important for knowledge as the establishment of results by laws and formulas.[18]

However askance Ostwald's energetics was viewed by his chief critics within the physics community, his insistence on the lack of experimental access to the molecular realm seems not to have been scorned to the same extent. Rather, it seems to have been widely acknowledged.

Among the four seminal figures within physical chemistry from 1890 on—Ostwald, van't Hoff, Arrhenius, and Nernst—only Ostwald championed energetics. Nevertheless, as we noted in quoting from van't Hoff's Nobel Prize lecture of 1901 in Chapter 2, as well as from many passages in Nernst's *Theoretical Chemistry* in that chapter and others, the other three continued to acknowledge the hypothetical status of molecules while remaining uncompromisingly focused on research into the molecular realm. So, we grant that Ostwald's conversion made little difference to their research, though it must have pleased them that this element of difference within their otherwise mutually supportive group had finally ceased.

Ostwald's conversion appeared in print in his *Grundriss* of 1909, the same year that he, perhaps not entirely coincidently, became the third of the four to be awarded the Nobel Prize. Perrin's first paper announcing results on Brownian motion (and a value for N_0 of 6.7×10^{23}) had appeared in May 1908, yet the only one of his papers from that year that Ostwald unmistakably cites in the book, as implied by its announced value of 7.1×10^{23}, appeared in October, the month preceding the date on Ostwald's Preface announcing his conversion. The departures of this new edition from the 1899 edition are far too extensive not to have been conceived and almost entirely put in place before Perrin's results had appeared in print. Perrin is cited in only two places in the book: a single sentence in the chapter on kinetic theory saying that 7.1×10^{23} is the most probable, immediately following Ostwald's derivation from van der Waal's formulation of the gas law of a value of 8.5×10^{23}, "or in round numbers 10^{24}";[19] and a single paragraph immediately following Ostwald's comparisons of Svedberg's 1906 Brownian motion

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results with calculations based on Einstein's and Smoluchowski's formulas[20]—comparisons that were providing evidence for the conclusion Ostwald had stated just before them:

> If the movements of a small particle suspended in a liquid are calculated on the basis of this hypothesis, the agreement with the movements actually observed is so close that we are compelled to regard this agreement as a fairly satisfactory proof of the kinetic nature of heat.

Though we are not in a position to insist on the point, both of these citations of Perrin's results appear to have been inserted into a finished manuscript, if not after the book was already in press. Added support for claiming this comes from the opening sentence of the second of the two: "Still more definite confirmation of the agreement of Brownian movements with the hypothesis of thermal agitation of the smallest particles has been afforded very recently by the measurements of Perrin (1908)."[21]

Our conclusion, accordingly, is that Ostwald's "conversion" and virtually all of the new edition reflecting it were already in place when he first learned of Perrin's results. He nevertheless would have had strong reasons for inserting the two citations of Perrin's October 1908 paper. Recall the demand Ostwald had made of molecular theory in the *Monist* translation of his 1907 article:

> When expressions or notations of magnitudes which can not be observed and measured and for which we can substitute *no definite and empirically determinable value*, occur in a formula by means of which some physical relations are to be represented, we have to deal with an hypothesis. For the task of the exact sciences is to establish the reciprocal relation of measurable and demonstrable quantities, or in other words to find the mathematical forms or functions by which these quantities are interrelated, so that one of them can be calculated when the others are given. In order to establish experimentally such a functional relation it is therefore necessary to measure singly all variable or constant magnitudes which appear in such an equation.[22] [emphasis added]

As we are about to explain, Ostwald had determined independent values for all three of the quantities in the equation eN_0 = Faraday's constant, before he had known of Perrin's October 1908 paper. This paper ends with values for N_0, e, and further molecular magnitudes, all inferred from the vertical gradient in Brownian motion; these values had two striking virtues from Ostwald's point of view: (1) they were independent of his determinations; (2) they were not only consistent with his determinations, but had clear claim to being more precise than his.

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Throughout our monograph we have been treating Ostwald's *Grundriss*, in all its editions, as a book-length review article, not as representing his view alone. This is the way those in chemistry at the time would have read not only the earlier editions on which we relied in Chapter 2, but also the 1909 edition. That Ostwald's "conversion" occurred before he knew of Perrin's results would not have been all that obvious to them. The conclusion we have reached now, however, makes the question of what led him to convert a question about Ostwald as an individual, not as a representative of the community. We shall answer it in this light, along with his view of the new standing of molecular theory, yet end by arguing that at least the latter can still be regarded as representative of the community.

Suppose, then, we assume that Ostwald had completed the manuscript for the new edition before he knew of Perrin's results and ask how the argument for the reality of molecules proceeded in it without the two passages citing him. Ostwald's new support for the kinetic theory of heat did not hinge on Svedberg's 1906 results supporting Einstein's and Smoluchowski's formulas. To begin with, the support was at best limited: "The paths calculated from Einstein's formula are roughly one-fourth the observed paths, from Smoluchowski's formula one-third," so that the agreement was only one of "approximate proportionality" as the solvent and hence its viscosity were varied.[23] More importantly, Ostwald reaches his key conclusion immediately prior to the section, "The Brownian Movements," in the chapter new to the 1909 edition, "Disperse Systems":

> It is highly significant that there is a continuous transition from suspensions and emulsions, which are readily recognisable as such, to true solutions (for which the laws of solutions hold), this fact indicating that the nature of both systems is the same. Or, in other words, the facts here discovered indicate that true solutions are really heterogeneous, and may therefore be regarded as mixtures of different kinds of particles in the sense of the molecular hypothesis. If we add to this the result of the investigations into the electrical discharges of gases (see later, p. 533), in which are recognisable discontinuities or a grained structure in the submicroscopic region, it appears that *the final proof of the grained or atomistic-molecular nature of matter has been obtained*, after a fruitless search during a whole century. Here we do not have relations which lead indirectly to atomic methods of representation, but which can also be explained in a more general, and therefore, more scientifically justifiable manner, and thus afford no decisive support to the hypothesis. They are experimentally demonstrated differences in the nature of matter, which exclude the possibility of a continuous occupancy of space. This discontinuity is below the limit directly reached by the ultramicroscope, but is not far removed from it. The "molecular dimensions" calculated from the surface tension lies roughly between 6×10^{-9} cm. and $6 \times$

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> 10^{-8}, while the ultramicroscope goes down to 5×10^{-7}. The gap to be bridged is only a tenth power, which is no great distance.[24] [emphasis in original]

Ostwald's conclusion about the granular structure of matter, at least as he presents it, thus relies on observations made possible by the ultramicroscope, not on any specifics of Brownian motion.

Because of Ostwald's own use of emphasis in the paragraph, we have not italicized the two sentences that seem to us most important: "*Here we do not have relations which lead indirectly to atomic methods of representation, but which can also be explained in a more general, and therefore, more scientifically justifiable manner, and thus afford no decisive support to the hypothesis. They are experimentally demonstrated differences in the nature of matter, which exclude the possibility of a continuous occupancy of space*." In other words, the grounds for Ostwald's conclusion is not its explanatory power, but "experimentally demonstrated differences" that allow only the possibility of discontinuous matter. The experimentally demonstrated differences in question are ones involving separation, filtration, and, most of all, optical phenomena presented on the preceding pages—*plus* electrical phenomena in gases, to be presented in the chapter, "Conduction in Gases and Radioactivity," that begins on p. 533. We leave it to readers to assess the strength of the evidence Ostwald gives on those pages. The key point is that he is appealing to experimental results that colloids and solutions have in common, on the one hand, and results on ions in gases and their relation to ions in solutions, on the other, to infer the discontinuity of matter below the level of visibility even in an ultramicroscope.

The new chapter on "disperse systems" ends with a section on "the size of disperse particles" that cites Zsigmondy's 1906 results in concluding that "the isolated existence of still smaller particles, which was to be expected considering the accidental nature of the limit [viz. that imposed by the intensity of the available light], has been demonstrated in other ways."[25] After repeating the point about the "unbroken continuity" down to the 10^{-8} cm called for by molecular theory, the chapter ends: "The limit of spatial discontinuity which is physically observable is still lower, for electrical phenomena take us at least a further power of ten."[26] We shall return to this and the reference to electrical phenomena in the preceding long quotation in a moment.

The section on "Brownian movements" immediately following the preceding long quotation addresses a topic never mentioned in the 1890 and 1899 editions. It concludes, "If the movements of a small particle suspended in a liquid are calculated on the basis of this [kinetic theory] hypothesis, *the agreement with the movements actually observed is so close that we are compelled to regard this agreement as a fairly satisfactory proof of the kinetic nature of heat*."[27] Ostwald then gives the comparison of Svedberg's observations with the calculations based on

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Einstein's and Smoluchowski's formulas. This is followed by a single paragraph (in German, two in the English translation) describing Perrin's 1908 vertical-gradient results. It concludes that Perrin "has developed the method to such an extent that it is probably the most accurate value for the molecular constant yet obtained."[28]

The chapter on the kinetic theory of heat, which had appeared very early in the book in 1890 and 1899, occurs some 20 pages after the chapter on disperse systems. It nevertheless remained virtually word-for-word the same as in the earlier editions until near the end, where two sections are added, the first entitled "Number of Molecules," and the second, "The Kinetic Theory of Liquids and Solids." The latter's opening sentence summarizes it: "While the kinetic theory of gases has led to a very remarkable series of conclusions, the later confirmation of which has given us some confidence in its expediency, the development of the corresponding theory of the two other states of aggregation is still in its infancy, though hopeful beginnings have been made."[29] The conclusion reached in the new section on the number of molecules, based on their size inferred from viscosity and van der Waals's formula, we quoted earlier: 8.5×10^{23} per mole, "or in round numbers 10^{24}." This is followed by a two-sentence paragraph, which we have concluded was inserted after the manuscript for the new edition was complete, noting first that other calculations "yield values which are a little higher or a little lower," and then giving Perrin's 7.1×10^{23} as the most probable value.[30]

We have gone into detail here in order to highlight the round-number value 10^{24} that we think Ostwald had settled on before he saw Perrin's October 1908 paper. Some 30 pages later (in both the German edition and the translation), Ostwald describes J. J. Thomson's isolating and counting ions and their charge in what we now would call a cloud chamber. He concludes, "*The electric charge on an electron is equal to the unit charge of an electrolytic ion, namely about 10^{-19}coulombs.*"[31] Ostwald does not cite a source, but this value corresponds so closely to the value Harold Wilson had announced from water-droplet experiments in 1903, 3.1×10^{10} esu,[32] that he might as well have cited him. More importantly, the product of the two round-numbers, 10^5, corresponds closely to a number Ostwald had emphasized as fundamental in his account of electrochemistry in all three editions, namely the value of Faraday's constant, 96,540 (coulombs per mole).[33] Ostwald had expressed the significance of this three pages before the sentence in italics we just quoted:

> If the method (p. 509) of calculating the number of molecules from the mean cross-section (which can be deduced from diffusion experiments as well as from viscosity) is applied to gas ions, and if the total electric charge is determined at the same time, values are obtained for the number of ions in a mol, and for the charge on a single ion. The former agrees with that given on p. 510,

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> and the latter with the value given on p. 538. Or, gas ions behave from the point of view of the kinetic theory ~~of~~ molecules ~~like~~ ordinary gases, and each ion carries the same charge as a univalent electrolytic ion.[34]

So, at least in terms of round numbers, Ostwald had begun to meet his announced requirement with independently measured, mutually consistent values for the triad of N_0, e, and Faraday's constant before he saw Perrin's paper and had thereby tested at least to first order the claim eN_0 = Faraday's constant. Perrin's value for N_0 then became somewhat of a two-edged sword for him: it proffered greater precision, but at the same time threw into some question the value for e that Ostwald appears to have taken from Wilson and Thomson.

Ostwald nowhere cites Perrin in his new chapter (that begins on p. 533), "Gaseous Conduction and Radioactivity," from which the preceding quotations are taken. The sections on radioactivity recount what was known as of 1905, qualified by the remark, "the facts are still in course of evolution, and a fairly comprehensive review is not yet possible."[35] Nothing in the discussion indicates that the phenomena contributed to Ostwald's conversion, though of course he could scarcely have missed their consistency with atomism. Beyond the previous two quotations and the passages leading up to them, the sections of the chapter on J. J. Thomson's findings consist mostly of proposals about the implications of "electron theory." One of these is worth noting because it concerns the structure of atoms:

> The simplest application of this [electron] theory is to electrolytic ions, which, from this point of view are compounds of chemical atoms with as many electrons as its valency amounts to. These compounds are separated by electrolysis, the chemical atom remaining behind in the electrolyte or on the electrode, while the electrons travel further into the metal. Metallic conduction must be regarded as a migration of electrons in the metal.
>
> It must be remembered that positive electrons are not known, but only positive atoms, or gas ions and kations. We may either assume that positive electrons similar to negative electrons exist, but have not yet been isolated, or we may consider the positive state to be produced by the loss of a negative electron in the system. In view of the want of perfect symmetry between positive and negative electricity, as shown in certain electrical phenomena, the second assumption (which corresponds to Franklin's old theory of electricity, only the signs are reversed) is preferable.[36]

Ostwald goes on to remark, "it is more difficult to apply the theory to chemical compounds."[37] In other words, he leaves the question of molecular structure

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entirely open in the chapters new to his 1909 edition, much as Nernst continued to do in his 1909 edition.

Our conclusion, then, is that Ostwald's conversion came about from the combination of four sets of experimental findings primarily during the period from 1903 to 1907: (1) the counting of gas ions and, with it, the discovery of the electron, the fundamental negative charge, and its asymmetry with the positive charge—mostly under J. J. Thomson between 1897 and 1904; (2) the related determination of an order-of-magnitude value for the fundamental charge that, in combination with an order-of-magnitude value for Avogadro's number based on van der Waals's gas law, was fully consistent with the established value of Faraday's constant; (3) the observations of extremely small particles in constant rapid motion under the ultramicroscope reported by Zsigmondy in 1905 and 1906; and (4) Svedberg's 1906 tests of Einstein's and Smoluchowski's formulas for Brownian motion showing that "agreement with the movements actually observed is so close that we are compelled to regard this agreement as a fairly satisfactory proof of the kinetic nature of heat."[38] Insofar as Ostwald was still insisting on the mere hypothetical standing of molecules in an article published in German early in 1907, and his *Fundamental Principles of Chemistry: An Introduction to All Text-Books of Chemistry* totally eschewing atoms and molecules had been published later that year, the fourth of these appears to have been a tipping point.

Given the amount of attention that he had devoted to ions in solution over the preceding two decades, however, being able to determine the charge per ion in liquids and to count their number had to have been giving him reason to reconsider for a few years before this. (In his Faraday Lecture of 1904 he remarked, "I think there is no word that I have oftener spoken or written than the word 'ion' ";[39] yet Thomson's research is not mentioned in this lecture, nor in his 1907 *Fundamental Principles*, much less the 1899 edition of his *Grundriss*.) Subsequently encountering Perrin's October 1908 paper and its far more precise values for the molecular magnitudes either just before or while the new edition was in press must have markedly reassured him—so much so as to add the citations to it.

Finally, we turn to the question of the new standing of molecular and molecular-kinetic theory indicated by the 1909 edition of Ostwald's *Grundriss* and expressed in his Preface as its having been "raised to the position of a scientifically well-founded theory." The independent values, 10^{24} and 10^{-19} coulombs, for N_0 and e that agreed to their level of precision with the known value of Faraday's constant were not sufficiently precise to meet Ostwald's stated requirement for grounding the theory—namely, definite values that can be substituted "in a formula by means of which some physical relations are to be represented . . . in order to establish experimentally" the functional relations in question. Perrin's more precise value of 7.1×10^{23} for N_0 was a step in the right direction, but the value for

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e he inferred from it, 4.1×10^{-10} esu (i.e., 1.4×10^{-19}), was too far removed from the Wilson-Thomson value of 3.1×10^{10} esu to give Perrin's 7.1×10^{23} much of any claim to finality. *So far as grounding of molecular theory, then, the most that the 1909 edition indicated was sufficient progress toward independently measured values of the relevant molecular magnitudes to license the conclusions that definite values exist and hence too that physical relations are indeed represented by the formulas involving these magnitudes.* As such, having "the position of a scientifically well-founded theory" was tantamount to giving unqualified warrant to continuing experimental pursuit of increasingly precise values for the magnitudes.

Moreover, the two magnitudes in question both involved counts of discrete items, the number of molecules in a mole and the number of elementary charges comprising the amount of electricity needed to yield a mole of a monovalent substance in electrolysis. To give unqualified warrant to pursuit of increasingly precise values for them therefore amounted as well to unqualified warrant to *take* both molecules and discrete elementary charges to exist while engaged in this pursuit. Short of systematic failure to achieve more definite values, the question of whether molecules exist had therefore become closed. None of this was to deny Nernst's point that the question about what binds atoms together in molecules and hinders their flying apart remained entirely open, as did questions about the shapes and sizes of molecules of different kinds. These questions, however, had gained unqualified legitimacy, and the emerging experimental results on ions in gases and solutions and on radioactivity were showing promise of providing experimental means to begin addressing them. Here too, then, having "the position of a scientifically well-founded theory" was tantamount to giving unqualified warrant to questions that before had lacked it because of the failure to find experimental means for addressing them.

7.4. The New Standing of Molecular-Kinetic Theory: Post-Solvay

We shift perspective now to late 1912 or early 1913, following the publication in French in 1912 of the proceedings of the 1911 Solvay Conference.[40] The obvious advance since 1909 was the number of alternative independent determinations of each of N_0 and *e*. Perrin's values for the former ranged from 65×10^{22} to 69×10^{22}, implying a value for *e* from Faraday's law ranging from 4.45×10^{-10} to 4.19×10^{-10} esu, added to which was the Boltwood-Rutherford value of 60.3×10^{22}, implying a value for *e* of 4.80×10^{-10} esu. The various new determinations of *e* independent of those for N_0 ranged in values from 4.65×10^{-10} to 4.891×10^{-10}, implying a value for N_0 ranging from 62×10^{22} to 59.2×10^{22}. Within those bounds were as well the two post-Solvay values for N_0 cited by Eucken that had

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been derived from measurements of k in blackbody radiation, 59.7×10^{22} and 62.0×10^{22}.

Ignoring the discrepancy between Perrin's values and all the rest, the two sets combined provided strong evidence that definite values exist for N_0 somewhere between 59×10^{22} and 69×10^{22} and for e somewhere between 4.1×10^{-10} and 4.9×10^{-10} esu. Insofar as other molecular magnitudes were derivable from these—in particular, energy per molecule per degree Kelvin and absolute masses of atoms and molecules with known chemical formulas—molecular and molecular-kinetic theory had to a greater extent than a decade before become "grounded," in van Fraassen's sense.

Put another way, sufficiently precise values for these molecular magnitudes had become established to warrant not just their being taken for granted, but for their being employed *constitutively* in ongoing research. Anyone doing this, however, had to choose a specific value, with or without error-bars. Thus, for example, in the summer of 1913, Bohr, needing a value for e in order to obtain a theoretically derived value for Rydberg's constant to test against the known value, chose 4.7×10^{-10} esu;[41] the value for N_0 corresponding to this according to Faraday's law is 61.6×10^{22}. At almost the same time that summer, the Braggs, faced with having to derive the distance between sodium and chlorine atoms in the crystal lattice of salt, chose a value for the mass of a hydrogen atom of 1.64×10^{-24} grams;[42] we assume they obtained this from a value for N_0 of 61.5×10^{22}. Perrin himself, who was not at the time using his values for molecular magnitudes constitutively in this way in his ongoing research, continued to insist in his *Les atomes* of 1913 and subsequent editions that his notably larger values for N_0 were to be preferred. Nevertheless, by the summer of 1913, leading figures at the forefront of ongoing research on atoms and molecules were departing from him.

The link between Avogadro's number and the fundamental unit of charge mediated by Faraday's constant is associated with molecular theory, not molecular-kinetic theory. The other such link that we emphasized in Chapter 6—between Avogadro's number and Boltzmann's constant—is, by contrast, entirely associated with molecular-kinetic theory through the relationship $N_0k = R$, where, according to that theory, the mean translational kinetic energy per molecule per degree K, designated α by Perrin, is $3k/2$. In this case, too, the independently measured values for k as of 1912 were not in full agreement. The value from blackbody radiation that Planck gave that year was 1.34×10^{-16} ergs per degree K, implying a value for α of 2.01×10^{-16} ergs per degree K.[43] Perrin's values for N_0 had come from assuming that his measured values for the mean translational kinetic energy of granules in Brownian motion correspond to the mean translational kinetic energy of molecules at the same temperature. Based on the values Perrin announced for N_0 at the Solvay Conference, his values for α ranged from

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1.81×10^{-16} to 1.91×10^{-16} ergs per degree—that is, between 5 and 10 percent less than Planck's.

In this case, the question of a preferred value did not arise with the same force as it did with e and N_0, for α (and hence also k) had less of an immediate role to play in ongoing research. Nevertheless, here were two robust values, 2.01×10^{-16} from by then established blackbody measurements, and $(1.86 \pm 0.05) \times 10^{-16}$ from highly developed Brownian motion measurements that, according to the respective theories, were values for the same physical quantity, yet did not match to the level of precision thought to hold for them. Planck's derivation of the blackbody law was linked to kinetic-molecular theory only through Boltzmann's account of entropy, and even then his appeal to that account had been in some respects idiosyncratic. Perrin's various derivations, by contrast, were linked to the core of kinetic-molecular theory, and his assumption of the identity between granule and molecular energies seemed in no way idiosyncratic. Were it not for the fact that Planck's implied value for N_0, 62.0×10^{22}, was so much closer at the time to the values other than those of Perrin, the difference might have given grounds for questioning whether Planck's k really does amount to a molecular magnitude. Regardless, we have found no one at the time who instead proposed that the mean translational kinetic energies of granules in Brownian motion are in fact simply less than the mean translational kinetic energies of the molecules in the ambient liquid.

These concerns over the less than desired agreement among values are not why we included the qualifier "to a great extent" in our claim three paragraphs ago about molecular-kinetic theory having become grounded. The molecular magnitudes N_0, e, and α are all three generic, not ones that vary from one kind of molecule to another. The only molecular magnitudes determined by them that vary from one kind of molecule to another are the absolute masses, and they scarcely enter at all into kinetic-molecular theory. The prominent macroscopic phenomena that do vary from one kind of molecule to another were viscosity, diffusion, thermal conduction, and deviations from the ideal gas law; and the molecular magnitudes with which they vary, according to kinetic theory, are the sizes and shapes of different kinds of molecules. No less than before the grounding of it that we are here considering, kinetic theory continued to deal with viscosity, diffusion, thermal conduction, and deviations from the gas law by postulating that molecules are spherical, and then inferring sizes for different kinds of molecules from these phenomena. Yet, even more so after the values for N_0 between 59×10^{22} and 69×10^{22} had become available than before, it was clear that (in the words of Jeans from 1916) if "the molecules are *assumed* as a first approximation to be elastic spheres, experiment leads to discordant results for the diameters of these spheres, shewing

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that the original assumption is unjustifiable."[44] One must accordingly be careful not to overstate the extent to which molecular-kinetic theory had become grounded by the end of 1912.

In point of fact, some progress was made on the shape of molecules in 1911 and 1912, much of it in Nernst's laboratory by such figures as Arnold Eucken. They and others had produced strong experimental evidence supporting the Einstein-Debye theory of specific heats of solids at low temperatures. This amounted to evidence that degrees of freedom "freeze out" as the temperature drops, or in other words, that energy thresholds have to be crossed in order for some degrees of freedom to absorb energy.

Eucken had also shown that the ratio of the specific heats of hydrogen does not remain at its room temperature value of 1.4 as the temperature is lowered, but increases, appearing to approach the 1.67 value of monoatomic gases as the temperature approaches absolute zero. This then was evidence that diatomic molecules like hydrogen, oxygen, and nitrogen have five degrees of freedom at room temperature, but as the temperature is lowered, two of these degrees of freedom "freeze out"; diatomic molecules like chlorine for which the ratio of specific heats is less than 1.4 at room temperature, on the other hand, have a vibrational degree of freedom already absorbing energy at room temperature. All of this not only began to address the long-standing quandary over the specific heats of gases, but also provided evidence that diatomic molecules have a dumbbell shape, with three translational degrees of freedom, plus two rotational once the requisite energy threshold is exceeded, and various vibrational at still higher energy levels.[45]

Suppose then that diatomic molecules are taken to have a dumbbell shape, and consider the problem of incorporating this into the kinetic theories of viscosity, diffusion, thermal conduction, and deviations from the ideal gas law for just the gases in question. The pivotal parameter in the molecular-kinetic theories of these phenomena is the *mean free path* of a molecule between its encounters with other molecules. Consider now the complexity of the problem of defining mathematically such a mean free path in the case of rotating dumbbell-shaped molecules, taking their number per unit volume and the statistical distribution of their translational velocities as given; and then using measured values of the viscosity, diffusivity, thermal conductivity, and deviations from the ideal gas law for the different gases in question to infer the sizes of their respective molecules, or even their respective effective "spheres of influence." One should scarcely be surprised that Jeans, after denying that molecules are spherical and subsequently noting the recent results on specific heats, nevertheless stayed with spherical molecules in the 60 pages of the 1916 edition of his *Dynamical Theory of Gases* devoted to viscosity, thermal conduction, and diffusion; nor that the 1916–1917 mathematical theories of viscosity, heat conduction, and diffusion

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in non-uniform gases developed independently by Enskog and Chapman concerned gases composed of spherical molecules. Simply put, notwithstanding the grounding provided by the values of N_0, e, α, and the masses of a huge range of molecules as of 1913, the molecular-kinetic theories of the phenomena of viscosity, diffusion, thermal conductivity, and deviations from the ideal gas law remained theories about hypothetical—or, more accurately, fictional—spherical molecules.

These considerations are reminiscent of what we regard are the most provocative remarks van Fraassen makes in "The Perils of Perrin, in the Hands of Philosophers":

> It is still possible, of course, to also read these results as providing evidence for the reality of molecules. But it is in retrospect rather a strange reading—however, much encouraged by Perrin's own prose and by the commentaries on his work in the scientific and philosophical communities. For Perrin's research was entirely in the framework of the classical kinetic theory in which atoms and molecules were mainly represented as hard but elastic spheres of definite diameter, position, and velocity.[46]

Leaving aside our claims about the nuanced relationship between Perrin's research on Brownian motion and molecular-kinetic theory, the considerations prompting our here quoting van Fraassen once again are reminiscent as well of part of the argument we made in section 2.8 of Chapter 2 for why molecular theory still had the standing of merely a hypothesis in 1900:

> Similarly, the statistical mechanical formulation of the molecular-kinetic theory gave generic qualitative explanations of the phenomena of viscosity and its independence of density in gases, yet absent any determination of the sizes and shapes of molecules of different kinds, it offered only a hand-waving explanation of the specific differences in viscosity from one gas to another or the specific systematic variation of viscosity with temperature in each gas. The inability to determine the sizes and shapes of molecules of different kinds resulted in the theory of viscosity and diffusion being a theory about hypothetical spherical molecules, and even then, efforts to determine their sizes had failed to yield determinate values. In short, the molecular-kinetic theory of heat, even with everything that could be said in its favor, remained a theory about hypothetical discrete entities.

We have now conceded that the grounding molecular-kinetic theory gained from the comparatively precise values determined for N_0, e, α, and the masses of a large range of molecules still had not met the requirement called for in this

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paragraph. Are we, then, to conclude, in agreement with van Fraassen, that the reality of molecules still had merely a hypothetical status in 1913?

We claim no, yet at the same time concede that the reality of molecules had not yet become established once and for all. The new standing of the molecular hypothesis, we claim, derived from adequate grounds having been given for *taking* molecules to exist for purposes of ongoing research, thereby closing the question of their existence—or, perhaps better put, eliminating the question from the scope of that research. The word "take" we are lifting from Newton's fourth rule of reasoning:

> In experimental philosophy, propositions gathered from phenomena by induction should be taken to be true, either exactly [*accuraté*] or very nearly [*quamproximé*], notwithstanding any contrary hypotheses, until yet other phenomena should render such propositions either more exact [*accuratiores*] or liable to exceptions.[47]

The propositions to be taken to be thus true as of 1913 were: (1) All moles of any substance contain the same definite number of chemically demarcated molecules, which to current approximation is around 64×10^{22}; (2) the molecules in all gases have the same definite mean translational kinetic energy per degree of absolute temperature, which to current approximation is around 1.95×10^{-16} ergs per degree Kelvin; and (3) there exists a definite fundamental, universal unit of negative charge, which to current approximation is around 4.5×10^{-10} esu, and it is always joined with a definite rest mass, the combination constituting the *electron*, with the number of them required to liberate a mole of molecules of any monovalent substance in electrolysis always the same. These propositions are to be presupposed as fully established in ongoing research—that is, they are licensed to enter *constitutively* into the design of experiments and rules for measurement—until yet other phenomena should render them either more exact or liable to exceptions. Keep these three propositions in mind, for we shall be returning to them repeatedly in the next section.

Our first chapter included the following quotation from Stephen Brush that, in effect, was rejecting the claim we have just made:

> For the next few years, he [Perrin] seems to have devoted much of his time to popularizing the significance of his work on Brownian motion, in particular the idea that atoms have now been proved to exist. In this he was surprisingly successful. In fact, the willingness of scientists to believe in the "reality" of atoms after 1908, in contrast to previous insistence on their "hypothetical" character, is quite amazing.

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> The evidence provided by the Brownian-movement experiments of Perrin and others seems rather flimsy, compared to what was already available from other sources. The fact that one could determine Avogadro's number and the charge on the electron by one more method seems hardly sufficient to justify such profound metaphysical conclusions. Several independent methods of determining these parameters had been known since 1870 or before, to say nothing of the many successes of kinetic theory in predicting the properties of gases.[48]

The obvious question, therefore, is how, according to us, the situation had changed between 1900 and 1913. The answer is simple. ***All*** of the determinations of Avogadro's number and the mean translational kinetic energy of molecules per degree Kelvin before 1908 (save for Planck's) had presupposed that molecules are spheres of a definite diameter, or at least have a definite sphere of influence. ***None*** of the determinations that emerged between 1908 and 1913 presupposed anything about their shape or size. In other words, before 1908 all the determinations of Avogadro's number were giving the number of hypothetical spherical molecules per mole, while the series of new determinations between 1908 and 1913 were not of the number of hypothetical spherical molecules, but rather simply of the number of molecules per mole. Questions about their sizes and shapes remained entirely open, to be pursued in further research taking full advantage of the newly determined molecular magnitudes.

We take this point to be grounds for dismissing van Fraassen's contention quoted earlier, as well as Brush's. There was nothing at all strange in reading the results of the new mutually complementary theory-mediated measurements of the quantities in question as providing evidence for the reality of molecules of a sort that had never been provided by their efficacy in conceptualizing and explaining phenomena. This efficacy had never yielded anything specific that could enter constitutively into ongoing research without begging questions and hence without the results of any such research having a merely hypothetical standing.

Because the distinction we are invoking with the notion of *take to be true* is generic, and not specific to molecules, we are saving until the next section of the chapter a more developed account of it. We shall end this section with a few brief points in response to questions that are likely to have arisen about the case at hand.

First, what difference did it make whether the molecular and molecular-kinetic hypotheses were taken to be true at the time versus simply continuing to grant that they were offering the only viable explanation of the chemical and thermodynamic phenomena? Part of the answer we gave in our discussion of Ostwald's conversion: questions about the specifics of the many kinds of molecules and

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what binds atoms together in them came to have not just a legitimacy, but beyond this an immediacy that they had not had before. Such questions had come to occupy the cutting edge of research precisely because further progress hinged on finding experimental means for answering them. Still more important than this, moreover, was that taking them to be true carried with it license to take as given in pursuing such experimental means the three claims we listed earlier concerning the number of molecules per mole, their mean translational kinetic energy per degree Kelvin, and the fundamental negative charge. Theory is needed to gain experimental access to regimes we cannot observe. Ampère's law and the galvanometer it licensed as a means of measuring it provided a degree of experimental access to electric current that had not existed before them. Taking these three claims as now given for purposes of research amounted to taking them to be providing means for gaining a degree of experimental access to the microphysical realm that had not existed before.

This, however, invites the question Brush raised: What stood in the way of the values for N_0, α, and e that had been determined by 1900 serving this same purpose? The answer is twofold. First, those values for each of these three had not converged remotely to the extent that the values in 1912 had. Tables 2.1 and 2.2 in Chapter 2 display this lack of convergence in the case of the supposedly universal number of molecules per mole or per cubic centimeter at standard conditions; the only way, then, to obtain values for α were via values for N_0, and hence these values too were not remotely convergent; and the values for e consisted of Townsend's 3.1×10^{-10} esu for the negatively charged and 2.8×10^{-10} esu for the positively charged hydrogen ion and Thomson's 6.5×10^{-10} esu for ions formed by x-ray in air and roughly the same for ions in hydrogen.

We noted earlier that an issue in the case of the values as of 1913 was which specific ones to adopt to incorporate in ongoing research. The risk involved in this choice was one of degree of precision, in particular at the time between a second and a third significant figure. By contrast, to select from the ranges available any specific values for purposes of ongoing research in 1900 was to render any results from that research at best precarious, just as Gibbs had said in 1901.[49]

Moreover, all the values for N_0 and α as of 1900 had been obtained from phenomena covered by kinetic theory under the assumption that molecules are spherical, or at least have definite spheres of influence in their interactions with one another. Had the values for N_0 in 1900 converged to the extent that they had by 1913, there would have been grounds for concluding that this assumption, despite the many misgivings that had been expressed in adopting it, had some merit. The failure of convergence, however, made it still more suspect. By contrast, the only assumptions that the several 1913 values for N_0 and α had in common were the reality of molecules and Avogadro's hypothesis concerning the number of molecules per unit volume at standard conditions or, as recast

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in the last decades of the nineteenth century, the number per mole. The combination of the convergence of the different 1913 values plus the diversity of phenomena from which those values had been obtained constituted strong evidence that both of these assumptions, even if not true, were safe to adopt at least for purposes of attaining robust theory-mediated measurements.

Still, granting all of this, the point we stressed in section 2.8 of Chapter 2 about the failure as of 1900 to gain experimental access to specific contrasting features of molecules of different kinds that could be correlated with contrasting features of macroscopic phenomena still seemed true at the end of 1912. Why wasn't that still giving reason to question whether experimental access had really been gained to molecules?

The concern as of early 1913 was not whether sufficient experimental access to the microphysical realm had been gained to determine the characteristic sizes and shapes of molecules of different kinds and the forces that bind the atoms forming them together. Clearly it had not. The concern was rather whether the experimental access that had been gained, as reflected in the values of N_0, α, and e that had then been determined, provided a basis for further experimental developments that would ultimately enable questions about the specifics of molecules beyond just their individual masses to be resolved. This concern hinged on whether the promise of such progress adequately outweighed the risk that research constitutively presupposing the values in question would ultimately have to all be thrown out should those values turn out not to have been well-founded after all. We leave to the reader whether the evidence we reviewed in Chapter 6 supporting the values adequately addressed this risk.

Needless to say, we have to eliminate from consideration the subsequent history of the research predicated on those measured values if we are to assess the potential they offered at the end of 1912—that is, before Bohr and the Braggs had published their papers and while von Laue's 1912 breakthrough in x-ray diffraction was still being assimilated. So, let us offer an example entirely of our own devising to indicate the sort of promise they were offering.

In section 2.8 we appealed to the phenomenon of NO_2 gas progressively transforming into a mixture of it and nitrogen peroxide, N_2O_4, as the temperature is lowered as an illustration of something about the specifics of molecules that molecular theory had not begun to address. Consider this phenomenon in the light of the measured values of N_0 and α and Perrin's verification of the Gaussian character of Brownian motion. As the temperature is diminished, the rate of change of mole density can be measured. But then so too can the number of NO_2 molecules per unit volume that have joined to form N_2O_4 molecules. The mean translational kinetic energy of both kinds of molecules is the same, and its variation with temperature can be calculated from α. From this and the masses of the two kinds of molecules, their mean translational

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velocities can be calculated as a function of temperature, with those for N_2O_4 roughly 70 percent of those for NO_2. Finally, taking the distribution of velocities among the NO_2 to be Gaussian, at least estimates can be made of the number of NO_2 molecules per unit volume with velocities close to the mean velocities of N_2O_4. All of this points to the possibility of finding some systematic correlation between the number of NO_2 molecules per unit volume with velocities in various ranges and the rate versus temperature at which they join to form N_2O_4 molecules. As conjectural as this is, its promise for revealing something instructive was notably greater at the end of 1912 than it had been before 1900.

Even granting the virtue of having comparatively specific values for these molecular magnitudes, however, if so many questions about the specifics of molecules remained open at the end of 1912, how was research predicated on taking the claims associated with these values to be true any different from research predicated on them in which they continued to be taken as mere hypotheses? Not any difference to speak of, so long as the research in question is making sustained progress. There are three ways in which it can fail to be making sustained progress. First, central questions can remain recalcitrant, beyond the reach of all experiments. This was true throughout the nineteenth century in the case of questions about the forces binding atoms together in molecules, and it continued to be so for more than a decade after 1913. Second, the results of experiments can fail to provide sufficiently unequivocal evidence to resolve questions. The experimental attempts predicated on kinetic theory reported in Meyer's 1899 book to determine the sizes of molecules and, with them, a constant value for Avogadro's number, provide an example, one that again continued, in the case of the sizes, for more than a decade, even after the value of Avogadro's number was available. Third, clear results can be obtained, but only by adopting further hypotheses that seem irreconcilable with established physics. The violation of classical physics by the Rutherford-Bohr model of the atom is an example.

In each of these cases, when the research is predicated on a claim that still has the status of a hypothesis, its continuing failure puts the hypothesis itself into dispute. When, by contrast, it is predicated on a claim for which evidence has provided adequate grounds for it to be taken to be true, it does not. It then has a status like the legal presumption of innocence until proven guilty beyond reasonable doubt. Decisive evidence is required even to raise doubts about its truth. This, of course, was the whole point of Newton's introducing the phrasing, *take to be true, either accuraté or quamproximé, notwithstanding any contrary hypotheses,* in his fourth rule. It was also the point Euler was making when, in 1752, he announced, in the introduction of his *Recherches sur les irrégularités du mouvement de Jupiter and de Saturne*:

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> since M. CLAIRAUT has made the important discovery that the movement of the apogee of the Moon is perfectly in accord with the Newtonian hypothesis . . . , there can no longer remain the least doubt about this proportion. . . . One can now maintain boldly that the two planets Jupiter and Saturn attract each other mutually in the inverse ratio of the squares of their distance, and that all irregularities that can be discovered in their movement are infallibly caused by this mutual action. . . . *And if the calculations that one claims to have drawn from this theory are not found to be in good agreement with observations, one will be always justified in doubting the correctness of the calculations, rather than the truth of the theory*.[50] [emphasis added]

Our answer to the question of the standing of the molecular and molecular-kinetic hypotheses as of early 1913 is that a remark to the same effect held for them. The reception of Bohr's model a few months later—when it clearly did not work for atoms with two or more electrons, and the only hard evidence for it was the agreement to within 6 percent between the calculation drawn from it of Rydberg's constant and the observed value—is one of several illustrations of these hypotheses then having this standing.

7.5. On the Standing of Hypotheses

The principal distinction Newton was making with his fourth rule was between "hypotheses," for which he had long had a low regard (to say the least), and "propositions gathered from phenomena." As any remotely knowledgeable reader at the time realized, the hypothesis he had in mind was the view that planets are carried around in their orbits by fluid vortices, hypothesized to provide a contact mechanism to counteract their centrifugal tendency;[51] and, needless to say, the propositions gathered from phenomena with which he was contrasting this hypothesis were his law of gravity and its account of planetary motions according to which they are perturbed departures from Keplerian motion. He had come to use the phrase, "take to be true," between the first and second editions of the *Principia*, initially in licensing the inductive leap from the evidence that his law of gravity holds for the planets to taking it to hold universally for all matter. Two considerations appear to have led him to do so: the realization that in any such leap to a sweeping generalization, one is always simply *taking* the general claim to be true; and his coming to appreciate the potential for much stronger evidence for his theory of gravity than he could muster in the *Principia* if others in the future were to use it, instead of relying on observation, to pin down the enormous systematic complexities of planetary motion that result from their mutual interactions.[52]

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Smith has invoked Newton's phrasing elsewhere to argue for a narrower than customary conception of what it is for a scientific community to accept a proposal.[53] His main point was to distinguish between the evidence that warrants its initially taken to be true for purposes of predicating ongoing research on it and the evidence that can accrue to it from the success of such research over extended periods of time. His aim there was to respond to historians and sociologists of science who had challenged the epistemic standing of the advanced sciences by emphasizing the extent to which many of their most fundamental tenets had become an integral part of those sciences on the basis of quite limited and sometimes even misdirected evidence. In response, Smith argued that this epistemic standing does not derive from canons of initial acceptance, but from the history of evidence that accrues to propositions from their continuing to be tested, often tacitly and *en passant*, by research constitutively predicated on them. Correlatively, he argued that philosophers assessing the epistemic standing of the sciences need to look carefully at the history of evidence that accrues to fundamental claims in this way over decades and, in some cases, centuries. The extraordinary evidence supporting Newtonian gravity—that is, the evidence that Einstein felt he somehow had to accommodate—was not to be found in the *Principia*, but in the subsequent two centuries of research in celestial mechanics.

Our aim in invoking Newton's phrasing here, by contrast, is to elucidate a distinction between, on the one hand, evidence taking broadly the form of inference to the best explanation that leads at least substantial fractions of a scientific community to believe a hypothesis and, on the other hand, evidence that leads a community to grant it a presumption of inviolability as a constitutive element in ongoing research. We say "elucidate" because we claim that the distinction by no means originates with us, but rather has been part of the practice of scientific communities since at least as long ago as the late seventeenth century. The quote from Euler in 1752 at the end of the preceding section is evidence of this. So too is our quote in Chapter 1 from the 1909 edition of Nernst's *Theoretische Chemie*, which immediately follows a list of values for Loschmidt's number: "the atomistic conception begins to lose its hypothetical character."[54]

An especially clear statement of the distinction can be found on the first page of the 1904 and 1916 editions of Jeans's *The Dynamical Theory of Gases*. In the first edition he says,

> It need hardly be said that this identification of heat and motion is only a hypothesis; it never has been, and from the nature of things never can be, proved. At the same time this hypothesis shows an ability to explain and even to predict natural phenomena, such that there can be little doubt that it rests upon a foundation of truth.

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As we have quoted earlier, in the second and all subsequent editions he says,

> This hypothesis was for long regarded as pure conjecture, incapable of direct proof, and probable just in proportion to the number of phenomena which could be explained by its help. In recent years, however, the study of the Brownian movements has provided brilliant visual demonstration of the truth of this conjecture, and the actual heat-motion of molecules—or at least of particles which play a role exactly similar to that of molecules—may now be seen by anyone who can use a microscope.[55]

(Jeans's statement notwithstanding, we do not mean to be suggesting here that visual access is in any way indispensable to the distinction we are elucidating.)

The distinction is in the same spirit as Poincaré's distinction between "indifferent hypotheses" and "real generalizations," where, in the case of the former, conclusions reached would be the same with or without them.[56] The sole reason for conducting a series of experiments may be to test whether the results are consistent with a hypothesis, and hence that it can explain them, yet the results of the experiment would be the same whether they are consistent with it or not. We saw an example of this with Perrin's Brownian motion experiments in Chapter 4. His reason for conducting the experiments, and Einstein's reason for developing his account of such motion, was to support the molecular-kinetic theory of heat as the sole viable explanation of the phenomenon. Perrin's measured values of the mean kinetic energies of his granules nevertheless did not logically presuppose either that molecules exist or that he was measuring their mean kinetic energies. Our analysis in Chapter 4 showed this as a matter of logic. The subsequent history showed it as well when the discrepancy between Perrin's values for α and those obtained through other approaches raised questions about whether his value for granules really does match the molecular value.

We are not belittling the importance of inference to the best explanation in the history of the physical sciences. We are only calling attention to the difference between evidence of that form and evidence that warrants granting hypotheses the status of a presumption of inviolability as a constitutive element in ongoing research. As we have noted more than once, "inference to the best explanation" initially entered the philosophic literature in the late nineteenth century as the name of a fallacy, specifically the fallacy of taking a conclusion to be established in spite of not having eliminated unconceived alternatives.[57] It is, of course, this very possibility that lies behind Jeans's distinction in the preceding quote between "direct proof" and "probable just in proportion to the number of phenomena which could be explained." Equally, we are not impugning the legitimacy of the widespread acceptance of the molecular and molecular-kinetic hypotheses among physicists in the years before the period we have examined.

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It did indeed have enormous heuristic value, just as Ostwald himself acknowledged in so many places in the 1889 edition of his *Grundriss*.

We do, however, remind readers that a material ether was similarly widely accepted as the only viable explanation of the oscillating transfer between kinetic and potential energy required for sustained wave motion. In his *Encyclopedia Britannica* on the ether, Maxwell reviews the failures of attempts to measure any specific aspect of the ether, including the pre-Michelson-Morley attempts to measure the velocity of the earth through it. He still, however, concludes,

> Whatever difficulties we may have in forming a consistent idea of the constitution of the aether, there can be no doubt that the interplanetary and interstellar spaces are not empty, but are occupied by a material substance or body, which is certainly the largest, and probably the most uniform body of which we have any knowledge.[58]

Again, we are not impugning the acceptance of the existence-of-the-ether hypothesis by Maxwell, Lorentz, and a great many others during the nineteenth century. We are only calling attention to the difference between his grounds for concluding that the ether exists and the grounds he might have had if any of the efforts to determine specific properties of it that he considers in the article had succeeded.

So much for one side of the distinction in evidence that we are elucidating; what about the other? As a step toward giving more content to it, consider the experimental efforts presented by Meyer in his 1899 *Kinetic Theory of Gases* to leverage measurements of the different sizes of molecules from phenomena for which molecular-kinetic theory offered an explanation, arguably the only viable explanation at the time. The molecular hypothesis *did* enter constitutively into those efforts, specifically the spherical molecule hypothesis, but scarcely with a presumption of inviolability. The failure of the different approaches to yield convergent values for the sizes, or even for the individual approaches to meet the cross-check of yielding a constant value for Loschmidt's number,[59] gave more reason to question the spherical molecule hypothesis than to accept it. In the end, Meyer nevertheless takes 2×10^{-8} cm to be "the average diameter of a gaseous molecule," prefacing his choice with the remark, "although we may not pretend to see [in it] an exact evaluation of the size."[60] Granted that this was at best only an order-of-magnitude value, but the question still arises why it was not grounds for taking the existence of molecules to be a settled question, yet the 1912 values for N_0, α, and e were.

One difference is that Meyer's 2×10^{-8} cm has virtually no claim to being a physically specific value. Consequently, were it to have been taken to be true and presupposed constitutively in further research, the most it could have contributed

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to measurement or evidence would have been an order-of-magnitude cross-check on results reached without it. Hence it would have added little more, if anything, toward advancing research than just the bare claim that molecules exist.

More generally, not much is to be gained from taking a claim to be true and presupposing it constitutively in further research unless it has some form of specificity. Specific claims have far more potential for being leveraged into something more. We hazard the guess that all that would have been needed for the existence of the ether to have been taken to be true in our sense was a reasonably specific value of the speed of the earth through it—that is, a well-behaved positive result from the Michelson-Morley experiment or its predecessors—even though all the other questions about it that Maxwell reviewed in his *Britannica* article would have remained totally open. Regardless, the claims about N_0, α, and e as of 1913 were not only comparatively specific, but beyond this, their claim to specificity was strengthened by the fact that a specific value for each of the three quantities had been determined that was independent of all of the convergent determinations of the other two. This, and the fact that there were the three of them, gave them a potential for being leveraged into something more, even though virtually all the questions about the specifics of molecules themselves remained open. What is the point of taking a claim to be true and adopting it constitutively in further research if it lacks such potential?

A second difference is that whatever claim to specificity that Meyer's 2×10^{-8} cm might have been taken to have was undercut by its presupposing something specific about molecules themselves, namely that they are, at least to high approximation, spherical in shape, or if not, that they have well-defined "spheres of influence" in their interactions with one another. So, any research predicated constitutively on this value would at the same time have been predicated on a claim about molecules that had nothing to recommend it except mathematical tractability. The whole point of taking a claim to be true and adopting it constitutively in further research is to pursue experimental answers to open questions. To presuppose an answer to one of the principal open questions in doing so puts all the answers that might emerge from the further research at risk of begging the question. By contrast, none of the assumptions we listed in section 6.5 of Chapter 6 that bridged the gap between macrophysically measured quantities and N_0, α, and e presupposed anything specific about molecules, and hence answers to further questions achieved by research predicated on them did not risk begging the question in this way.

The two preceding paragraphs have taken for granted that the step from a claim's having a merely hypothetical standing to its being granted a presumption of inviolability has to turn on risk versus potential gain. The risk is not that the research will not yield robust results, but that those results will all have to be discarded at a later point when the claim on which they are predicated turns out

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not to be true. Its later turning out not to be true, however, does not automatically entail that the results predicated on it will have to be discarded. Newton's law of gravity has now given way to Einstein's field equations, yet the two centuries of results concerning which details in our solar system make a difference and what differences they make that presupposed it were not discarded, thanks to Einstein's showing that it holds asymptotically in the static, weak field limit of his equations. Sheldon Glashow has generalized this point in a trenchant sentence: "*A physical principle cannot be said to be true until it has been shown to be false and its envelope of applicability delineated*."[61]

The question, then, is what requirements a claim presumed to be inviolable in ongoing research has to satisfy in order for the results that presuppose it to survive its later turning out not to be true. The answer we propose, in the spirit of Glashow's remark, is that it has to satisfy some combination of three requirements: (1) it has to hold at least to an appropriately high level of approximation; (2) it has to hold, if not universally, then at least over a domain that reasonably approximates the one the research on which it is predicated is trying to cover; and (3) the terms in which it is stated, if not entirely adequate in their own right, will at least be effective proxies over the domain in question, especially for purposes of measurement, for terms that allow it to better meet the first two requirements. In the preceding section of this chapter we proposed that three claims, interrelated by the convergent measurements reviewed in Chapter 6, had by the end of 1912 come to have a standing of presumed inviolability for purposes of ongoing research: two, stated in terms of molecules, centered on the measured values of N_0 and α, and a third, stated in terms of electrons, centered on the measured values of e. We here propose that the evidence supporting these three should be assessed with respect to the three requirements we have just listed.

In Chapter 6 we concentrated on the convergence of the measured values for N_0, α, and e as evidence. In doing so, we somewhat understated the strength of the evidence that in each case a definite physical quantity exists for which approximate values were being determined in the measurements. Values for e were being determined from two separate phenomena, ionization in gases and α-particles. Values for α were also being determined from two separate phenomena entirely independent of those from which values for e were being determined, namely blackbody radiation and Brownian motion. Even in the case of N_0, values had been determined from one phenomenon entirely independent of those from which values for e and α were being determined, namely the collection of a countable number of α-particles to form a given fraction of a mole of helium gas. Moreover, as Perrin had stressed, thanks to the established values of the gas constant and Faraday's constant, each determination of a value for any one of e, α, or N_0 implied a value for each of the other two as well. So, every determination

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of a value for any one of the three was subject to cross-checks not only from other values determined for it, but from all the values determined for the other two.

All this, together with the degree of convergence of the measured values for all three between 1908 and 1913, gave strong grounds for concluding that definite physical values exist for (1) the number of molecules in a mole, (2) the mean translational kinetic energy per degree Kelvin of the molecules, and (3) the fundamental unit of charge—that is, definite values at least over the ranges of the various phenomena for which determinations had been made; and these definite values lie within the ranges of the announced measured values from the most reliable experimental determinations of each of the three. Moreover, the strong grounds that a definite number of molecules exist per mole were grounds too that molecules exist, or (using Jeans's phrasing) something akin to them that within this evidence "plays the same role"; and correspondingly for the existence of a fundamental unit of charge and with it the electron. Beyond this, the range of the measured values for N_0 implied a corresponding range of values for the absolute masses of all atoms for which atomic weights had been established, and hence too a range of values for the masses of all molecules for which chemical formulas had been established in terms of these atoms. Finally, insofar as a comparatively precise value had been determined for the charge-to-mass ratio of the electron, the range of the measured values for e gave a range for its mass as well.

In short, notwithstanding how many questions about molecules and, for that matter, electrons remained entirely open, the evidence coming out of the theory-mediated measurements considered in Chapter 6 had licensed quite a range of interrelated propositions to be taken to be true and presupposed constitutively in continuing research. We are not denying that the evidence for the existence of molecules was limited, for so many questions about them remained open. The evidence for the claims listed in the preceding nevertheless gave good reason (1) that taking them to be true would enhance the pursuit of answers to those questions and (2) that experimental results constitutively presupposing them would not themselves be undercut should those answers require the claims to be reconsidered.

One way to avoid an extended garden path of research predicated on a claim that has been taken to be true on limited evidence is to demand a great deal of that research right away. In some respects, the 1913 Bohr model did this in the case of the charge and mass of the electron, and since the charge of the electron had become interlinked with the number of molecules in a mole and their mean kinetic energy, for them too. The Bohr model offers a lesson of a different sort as well. After the Bohr model was shown to hold for the helium ion and Sommerfeld had accounted for a doublet in the Balmer series by adding elliptical to Bohr's circular orbits, the tenet that electrons describe definite orbits around nuclei was accorded just the sort of privileged status in the case of ongoing

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quantum research that we have been highlighting.[62] Yet it was just this tenet that Heisenberg, a decade later, rejected as having been a continuing impediment in his watershed paper of 1925—that is, the paper in which he insisted on restricting quantum theory to what he called "observable quantities": "it is necessary to bear in mind that in quantum theory it has not been possible to associate the electron with a point in space, considered as a function of time, by means of observable quantities."[63] We cite this example to emphasize again that the presumption of inviolability of a tenet that has been taken to be true in our sense is provisional. Notice, too, that the distinction marked by Bohr orbits—between stable energy states for which they turned out to have been a surrogate—survived.

The point of this section has been to clarify our distinction between a hypothesis having been generally accepted within a community, yet still having the standing of a hypothesis, and the standing we claim the molecular and molecular-kinetic hypotheses had come to have at the end of 1912 and the beginning of 1913. This moment corresponded to the time immediately after the results of the Solvay Conference became available to the community at large with the 1912 publication of the proceedings in French. Our choice of this moment is nevertheless rather arbitrary insofar as so many groundbreaking results of research at least bearing on, and usually constitutively presupposing, these hypotheses had already appeared or were about to. Experimental results supporting Debye's version of Einstein's theory of specific heats of solids had been published in 1912, as had both results on the specific heats of diatomic gases at low temperatures and von Laue's breakthrough with x-ray diffraction. The Braggs' paper on the crystal structure of halides appeared in July 1913, followed by further papers, including one on the structure of diamond, in September. The first of Bohr's three papers also appeared in July 1913, and the first of Moseley's two papers tying x-ray spectra to atomic number appeared in the same journal five months later. By the end of 1913, therefore, significant new evidence had already accrued to the claims associated with the values for N_0, α, and e from groundbreaking results presupposing them. The time at the end of 1912 and beginning of 1913 for which the molecular and molecular-kinetic hypotheses had just the standing we are attributing to them, and nothing beyond it, was a fleeting moment.

However brief the moment was, we nonetheless contend that the best way to understand the new standing the molecular and molecular-kinetic hypotheses had acquired as a consequence of the measurements of N_0, α, and e between 1908 and 1913 is ours—namely, a middle stage between their being widely accepted yet still having the standing of a hypothesis, and a subsequent stage in which they became increasingly entrenched by virtue of the results of research constitutively presupposing them.

An added virtue of viewing the new standing in this way is the incisive rejoinder it provides to van Fraassen's remark, that to "read these results as

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providing evidence for the reality of molecules . . . is in retrospect a rather strange reading."[64] For it justifies this reading without implying that the evidence for the reality of molecules in 1913 was in any respect definitive. From the first edition of his *Theoretische Chemie* in 1893 to the "eighth-tenth" edition in 1921, Nernst had said, "At present scarcely anything definite is known about the nature of the forces which bind atoms together and which hinder them from flying apart."[65] Nothing in the results for N_0, α, and e amounted to any advance on this. Worse, so far as the molecular-kinetic hypothesis is concerned, the only specific variation from one kind of molecule to another that had been determined was their masses. No advance had been made on the specific variations in their sizes and shapes needed to link different kinds of molecules to the macroscopic differences they display in phenomena of viscosity, diffusion, heat conduction, and deviations from the ideal gas law, not to mention chemical kinetics. The referent of the term "molecule" remained experimentally thoroughly underdetermined.

We noted at the beginning of this section that Smith has elsewhere invoked Newton's "take to be true" phrasing to contrast the standing that the law of gravity had on the basis of the evidence presented in the *Principia* with the increasingly entrenched standing that it came to have over the course of the next two centuries as a consequence of evidence accruing to it from the research that presupposed it. This contrast is no less apropos in regard to the reality of molecules. Anyone asking now what the evidence is that molecules exist is misconceiving the way evidence works in physics if they focus on the time before 1913. This question can be meaningfully answered only through a critical review of *all* the evidence bearing on molecular reality that has been, and is still being, generated since 1913.[66]

7.6. 1905–1913 within the History of Theory-Mediated Measurement

We have noted in earlier chapters that theory-mediated measurement in what became known as the physical sciences goes back at least as far as Ptolemy's *Almagest*. The developments during the years 1905 to 1913, with Perrin at the center, were a central part of a major transition, initiated by J. J. Thomson in 1897 and Planck in 1900, in how theory-mediated measurement can contribute to evidence in microphysics. The primary contribution to evidence before then was to confirm that various quantities that according to theory should be constant actually do have determinate single values. We have more than once cited the effort that went into determining such values in the case of Faraday's constant from electrolysis and the ideal gas constant. A further example from the second half of the nineteenth century was the series of measurements of the gravitational

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constant G from a variety of variations of the Cavendish experiment.[67] A review article of 1901 cites the agreement among the values obtained from Cavendish forward as the only unqualified evidence that gravity varies linearly with the mass of the attracting body—evidence that took the form of the constant of proportionality in the law of gravity being verified, within the limits of experimental precision, to be constant.[68]

A new era began with J. J. Thomson's measurements of the mass-to-charge ratio, m/e, of the constituents of cathode rays in 1897.[69] This value turning out to be more than three orders of magnitude smaller than the smallest known value of m/e, of the hydrogen ion in electrolysis, led him to conclude, "Thus on this view we have in cathode rays matter in a new state, a state in which the subdivision of matter is carried out much further than in the ordinary gaseous state."[70] Two years later, after finding the same order of magnitude for m/e for the discharges in both the photoelectric effect and thermionic emission, Thomson concluded, "From what we have seen, this negative ion must be a quantity of fundamental importance in any theory of electrical action; indeed, it seems not improbable that it is the fundamental quantity in terms of which all electrical processes can be expressed."[71] Thomson's "discovery of the electron" has properly been termed the birth of microphysics. It also has claim to being the first successful measurement of any microphysical quantity. And, as such, it opened a new era in theory-mediated measurement by virtue of such measurement of various microphysical quantities being the primary way of gaining experimental access to this realm.

In a paper of 1898, midway between the two papers cited in the preceding paragraph, Thomson published his first measured values for the charge of ions in air, 6.5×10^{-10}. He then inferred from this value and the known value of the total quantity of electricity per cubic centimeter of hydrogen released in electrolysis—tantamount to Faraday's constant—a value for Loschmidt's constant, 20×10^{18}, which corresponds to a value of 45×10^{22} for Avogadro's number.[72] Thomson then cited a value of Loschmidt's number, 20×10^{18}, that he said was from experiments on the viscosity of air, as evidence supporting his 6.5×10^{-10} value for the charge. In a related paper the next year, John Townsend provided evidence from experiments on the diffusion of gaseous ions to conclude, independently of the specific value for either the charge or Loschmidt's number, that the charge per ion generated by x-rays is the same as the charge on the hydrogen ion in electrolysis.[73] With these steps in evidential reasoning, the putative link provided by Faraday's measurements in electrolysis between the charge on ions and the number of molecules in a mole became a pivotal source of evidence for claims in microphysics.

Thomson's point in deriving a value for Loschmidt's number was to confirm the value for the charge of gaseous ions he had obtained, under the assumption that

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this charge is the same as the charge on hydrogen ions in electrolysis. Townsend's point was to confirm this assumption. As we have seen, the existing value for Loschmidt's number with which Thomson compared the value he had obtained was somewhat arbitrarily chosen among the many available, resulting in not much more than order-of-magnitude confirmation. This became clear five years later when Harold Wilson and he obtained a value for the charge from water droplets in an electric field less than half of the value from 1898. Our main point here, however, is that Thomson's goal in invoking Faraday's constant was simply to confirm his theory-mediated measurement of the microphysical charge.

Planck's derivation in 1900 of the blackbody radiation formula added a further step in the endeavor to extract evidence from theory-mediated measurements of microphysical quantities. This derivation had yielded a coefficient of temperature k in the formula that he claimed to be Boltzmann's constant, a measure of the mean kinetic energy per molecule per degree Kelvin of a gas. In tandem with a value for what became known as Planck's constant, blackbody radiation determines a comparatively precise value for k, but there was no existing value for it with which this value could be compared. So, from the measured value of k Planck derived values for Avogadro's number, $N_0 = R/k$, Loschmidt's number, and "the elementary quantum of electricity e," the last of these through eN_0 = Faraday's constant. His paper then ends with a paragraph that we quote once again, for it marks a landmark in the history of evidence deriving from theory-mediated measurement:

> If the theory is at all correct, all these relations should not be approximately, but absolutely, valid. The accuracy of the calculated numbers is thus essentially the same as that of the relatively worst known, the radiation constant k, and is thus much better than all determinations up to now. To test it by more direct methods should be both an important and a necessary task for further research.[74]

As we have noted earlier, Planck was invoking these other constants not to confirm his value for k, but to lend support to his derivation. In other words, Planck's goal in invoking the gas constant and Faraday's constant was not to confirm his value for k, but for other independent theory-mediated measurements of N_0, Loschmidt's number, and e to provide evidence for his theory of the blackbody formula.

The trouble, of course, was the absence at the time of values for these other constants of remotely the precision as that of the value for k, or for that matter for h, obtained from blackbody radiation. We can do no better than Planck's summary in his Nobel Prize lecture of 1920 of the situation in 1900 and what then ensued over the course of the next few years.

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> The first constant is of a somewhat formal nature; it is connected with the definition of temperature. If temperature were defined as the mean kinetic energy of a molecule of a perfect gas, which is a minute energy indeed, this constant would have the value 2/3. But in the conventional scale of temperature the constant assumes (instead of 2/3) an extremely small value, which naturally is intimately connected with the energy of a single molecule, so that its accurate determination would lead to the calculation of the mass of a molecule and associated magnitudes. This constant is frequently termed Boltzmann's constant, although to the best of my knowledge Boltzmann himself never introduced it (an odd circumstance, which no doubt can be explained by the fact that he, as appears from certain of his statements, never believed it would be possible to determine this constant accurately). Nothing can better illustrate the rapid progress of experimental physics within the last twenty years than the fact that during this period not only one, but a host of methods have been discovered by means of which the mass of a single molecule can be measured with almost the same accuracy as that of a planet.
>
> While at the time when I carried out this calculation on the basis of the radiation law an exact test of the value thus obtained was quite impossible, and one can scarcely do more than test the admissibility of its order of magnitude, it was not long before E. Rutherford and H. Geiger succeeded, by means of a direct count of α-particles, in determining the value of the electrical elementary charge as 4.65×10^{-10}, the agreement of which with my value 4.69×10^{-10} could be regarded as a decisive confirmation of my theory. Since then further methods have been developed by E. Regener, R. A. Millikan, and others, which have led to a but slightly higher value.[75]

Although Planck did not mention Perrin, and Perrin's values had by then been superseded, what he was referring to was the 1905–1913—or, perhaps better, 1908–1913—period in which Perrin was in many respects the central figure. The list of measured values of N_0 (and e) in our Table 6.1 in Chapter 6 is intended to encapsulate the advances to which Planck was referring that were made in this period. During these years, there was a quantum jump, so to speak, in both the kind and the strength of the evidence that was coming out of theory-mediated measurements of microphysical quantities. Hopefully we have said more than enough about this already. Here we need only remind readers that the progress made in the measurements themselves during this period amounted to an increase in reliability from order-of-magnitude to Poincaré's first significant figure—or perhaps better, paraphrasing Planck's 1913 assessment, to values for N_0 between 58 and 70 times 10^{22}, if not between 59 and 69 times 10^{22}, and correspondingly for e at least between 4.1 and 4.9 times 10^{-10} esu. Even at that level of

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reliability, we have seen how much impressive evidence was marshaled out of the various measurements of the quantities in question.

We chose to bound our study by the publication of Bohr's first paper of his 1913 trilogy not merely because it opened the way for the first time to theory-mediated measurements of quantities pertaining to the structure of atoms and hence changed the entire character of molecular research, but no less so because it happens also to have been a landmark event in the history of theory-mediated measurements of microphysical quantities generally. Bohr's proposal that electrons in a hydrogen atom emit radiation only when they jump from one quantized energy state to another was, needless to say, a radical departure from classical physics. That it then allowed the energy differences between such states to be specified by the energy *hv* corresponding to lines in the spectra allowed the spectra, for the first time ever, to provide more empirical information than just a fingerprint, so to speak, of the atom or molecule emitting or absorbing radiation. Beautiful and promising though that was, it did not in and of itself provide any empirical evidence for the Bohr model, only evidence of its promise. The sole empirical evidence to support the model in Bohr's first paper was his derivation of the value of the Rydberg frequency from values of *e, h*, and *m/e* of the electron. The agreement Bohr's calculation achieved, 3.1×10^{15} versus the known 3.29×10^{15} value, was, he said, "inside the uncertainty due to experimental errors in the constants entering into the expression for the theoretical value."[76] What more could anyone have asked for?

The answer, of course, was for the value Bohr used for *e*, 4.7×10^{-10}, actually to have been "inside the uncertainty due to experimental errors." As we noted at the end of section 6.2 in Chapter 6, *e* entered Bohr's expression for the theoretical value of the Rydberg frequency, as calculated, to the fifth power. Had he used the value for *e* that Perrin was defending in 1913, 4.21×10^{-10}, the discrepancy in the calculation would have been 55 percent instead of merely 6 percent, and the evidence supporting Bohr's radical departure from classical physics would have been far less compelling than it was. Consequently, Bohr's appeal to Rydberg's constant as his source of evidence required far tighter error bounds on the value of *e* than Perrin had demanded of the various values of N_0 which he had cited as evidence for the existence of molecules at the end of his 1909 monograph, his 1911 paper, and his *Les atomes*. This is one respect in which Bohr's paper amounted as well to a landmark event in the history of theory-mediated measurements of microphysical constants: his evidence needed an error-band on the value of *e* not much greater than ±1 percent—or, in other words, a value in which the *second* significant figure, not just the first, was reasonably reliable. The demands on the precision of the theory-mediated measurements for purposes of evidence in ongoing research thus increased with Bohr's paper by virtually an order of magnitude.

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A second respect is that the test against Rydberg's constant linked h and e to one another, and, because the measurement of this constant was of a wave length, to the speed of light c as well. Theretofore, only Planck's derivation of the black-body radiation equation and Einstein's yet-to-be-confirmed proposal on the photoelectric effect[77] had linked h and e to one another and to quantities outside of microphysics. Over the course of the next decade and a half, as quantum theory was developing out of the Bohr model, several other such combinations of microphysical constants yielded calculated values of quantities that could be measured independently, and hence provided the same kind of test of theory as Bohr's appeal to Rydberg's constant. To give just a couple of examples, Sommerfeld's fine-structure constant, $\alpha = 2\pi e^2/ch$, gave the fine structure in the spectral lines of hydrogen that Bohr had ignored; and the so-called Bohr magneton, $eh/4\pi m$—that is, the elementary quantum of magnetic moment—entered into the calculation of magnetic properties of gases and, at a later stage, into electron spin. The force of the evidence provided by the calculations of these quantities depended on the level of agreement with independently measured quantities, and hence depended as well on the precision of the theory-mediated values for e, h, and m. Equally, however, having several different combinations of these three linked to independently measurable quantities opened the way to an increasing number of theory-mediated measures of the three, and hence to being able to play these off against one another in pursuit of preferred values.[78]

A milestone in the development of quantum theory, and with it a transition to a new stage—in effect, from quantum mechanics to the beginnings of quantum electrodynamics—occurred in 1928 with the publication of Dirac's relativistic extension of quantum mechanics in his two-part paper "The Quantum Theory of the Electron," and Darwin's "The Wave Equations of the Electron" confirming that the full spectrum of the hydrogen atom, fine-structure and all, plus the Zeeman effect, could be derived from Dirac's theory.[79]

Though only peripherally related to it, a transition to a new stage in the history of theory-mediated measurement occurred a few months later with the publication of Birge's "Probable Values of the General Physical Constants (as of January 1, 1929)."[80] Preferred values of the fundamental constants in orbital astronomy had been adopted in the last decades of the nineteenth century in order to eliminate differences in calculated planet locations arising purely from different ephemerides using slightly different values for the constants.[81] As preoccupation with precise agreement between theoretically calculated and observed values in microphysics increased, a corresponding need arose there as well. Thus Birge notes in his opening paragraph,

> Some of these constants can be evaluated by various methods. Each has been investigated by various persons, at various times, and each investigation

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> normally produces a numerical result more or less different from that of any other investigation. Under such conditions there arises a general and continuous need for a searching examination of the *most probable* value of each important constant. The need is general since every physical scientist uses such constants. The need is continuous since the most probable value of to-day is not that of to-morrow, because of the never ending progress of scientific research. These remarks appear to the writer so self-evident that the mere statement of them may be deemed superfluous. However, in spite of these facts, an investigation of the values of general constants in current use in the literature reveals a surprising lack of consistency, both in regard to the actually adopted values and to the origin of such values.[82]

Over the course of the next 40 years, Birge and his protégés took it upon themselves to issue updates to the values of the "fundamental" constants, the number of which kept increasing as a consequence of continuing research in microphysics. One of these updates, Birge's "The 1944 Values of Certain Atomic Constants with Particular Reference to the Electronic Charge," is of special interest in the context of the present monograph.[83] As we noted earlier, in the wake of Bohr's 1913 paper, measured values for e became primary, with preferred values for N_0 inferred from them via Faraday's constant. Birge's paper revisits these three constants, examining not only the precision with which each was being measured independently of the others, but also possible sources of systematic error in them in response to their imperfect consistency with one another. He concluded that x-ray crystallographic measurements of N_0 from five different crystals were most reliable, yielding a value of $(60.2338 \pm 0.00043) \times 10^{22}$; and the measurement of Faraday's constant was sufficiently precise and reliable[84] to give a value for e from it, $(4.8021 \pm 0.0006) \times 10^{-10}$, that was to be preferred over the most tenable value from measures entirely independent of the Avogadro and Faraday constants, $(4.803 \times 0.002) \times 10^{-10}$. In other words, by 1945 the situation had become the reverse of what it had been at the end of the period covered in our study: the measured value for N_0 had become primary, with the preferred value for e inferred from it.

The further development of quantum theory took a dramatic turn two years later with the announcement at the Shelter Island Conference of discrepancies between Dirac's theory and highly precise measurements first in the fine structure of hydrogen—the Lamb shift—and then in the hyperfine structure of hydrogen and deuterium—the anomalous magnetic moment of the electron.[85] Although nearly two decades elapsed, the new quantum electrodynamics that emerged after Shelter Island led to a new era in the history of theory-mediated measurement with the 1969 publication by three of Birge's protégés, Taylor, Parker, and Langenberg, of the landmark article, "Determination of e/h, Using

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Macroscopic Quantum Phase Coherence in Superconductors: Implications for Quantum Electrodynamics and the Fundamental Constants"[86]—the same year as CODATA, which had been created the year before by the International Committee for Science, formed its Task Group on Fundamental Constants. This Task Group of CODATA has thereafter assumed the responsibility that Birge and his protégés had been shouldering for the previous 40 years. What Taylor et al. announced in 1969 that helped to spur this change was a determination of a value for the fine-structure constant "that did not require the use of QED theory" and that thereby provided "a test of QED in which a priori information from QED itself is not essential."[87] With this, an approach to testing QED emerged involving comparison of measured values of fundamental constants at most weakly presupposing QED with ones strongly presupposing QED.[88]

In other words, the role in evidence of the theory-mediated determinations of microphysical constants expanded from one of supplying values needed for comparisons of theory and experiment to one of themselves comprising the immediate content of the evidence. The 2006 CODATA report lists six distinct theory-mediated values of the fine-structure constant, depending to different degrees on different aspects of QED, that agreed with one another to seven significant figures, that is, to better than 1 part in 10 million. The two with the smallest standard uncertainty, one strongly dependent on QED, the other weakly, diverge in the tenth significant figure, and even there by less than half of the standard uncertainty. To quote the report once again, "This is a truly impressive confirmation of QED theory."[89]

As significant as such increasingly stringent testing of QED has been, focusing on it takes us away from the history of measurement of the three constants Perrin singled out in 1908. From 1969 until 1998, CODATA published only two sets of recommended values, in 1973 and 1986. Starting in 1998, however, they decided that the pace of advances toward greater precision had increased to a point that adjustments to the values of the fundamental constants should be made every four years. To get a sense of the extent to which the precision was increasing, consider the estimated relative uncertainty in the values for N_0: 6.6 parts per million in 1969, 7.9 parts per 100 million in 1998, and 1.2 parts per 100 million in 2014; correspondingly, the uncertainties in values for e were 4.4 parts per million in 1969, 3.9 parts per 100 million in 1998, and 6.1 parts per billion in 2014.[90] In 2014 sufficient precision had been achieved with the constants singled out by Planck in 1900 and Perrin in 1908 to warrant their assigned values being the basis of a 2018 revision to the International System of Units (the SI) in which "the physical concepts underlying the definition of the kilogram, the ampere, the kelvin, and the mole have been changed."[91]

On our account, then, the history of theory-mediated measurement is to be thought of as having had five stages: (1) before 1897; (2) from 1897 to 1929;

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(3) from 1929 to 1969; (4) from 1969 to 1998; and (5) from 1998 to 2018. The second of these saw the successful development of convergent, stable theory-mediated measurements of microphysical quantities, N_0, α, and e in particular, from at best order-of-magnitude to values that had claim to being within ±1 to 2 percent. The present monograph has focused on the middle portion of this stage, from 1905 to 1913, during which convergent, stable measurements of these constants came to have claim to being within ±5 percent, that is, claim to Poincaré's one significant figure. Insofar as that step was so immediately superseded by a demand for values within ±1 to 2 percent, one might think it as not all that important to the history of theory-mediated measurement. Any such thought is wrong. Before the 1905–1913 period, the question of whether meaningful measurements of microphysical quantities could be achieved at all was still open. Achieving convergent, stable measurements of such quantities to within ±5 percent settled that question: ***it was clearly possible to gain sufficient experimental access to the microphysical realm by means of theory-mediated, mutually corroborating measurements to attain meaningful values for at least some highly important quantities, in particular N_0, α,* and *e***—that is, the three "universal constants" that Perrin had singled out in the opening paragraph of his October 1908 paper.

We list here the CODATA 2018 recommended values for these three constants, together with the two macrophysical constants that linked them to one another throughout our study and, for that matter, continuing to the present:[92]

Avogadro's number:	$60.2214076 \times 10^{22}$
e:	$1.602176634 \times 10^{-19}$ C (= $4.80320471 \times 10^{-10}$ esu)
α:	$2.0709735 \times 10^{-23}$ J/degK
Faraday's constant:	96485.33212 C/mol (= $2.892557488 \times 10^{14}$ esu)
R:	8.31446261 J/mol degK

For completeness, we include as well the 2018 values of the two constants of Planck's blackbody formula: $k = 2\alpha/3 = 1.380649 \times 10^{-23}$ J/degK and $h = 6.62607015 \times 10^{-34}$ J/Hz. As of 2018, all of these values have come to be stipulated as *exact* for purposes of defining the new SI units, specifically for the mole, the ampere, the kelvin, and the kilogram. The research into the values of these constants that Planck set in motion in 1900 thus came to a fulfillment of sorts in 2018, though doubtless not in a form that he had anticipated.

The preceding sections gave our answer to the question of how the standing of molecular-kinetic theory changed during the 1905 to 1913 period. Perhaps we should instead have been answering the question of how the standing of research into microphysics changed during this period. We have just given our answer in italics in the last sentence of the paragraph before last. On its face, that answer

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seems not to be that far removed from van Fraassen's claim that what changed was molecular-kinetic theory becoming "grounded." But thinking of it in that way is missing the point. For among the microphysical quantities that had come to be shown to be measurable were the number of molecules in a mole of gas, and by virtue of that their masses and hence their numbers in liquids and solids as well; the mean translational kinetic energy per degree Kelvin of the individual molecules in at least gases and liquids, if not solids as well; and the number of elementary charges in any macroscopic quantity of electricity, whether static or in electric current, and hence too the number of electrons in any aggregate of negative charge. On our view, this was the most important contribution Perrin and others during this period made to science. It is why we wrote this monograph.

Notes

1. See, most notably, the footnotes on pages 120 and 128 of Perrin (1990) and (1916) [(1913), pp. 171 and 182f, or (2014), pp. 218 and 225f], the latter of which even dismisses Svedberg's results on diffusion while granting that they support Brillouin's. Perrin does not cite Svedberg (1912) in which his diffusion results are elaborated in some detail (pp. 61–83), along with the value for Avogadro's number drawn from them of 58×10^{22}.
2. Jeans (1916), p. 8.
3. Perrin's high values for Avogadro's number of course imply that, had he derived a value for R from his results, he would not have confirmed its agreement with the established value of the gas constant. In his Nobel Lecture (Perrin, 1926, p. 6f) he offered an indirect argument, derivative from one by van't Hoff, that any osmotic pressure in his emulsions must be the same as the partial pressure of the granules on the wall of the container. As we noted in Chapter 4, however, even granting him this argument, he still would not have done what van't Hoff did, namely confirm that the constant of proportionality for the osmotic pressure is equal to the gas constant.

   We ignored Perrin's Nobel lecture in Chapters 4 through 6 because it fell outside the 1905–1913 period on which we were focusing. One other oddity of the lecture is nevertheless worth noting here. Oseen's presentation speech lists Perrin's values for Avogadro's number: 68.2×10^{22} from vertical gradients, 68.8×10^{22} from mean square displacements, and 65.0×10^{22} from rotation. In his lecture, by contrast, Perrin lists 68×10^{22} for the first, 64×10^{22} for the second on the basis of Constantin's preliminary results (which he did not identify as preliminary), and ends by stating "a crude average" of 64×10^{22} after citing some results on concentrated emulsions not involving Brownian motion. We can only wonder whether he ever acknowledged in print that his values for Avogadro's number from Brownian motion were notably higher than the 60.6×10^{22} Millikan value that had long since become standard.
4. W. Ostwald (1912), p. vi.
5. Ibid., p. 485f.

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6. Eucken (1914), p. 376.
7. Bohr (1913), p. 5.
8. Bragg and Bragg (1913), p. 437. The Braggs needed a value for the mass of hydrogen in order to infer the spacing between atoms in the crystal lattice, and with it the wave length of the x-ray. Their value for the mass had to have been derived from a value for Avogadro's number, specifically 61.5×10^{22}.
9. Jeans (1916). In the subsequent 1921 and 1926 editions, Jeans adopts a value of $(60.62 \pm 0.06) \times 10^{22}$, obtained from Millikan's 1917 value for e, 4.774×10^{-10}.
10. Perrin (1908a), (1908c), (1908d); Chaudesaigues (1908). At the time his new edition went to press, Ostwald appears to have read only the last of these notes by Perrin, which initially lists 7.1×10^{23} as the "approximate" value of Avogadro's number, the value Ostwald cited in his book.
11. Perrin's (1908d) ends with two short paragraphs on the top of the next page, the first pointing out that the masses of atoms and molecules can be inferred from his value for Avogadro's number "with the same precision," giving 0.45×10^{-22} grams for the mass of the oxygen molecule, 1.4×10^{-24} for the hydrogen atom, and 0.75×10^{-27} for the mass of the electron; and the second infers from the density of liquid oxygen a diameter for a molecule of oxygen of 2.6×10^{-8} cm and for a molecule of helium, 1.7×10^{-8}.
12. W. Ostwald (1912), p. 485.
13. Nernst (1911), p. 207.
14. Ehrenfest and Ehrenfest (1959), p. 37.
15. Mazo (2002), p. 9. The translation is Mazo's, as is the italicization of the word *experimental*.
16. We thank Paul Forman for repeatedly calling our attention to the low standing of Ostwald in the physics community in the first decade of the twentieth century.
17. The literature on the controversy and the critiques of Helm, Ostwald, and energetics is too large to cover here. A classic source is Hiebert (1971). More recently Robert Deltete has published a number of works on energetics, of which we found the following especially informative for our purposes: (2007a), (2007b), (2007c), (2010), and (2012).
18. Boltzmann (1995), p. 407. The original German edition appeared in 1896 and 1898.
19. W. Ostwald (1909), p. 571; in translation (1912), p. 510. The chapter on the kinetic theory of gases, by the way, reads more or less word-for-word as in the earlier editions up to the point that Ostwald derives a value for Avogadro's number near its end; it has, however, been moved to a much later point in the book, inserted within the new chapters on disperse systems.
20. W. Ostwald (1909), p. 544; (1912), p. 485, where the translator has chosen to split the German paragraph into two paragraphs.
21. W. Ostwald (1909), p. 544. The English translator chose to shorten this sentence: "Still more definite confirmation has been afforded very recently by the measurements of Perrin (1908)" (Ostwald, 1912, p. 485).
22. W. Ostwald (1907), p. 500.
23. W. Ostwald (1912), p. 485.

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24. W. Ostwald (1912), p. 483f; (1909), p. 541f in the German.
25. W. Ostwald (1912), p. 491; (1909), p. 550f in the German.
26. Ibid.
27. W. Ostwald (1912), p. 485; the emphasis is not included in the translation, but is present in the German (1909), p. 543.
28. W. Ostwald (1912), p. 485f; (1909), p. 544; as we noted earlier, the constant Ostwald cites in the paragraph is *R*, the gas constant, not Perrin's featured molecular value, Avogadro's number. This is one more sign that the paragraph was inserted in proof, allowing a fairly egregious mistake to escape unnoticed. In the subsequent citation of Perrin, in the chapter on kinetic theory, Ostwald correctly identifies the constant as Avogadro's number, specifically ".71 × 10^{24}."
29. W. Ostwald (1912), p. 510; (1909), p. 571. The conclusion Ostwald offers in the case of solids is worth quoting:

    > While the molecular motion in liquids must be assumed to be uniform in all directions, since they are isotropic in every property, crystalline solids exhibit ordered anisotropism, i.e. suitable properties are functions of the direction in accordance with the general laws of symmetry, which find their highest expression in the crystal form. Hence the assumption that the molecules have different properties in different directions, and that in crystals they are arranged either parallel to each other or at least in congruent relative positions (e.g. alternately perpendicular to each other), so giving rise to anisotropism.
    >
    > The immediate conclusion from this, that the form of the molecule, indicated by the chemical constitution, must find expression in the crystal form, has not yet led to any simple general results, in spite of the attention which has recently been devoted to the idea; in certain special cases, however, some relations of this kind have been recognized.

    The 1909 edition includes a chapter on crystals much earlier in the book, which remained largely word-for-word the same as in the 1899 edition; the latter, however, departs significantly from the corresponding chapter of the 1890 edition.
30. W. Ostwald (1912), p. 510.
31. W. Ostwald (1912), p. 538; (1909, p. 602). Emphasis in both.
32. H. Wilson (1903), p. 439.
33. W. Ostwald (1912), p. 383.
34. Ibid., p. 535f. A footnote attached to the end of the quote says, "Isolated instances have been noticed in which ions occurred with double charges; still the simple ions were apparently in marked abundance." This sentence must have been written before Ostwald had read the two Rutherford-Geiger papers of 1908 and the clear case they make that α-particles are doubly positively charged, and that a more reliable value for the basic charge is 4.65 × 10^{-10} esu (= 1.55 × 10^{-19} coulombs). Insofar as Perrin cites the Rutherford-Geiger results in the 1908 paper cited in two places by Ostwald, this is still further evidence that the citations in these two places were added either while the book was in press or after the manuscript had been completed.
35. Ibid., p. 545.
36. Ibid., p. 539. Here too Ostwald might as well have cited Thomson (1900), for the emphasis put on the asymmetry of electric charge and electric current consisting

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of the flow of negative, not positive, charges is little more than a summary of that landmark paper.

37. Ibid., p. 540.
38. Ibid., p. 485.
39. W. Ostwald (1904, p. 506).
40. The proceedings in German were not published until 1914, well after publication of the Bohr model.
41. Bohr (1913), p. 9. We shall discuss Bohr's reasons for this choice in section 7.6.
42. Bragg and Bragg (1913), p. 437. That it was atoms of sodium and chlorine occupying the lattice points in the crystal, and not salt molecules as generally thought before this publication, was notable. It must have been especially gratifying to the four seminal figures of physical chemistry.
43. Planck (1991), p. 172f. The original, *Wärmestrahlung*, was published in 1913, with a Preface dated November 1912.
44. Jeans (1916), p. 9. The passage continues, "The divergences arise not only from the fact that the shape of the molecules is not spherical, but also from the fact that the molecules are surrounded by fields of force, and in most experiments it is the extension of this field of force, rather than that of the molecules themselves, with which we are concerned."
45. These results were summarized by Eucken (1914) in the appendix he added, as editor, to the 1914 publication in German of the proceedings of the 1911 Solvay Conference. A summary of the progress that had been made by 1913 in confirming the Einstein-Debye theory of specific heats of solids can also be found in Jeans (1914), pp. 66–78.
46. van Fraassen (2009), p. 22f.
47. Cohen and Koyré (1972), p. 555 (p. 389 in the original Latin, p. 796 in the Cohen and Whitman translation [Newton, 1999]). Newton himself, by the way, appears to have maintained a distinction between the words *exacté* and *accuraté*— hence our including the Latin. The original Latin is, "*In philosophia experimentalis, propositiones ex phaenominis per inductionem collectae, non obstantibus contrariis hypothesibus, pro veris aut accurate aut quamproxime haberi debent, donec alia occurrerint phaenomena, per quae aut accuratiores reddantur aut exceptionibus obnoxiae.*"
48. Brush (1976b), p. 697f; originally published as Brush (1968).
49. Gibbs (1948), p. ixf.
50. Euler, from the Preface of his *Recherches sur les irrégularités du mouvement de Jupiter and de Saturne*, 1752; translation from C. Wilson (1980), p. 144.
51. The vortex theory was originally put forward by Descartes in his *Principia*, but then was formulated in a more mathematical version by Leibniz in his "Tentamen de Motuum Coelestium Causis," published in *Acta Eruditorum* in 1689, two years after the first edition of Newton's *Principia*. For a history of vortex theories, see Aiton (1972).
52. See Smith (2014); and "25 Years of Second Thoughts: Substantive Changes in the 1713 Edition of Newton's *Principia*," in preparation.
53. Smith (2010).

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54. Nernst (1909), p. 438. The sentence occurs in the 1911 English translation as well (Nernst, 1911, p. 437), but Nernst had meanwhile authorized the translator to add a paragraph on Perrin's results from Brownian motion that had not been included in the list in German of recent results for Loschmidt's number that was providing the basis for the quote. The list in the original was: 4.5×10^{19} from kinetic theory; 3.7×10^{19} from values for the charge per ion from the method "of cloud formation"; 2.78×10^{19} from the charge of α-particles (as per Rutherford and Geiger); and 2.76×10^{19} from blackbody radiation.
55. Jeans (1904), p. 1; (1916), p. 1.
56. Poincaré (1952), p. 152f.
57. Sidgwick (1884), pp. 250 and 270ff.
58. Maxwell (1965), p. 775.
59. See Table 2.2 in Chapter 2.
60. Meyer (1899b), p. 332; (1899a), p. 334 in the German original.
61. Glashow (1997), p. 276, italics added. Smith thanks the late Sylvan (Sam) Schweber for having called his attention to this remark.
62. See Kragh (2010) and (2003).
63. Heisenberg (1967), p. 263.
64. van Fraassen (2009), p. 22.
65. Nernst (1911), p. 280.
66. For a limited example of such a review that reveals how far from simple and straightforward that evidence can be (at least in the case of diatomic molecules), see Miyake and Smith (2020).
67. See Poynting (1894) for a thorough history of such measurements plus his own contribution to them.
68. Zenneck (2007). For those who find the claim curious, keep in mind that there is no way to test the claim in question celestially because all bodies at any point respond to the gravitational attraction toward any other body in the same way, regardless of their mass. So, as Zenneck notes, the claim that the attraction toward, for instance, the Sun varies as its mass presupposed that the third law of motion applies to gravity. The experiments Zenneck cites do not presuppose the third law of motion. In fact, the masses that appear in the equations of celestial mechanics are all derived from the strength of the gravitational attraction toward the bodies in question, and they are then deployed always in the form *GM*, that is, as an acceleration; hence those masses serve as nothing but placeholders for the field strengths, expressed as accelerations.
69. Thomson (1897).
70. Thomson (1897), p. 312.
71. Thomson (1899), p. 565. The paragraph that opens with the quoted passage ends with, "These considerations have led me to take as a working hypothesis the following method of regarding the electrification of a gas, or indeed of matter in any state"; what follows is the twentieth-century conception of electrical action. For a review of these developments, see Smith (2001b).
72. Thomson (1898).
73. Townsend (1900).

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74. Planck (1967), p. 89f.
75. Planck (1922), p.11.
76. Bohr (1913), p. 9. Bohr's formula, technically for Rydberg's constant times the speed of light, presupposes that the nucleus is of infinite mass. As research developed following this paper, the distinction among the Rydberg constant for this case and Rydberg constants for hydrogen and the helium ion became an important source of evidence historically. See Bohr (1914).
77. This relation was confirmed in 1916 by Millikan in his (1916).
78. Sommerfeld in one place went so far as to propose that the measured values for the Rydberg frequencies for the hydrogen atom and helium ion together with that of the fine-structure constant "provide us *with three determining equations for the three unknowns, e, m, and h*, that is, for the three most important universal constants of physical nature" (1923, p. 526).
79. Dirac (1928a), (1928b); and Darwin (1928).
80. Birge (1929). As others have noted, this was the first article in the journal that was subsequently retitled *Reviews of Modern Physics*.
81. See Seidelmann (1992), p. 616.
82. Birge (1929), p. 1f.
83. Birge (1945).
84. Birge's 1945 value for Faraday's constant, in abs.-coulomb units, was 96,487.7 ± 10. The CODATA preferred value, as of 2018, is 96,485.33212, taken to be exact in conjunction with the new SI units of 2019.
85. See Schweber (1994), Chapter 5, and (2015), pp. vii–lvi, especially pp. xvff.
86. Taylor et al. (1969a). This article appeared almost immediately in book form (1969b).
87. Ibid., p. 375.
88. For a 900-page discussion of such evidence, see T. Kinoshita (1990).
89. Mohr et al. (2008), p. 721.
90. See Taylor et al. (1969a), p. 477; Mohr and Taylor (2000), p. 447; and Mohr et al. (2016), p. 56.
91. Stock et al. (2019), p. 1. For details on the CODATA 2017 adjustment that formed the basis for the new SI, see Mohr et al. (2018).
92. https://physics.nist.gov/constants.

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